ALAUTOMATIC PLASTIC BOTTLE SEGREGATION SYSTEM

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by

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ABSTRACT

This study addresses the persistent issue of improper plastic bottle segregation, which reduces recycling efficiency and contributes to contamination in waste management systems. To mitigate this problem, the researchers developed an automated segregation system integrating image classification, sensor monitoring, mechanical sorting, and an online tracking interface. A dataset of 11,712 labeled images representing various plastic bottle types was used and divided into training and testing sets. The model achieved a validation accuracy of 98.6% and consistently demonstrated strong performance across all categories, with recognition rates exceeding 90%. The system reliably detected plastic bottles under varying environmental conditions and recorded real-time data through an online platform (bottlesort.online). Overall, the findings indicate that automated classification and monitoring significantly enhance sorting efficiency and support more effective waste management practices.

Keywords: Plastic Bottle Segregation, Machine Learning Classification, Automated Sorting, Waste Management Technology, Real-Time Monitoring, Image Detection.

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Chapter 1

INTRODUCTION

1.1. Background of the Study

The rapid increase in the use of plastic bottles has become a major global environmental concern. Commonly used for packaging beverages and various liquid products, plastic bottles significantly contribute to the growing volume of plastic waste generated each day (Geyer et al., 2017). Unfortunately, a substantial portion of these bottles is improperly discarded, eventually accumulating in landfills, waterways, and marine environments. This persistent mismanagement has intensified environmental degradation, with plastic pollution emerging as a serious threat to ecosystems—particularly marine life. Each year, millions of tons of plastic enter the oceans, posing severe risks to biodiversity and disrupting the balance of marine ecosystems (Jambeck et al., 2015).

The inefficiency of current waste-sorting and recycling processes further aggravates this issue. When recyclable plastic bottles are mixed with non-recyclable or contaminated waste, recycling becomes labor-intensive, costly, and less effective, resulting in low material recovery rates. This challenge is especially evident in developing countries that lack sufficient waste management infrastructure (Fidje et al., 2023). As a result, improper disposal practices such as open dumping and incineration remain common, creating serious environmental and public health hazards.

Additionally, landfills in urban areas are rapidly reaching full capacity. Improper disposal of plastic bottles worsens this situation by generating leachate that contaminates soil and groundwater. Furthermore, the incineration of plastic waste

releases toxic gases that pose significant health risks to nearby communities (Fang et al., 2023).

These concerns underscore the urgent need for improved waste-segregation strategies that can support sustainable and efficient plastic waste management.

In response to these challenges, advancements in modern technology have paved the way for automated systems designed specifically for plastic bottle segregation. Equipped with sensors, artificial intelligence (AI), and machine learning, these systems enhance sorting accuracy and efficiency. By minimizing the need for human involvement, they also reduce errors and labor costs, making waste-management processes more scalable and effective (Fang et al., 2023).

Consequently, the integration of AI and Internet of Things (IoT) technologies into waste disposal and management is gaining recognition as a promising solution. These intelligent systems optimize waste-management operations and provide a proactive approach to addressing the global plastic waste crisis.

Guided by these developments, this study aims to design and implement a reliable automated plastic bottle segregation system. By addressing key challenges in plastic waste management through innovative engineering and smart technology, the study seeks to promote environmental sustainability and encourage more responsible recycling practices.

1.2. Statement of the Problem

Insights gathered through an interview with Engr. Esmael Larubis, Environmental Sustainability Officer of La Salle University Ozamiz, revealed that plastic bottles are frequently disposed of together with plastic cups. This practice

increases the workload of maintenance personnel, who must manually segregate these materials. The situation stems from the repurposing of plastic items on campus—plastic bottles are sold to junk shops or sent to Barangay Pulot, Ozamiz City, for use as floaters, while plastic cups are reused in the university's tree-planting activities. This observation highlights the need for an automated system that can improve the speed and accuracy of waste segregation.

Among the types of plastic waste generated on campus, PET (polyethylene terephthalate) bottles are the most commonly discarded. Their prevalence is largely due to their use in packaging everyday consumer beverages such as bottled water and soft drinks, which are frequently purchased and consumed by Lasallians. The continuous generation of PET bottle waste reflects daily consumption habits and contributes substantially to the volume of plastic materials that waste-management personnel must sort manually. This ongoing trend underscores the necessity of adopting a more structured and efficient method of plastic waste segregation at the institutional level.

The issue is even more critical at the national level. The Philippines produces approximately 2.7 million tons of plastic waste annually, with nearly 20% of this entering marine environments (World Bank, 2020). Despite the increasing demand for recycled plastics, only 28% of PET bottles were recycled in 2019, marking a 72% shortfall in recycling capacity compared to neighboring Southeast Asian countries (FactsAsia, 2023). Factors contributing to this shortfall include weak segregation practices, insufficient infrastructure, high equipment costs, and competition from the informal recycling sector (Sustainability.ph, 2022).

Globally, the rising volume of plastic bottle waste poses severe environmental and health risks. Large quantities of plastic are improperly disposed of in landfills and oceans, harming biodiversity and polluting ecosystems (Jambeck et al., 2015; Geyer et al., 2017). In many regions, recyclable plastics are mixed with non-recyclable materials due to inadequate segregation systems, making recycling inefficient and costly (Fidje et al., 2023). Manual sorting remains the dominant method of plastic waste segregation, but it is labor-intensive, prone to human error, and increasingly unsustainable as waste volumes grow. Contaminated plastics further reduce the quality of recyclables and contribute to landfill overflow, while incineration releases toxic emissions harmful to the environment and public health (Fang et al., 2023).

Given these circumstances, this study seeks to address the urgent need for a more efficient solution by developing an automated plastic bottle segregation system. Such a system aims to enhance recycling operations, reduce reliance on manual labor, and support sustainable waste-management practices in the Philippines.

1.3. Objectives of the Study

The primary goal of this study is to design, develop, and implement an automated system capable of sorting plastic bottles based on their physical characteristics, thereby improving the efficiency of waste segregation.

Specifically, the study aims to:

1. Develop an image-processing and detection system capable of identifying the type of plastic used in bottles by analyzing physical attributes such as color, shape, texture, transparency, and labeling.

2. Design and construct an automated sorting mechanism that categorizes and separates plastic bottles according to their identified material type (e.g., PET, HDPE, LDPE, PP, or unclassified).
3. Evaluate and analyze the system's performance, reliability, and accuracy in real-world waste-sorting scenarios.

1.4. Originality of the Study

This study presents a novel approach to plastic bottle segregation by focusing on the physical properties of bottles, identified through advanced image-processing techniques, to enable automatic classification and sorting. While many existing waste-management systems rely on manual sorting or basic sensor-based detection, this project integrates sophisticated material-identification methods with automated sorting mechanisms to significantly enhance efficiency.

By combining advanced image processing with a computerized sorting mechanism, the study addresses current gaps in automated classification systems, particularly those that utilize visual and physical attributes—an area that remains underexplored.

Furthermore, this study contributes to the growing field of innovative waste management by developing a system that improves sorting accuracy while minimizing human error and labor costs. Segregating plastics based on physical characteristics enhances recycling efficiency, ensuring higher-quality recyclables. Ultimately, this study promotes sustainable plastic waste management practices and encourages responsible recycling behavior.

1.5. Significance of the Study

This study is expected to benefit the following groups:

Communities. By improving plastic segregation, communities can reduce pollution and encourage responsible waste-management practices. This contributes to a cleaner environment, enhanced public health, and greater environmental awareness.

Waste Management Personnel. The automated system can assist in accurately and efficiently segregating plastic bottles, reducing manual labor and the risk of human error. This improvement can enhance the overall recycling process within collection and sorting facilities.

Schools and Institutions. Educational institutions can integrate the system into their campus sustainability programs, promoting proper waste management and serving as a model of eco-friendly and innovative practices.

Students. The project increases student awareness of environmental concerns while demonstrating practical applications of technology, engineering, and sustainable innovation.

Future Researchers. This study provides a valuable foundation for further advancements in automated waste-segregation technologies, particularly in the fields of sensor development, plastic classification, and environmental engineering.

This study also aligns with key United Nations Sustainable Development Goals (SDGs):

SDG 9: Industry, Innovation, and Infrastructure. The integration of IoT, AI, and automation in plastic-bottle handling promotes technological innovation and supports the development of sustainable infrastructure.

SDG 12: Responsible Consumption and Production. The study advocates effective waste segregation, reduces plastic waste, and supports recycling efforts, all of which contribute to sustainable consumption and production practices.

SDG 13: Climate Action. By minimizing plastic pollution and promoting reuse and recycling, the study helps mitigate environmental impacts and supports global climate action initiatives.

Overall, this study contributes to both local and global efforts toward sustainability, offering a practical and innovative solution to the growing environmental challenges posed by plastic waste.

1.6. Scope and Limitations

This study focuses on the design and development of an automated system capable of detecting and classifying plastic bottles based on their physical characteristics. Using sensor technologies and image-based detection methods, the system aims to identify common types of plastic bottles, including polyethylene terephthalate (PET), high-density polyethylene (HDPE), low-density polyethylene (LDPE), and polypropylene (PP). The study concentrates on developing the detection, control, and sorting components of the system, as well as evaluating its accuracy and efficiency in controlled environments. The target performance of the system is an accuracy rate of at least 90%, ensuring reliable classification and sorting.

However, the study is subject to the following limitations:

1. **Image-Based Recognition Accuracy**

   The system identifies plastic types by analyzing visual characteristics such as color, shape, transparency, label information, and Resin Identification

Code (RIC) symbols. Its accuracy may decrease when processing bottles that are dirty, deformed, worn out, or partially obstructed, as these conditions obscure key visual features.

2. **Limited Waste Scope**

   The system is exclusively designed for plastic bottle classification. It does not detect or process other forms of plastic waste, such as cups, food containers, sachets, or wrappers.

3. **Environmental Sensitivity**

   The effectiveness of the image-recognition system may be influenced by environmental variables such as lighting conditions, camera angle, and background contrast. These factors can impact detection accuracy, particularly in uncontrolled or low-light settings.

4. **Waste Management Coverage**

   This study focuses solely on the segregation phase of waste management. It does not address post-segregation processes such as waste collection, transport, recycling, or disposal.

5. **Controlled Testing Environment**

   System development and testing were conducted in a controlled indoor environment. Its performance in real-world conditions—such as outdoor areas, large facilities, or public spaces—was not evaluated and may require further refinement and field validation.

6. **Size and Shape Constraints**

   The hardware is optimized for cylindrical or near-cylindrical plastic bottles. Bottles with irregular shapes or unconventional designs may not be

accurately identified or effectively processed. To maintain mechanical stability and consistent operation during testing, bottle dimensions were limited to a maximum height of 10.6 inches (27 cm), a diameter of 5.5 inches (14 cm), and a weight not exceeding 0.5 kilograms. Heavier bottles may hinder the movement of the servo-operated flap gate or place excessive strain on the stepper motor responsible for bin rotation.

1.7. Conceptual Framework

Figure 1

Conceptual Overview of the System

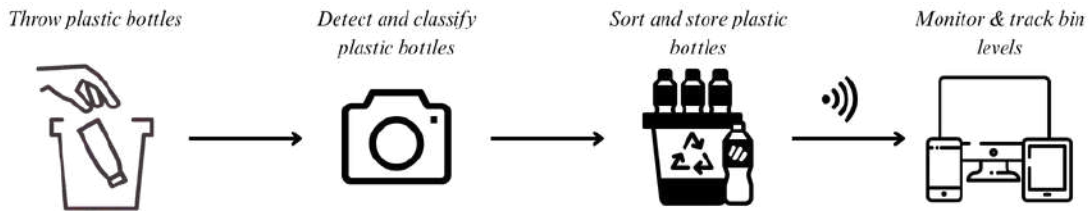


Figure 1 presents the conceptual framework for the automated plastic bottle segregation system. When a plastic bottle is inserted, detection sensors activate the system, sending the input to the Raspberry Pi for processing. The Raspberry Pi utilizes a camera-based classification module to analyze the bottle and determine whether it is PET, HDPE, LDPE, PP, or unclassified.

Upon classification, the system engages the mechanical sorting mechanism, which includes a stepper motor and a servo motor, to direct the bottle into the designated bin. The chosen bin receives the bottle, and an LCD shows the classification result.

Each bin is equipped with an ultrasonic sensor that monitors its capacity. The logs and status data from the bins are transmitted to a remote server for real-time monitoring. This feedback loop facilitates the timely collection of plastic bottles and enhances the efficiency and sustainability of automated plastic-bottle segregation.

1.8. Definition of Terms

The definitions below clarify how key terms are used within the context of this study.

Actuator. A device within a machine that produces movement or controls a mechanism or system.

Artificial Intelligence (AI). A field of computing that enables machines to imitate human cognitive functions such as learning, reasoning, and self-improvement.

Computer Vision. A subfield of AI that enables machines to interpret and analyze visual information from images or videos.

Controlled Environment. A testing setup in which conditions such as lighting, background, and object placement are carefully regulated to ensure consistent and precise results.

Convolutional Neural Network (CNN). A deep learning architecture widely used for processing and analyzing visual data, particularly in image recognition tasks.

Embedded System. A computer system designed around a microprocessor to perform dedicated control functions within a larger mechanical or electronic device.

Features / Output Features. Measurable characteristics or target variables that a machine-learning model aims to analyze, predict, or generate.

High-Density Polyethylene (HDPE). A petroleum-based polymer known for its durability and high impact resistance, commonly used in producing plastic bottles, pipes, and plastic lumber.

Image Recognition. A subfield of AI that enables software to identify objects, scenes, or actions within digital images.

Impulse. Refers to the sequential steps in a machine-learning pipeline, including data collection, processing, feature extraction, and model prediction.

Inferencing Time. The period between the input of a sample into a trained machine-learning model and the generation of the model's predicted output.

Learning Rate. A parameter that determines the magnitude of adjustments made to a model's weights during each iteration of the training process.

Low-Density Polyethylene (LDPE). A flexible and transparent thermoplastic commonly used in plastic bags, squeezable bottles, and film wraps.

Machine Learning. A branch of artificial intelligence that enables computers to learn from data and improve performance without explicit programming.

Microcontroller Unit (MCU). A compact integrated circuit designed to manage control functions in embedded systems.

Peak RAM Usage. The highest amount of RAM consumed by a system during its operation, representing its peak memory footprint.

Plastic Bottle. A durable plastic container with a narrower neck than its body, used for storing liquids such as water, beverages, and cleaning products. Common materials include PET, HDPE, LDPE, and PP.

Polyethylene Terephthalate (PET). A thermoplastic polymer from the polyester family, widely used in beverage containers, food packaging, and textile fibers due to its strength, clarity, and thermal stability.

Polypropylene (PP). A thermoplastic polymer used in packaging, textiles, and various industrial applications, noted for its chemical resistance, flexibility, and durability.

Precision. A metric in machine learning that measures the proportion of correctly predicted positive instances relative to all predicted positive instances.

Raspberry Pi. A low-cost, compact single-board computer commonly used for programming, automation, and electronics projects.

Real-Time Feedback. The immediate delivery of system outputs or responses as events occur within a process.

Resin Identification Code (RIC). A numerical symbol (1–7) found on plastic products that indicates the type of plastic resin used.

Servomotor. A type of actuator that provides precise control over position, speed, and acceleration in either rotational or linear motion.

Smart Waste Management. The use of technology and data-driven systems to optimize waste sorting, collection, and recycling processes.

Training Cycles / Epochs. The number of times a machine-learning model processes the entire training dataset during the training phase.

Transfer Learning. A method that enhances performance on a new task by applying knowledge gained from a previously learned, related task.

Ultrasonic Sensor. A device that emits ultrasonic waves and measures the time it takes for the waves to bounce back after hitting an object, allowing it to detect distance or object presence.

Visual Classification. The process of assigning images to predefined categories by analyzing visible features such as color, shape, and texture.

Waste. Unwanted materials or substances discarded after use in households, industries, or other environments.

Waste Segregation. The process of separating waste into specific categories to improve recycling efficiency and waste-management effectiveness.

Chapter 2

REVIEW OF RELATED LITERATURE

This section outlines and analyzes the existing research literature concerning plastic waste management, segregation practices, Machine Learning, IoT, and the technologies and systems supporting automated plastic bottle classification.

2.1. Global Plastic Bottle Consumption

Approximately 1 million plastic bottles—classified as single-use plastics—are purchased every minute worldwide. These single-use plastics represent the largest category of ocean plastic pollution by weight, accounting for nearly 70% of all plastic waste collected during coastal cleanup activities. This widespread pollution poses significant threats to wildlife, as animals may become entangled in plastic debris or ingest it, leading to injury or death. Additionally, the incineration and open burning of plastic waste release harmful pollutants into the atmosphere and soil, while plastic litter blocks waterways and drainage systems, exacerbating flooding and increasing the risk of disease outbreaks (World Wildlife Fund, 2019).

Rivers and coastlines also play significant roles in transporting bottle waste into the oceans. Each year, an estimated 8 to 10 million metric tons of plastic—bottles included—enter the marine environment, comprising approximately 80% of total ocean litter. Plastic bottles, often produced from polyethylene terephthalate (PET), persist in the environment for centuries. Over time, they break down into microplastics, which have been detected in marine organisms, freshwater ecosystems, drinking water supplies, and even human blood. These microplastics pose potential

risks to ecological and human health, underscoring the urgent need for improved waste management solutions (UNESCO Ocean Literacy, 2022).

2.2. Resin Identification Code (RIC)

The Resin Identification Code (RIC) system was established in 1988 by the Society of the Plastics Industry in the United States to support recycling efforts by providing a standardized method for identifying plastic resins used in consumer products (Agarwal et al., 2020). The system employs numbers from 1 to 7, each displayed within a solid triangle symbol. Each number corresponds to a specific plastic type:

- 1 – PET (Polyethylene Terephthalate): Commonly used in beverage bottles.
- 2 – HDPE (High-Density Polyethylene): Found in containers such as detergent bottles and milk jugs.
- 3 – PVC (Polyvinyl Chloride): Used in pipes and certain food packaging.
- 4 – LDPE (Low-Density Polyethylene): Used for plastic bags, films, and wraps.
- 5 – PP (Polypropylene): Found in bottle caps, food containers, and some laboratory equipment.
- 6 – PS (Polystyrene): Used in foam packaging and disposable cutlery.
- 7 – Other: Includes less common or multi-layer plastics like polycarbonate (ANSI, 2020).

Beyond consumer guidance, the RIC system remains relevant in research and modern waste management technologies. For instance, Bułkowska, Zielińska, and Bułkowski (2024) integrated the RIC framework into a smart waste management

system powered by blockchain, IoT, and artificial intelligence to enhance traceability and sorting efficiency. Likewise, Agarwal, Gudi, and Saxena (2020) used computer vision techniques to detect RIC symbols, achieving high classification accuracy, thereby highlighting the utility of RIC labels in automated recycling systems.

2.3. Characteristics, Applications, and Recycling of Plastic Bottle Materials

Polyethylene Terephthalate (PET)

Polyethylene terephthalate (PET) is one of the most widely used materials in beverage packaging due to its lightweight nature, clarity, durability, and strong resistance to breakage. These qualities make PET a practical and safer alternative to glass, particularly in settings where portability and safety are essential. Its transparency provides a marketing advantage by allowing consumers to view the product's contents. PET is also highly recyclable, making it a preferred choice for sustainable packaging initiatives (DLG Expert Report, 2016).

High-Density Polyethylene (HDPE)

HDPE, identified by RIC 2, is valued for its toughness, moisture resistance, and versatility. It is commonly used in shampoo bottles, milk containers, and cleaning product packaging. Although HDPE originates from non-renewable fossil fuels and can persist for centuries in landfills (Lansen, 2023), it is widely accepted in recycling programs and can be processed into new containers, pipes, and plastic lumber.

Low-Density Polyethylene (LDPE)

Low-Density Polyethylene (LDPE) is extensively used in flexible packaging due to its lightweight structure, chemical resistance, and durability. It is commonly found in squeeze bottles, food packaging, and dispensing containers (Rahman et al.,

2022). Despite being recyclable, LDPE poses significant challenges in waste recovery because it is difficult to manually distinguish from other plastics, leading to low recycling rates.

To address this issue, recent advancements have introduced vision-based systems and robotic arms capable of real-time sorting. Sundaralingam and Ramanathan (2024) demonstrated that such technologies can effectively differentiate LDPE from other polyolefins, even when visual differences are subtle. Artificial intelligence and deep learning methods have also shown high potential for identifying LDPE based on surface textures and varying shape patterns, particularly in post-consumer waste (Meeradevi et al., 2020).

On the policy side, national frameworks such as the Philippine Plastic Roadmap (Solid Waste Management Division, 2024) emphasize the importance of public engagement and investments in material recovery facilities to improve recycling outcomes for low-value plastics like LDPE.

Polypropylene (PP)

Polypropylene (PP) is widely used in packaging and constitutes a substantial portion of post-consumer plastic waste. Significant advancements have been made to improve PP recycling, including Project OMNI by TotalEnergies, Recycleye, and Valorplast, which integrates AI-driven computer vision, robotics, and decontamination cycles. This system achieves a selection rate of about 50% and a purity level exceeding 95% for food-grade PP, enabling high-quality recycled PP suitable for packaging (TotalEnergies, Recycleye, & Valorplast, 2023).

Technologies such as computer vision and near-infrared (NIR) spectroscopy have shown excellent performance in identifying aged and post-consumer PP, with

classification accuracy reaching up to 99% (Zhu et al., 2025). NIR-based methods have also proven highly effective for distinguishing polyolefins, achieving more than 95% accuracy (Sutliff et al., 2024). However, challenges remain in contamination control, maintaining high throughput rates, and ensuring the safety of recycled PP. Continued improvements in decontamination processes and system durability are essential to support reliable industrial-scale recycling operations (Batchelor, 2023).

Table 1

Material Properties and Visual Features of Plastic Bottle Materials by RIC

RIC	Plastic Material	Material Properties	How to Identify
1	PET (Polyethylene Terephthalate)	High strength, excellent moisture barrier, good gas barrier, recyclable	Transparent, glossy, often smooth surface
2	HDPE (High-Density Polyethylene)	Tough, chemical-resistant, impact-resistant, suitable for heavy-duty containers	Opaque, matte finish, solid colors,
4	LDPE (Low-Density Polyethylene)	Flexible, low tensile strength, high ductility, resistant to chemicals and UV	Very glossy, semi-transparent
5	PP (Polypropylene)	High fatigue resistance, heat-resistant, resistant to many chemicals	Slightly cloudy or translucent, smooth surface

Table 1 presents the visual characteristics and material properties of commonly used plastic bottle materials categorized by their Resin Identification Code (RIC). This classification is essential for supporting image-based plastic waste recognition systems and improving material-specific recycling efforts. The table

outlines both physical material properties and observable surface features, which are important for automated sorting technologies.

2.4. The Importance of Plastic Segregation

Poor plastic waste management can significantly pollute soil, air, and water, severely impacting ecosystems and human health (Geyer et al., 2017). When different types of resins are mixed, entire batches of plastic become unsuitable for material recovery facilities, often ending up in landfills or being incinerated, rendering recycling efforts ineffective (Burkasan, 2024). Proper segregation at the source—sorting plastics into PET, HDPE, LDPE, PP, and other resin codes—creates homogeneous streams, reducing energy consumption in downstream processing by up to 30% (Burkasan, 2024).

Contaminated loads also contribute to greenhouse gas emissions, as producing virgin resin requires significantly more energy than recycling existing plastics (U.S. Environmental Protection Agency, 2025). Moreover, transparent bin labeling and awareness campaigns have been shown to increase household participation by 20–35% in various case studies, further enhancing environmental benefits (WasteMission, 2025). Proper source segregation is therefore fundamental to a circular plastics economy, preserving material value while preventing pollution (Greyhound Recycling, 2025).

The consequences of poor plastic management extend beyond direct environmental pollution. Inefficient recycling processes, threats to public health, and long-term environmental degradation all arise from improper disposal. Plastics that are decomposed or incinerated release harmful chemicals such as dioxins and furans,

posing serious risks to both ecosystems and humans (UNEP, 2021). Prolonged exposure to these toxicants has been associated with respiratory illnesses, endocrine disruption, and even cancer (World Health Organization, 2023).

Improper segregation also increases contamination and diminishes material recyclability. Exposure to moisture or mixing with organic waste accelerates degradation into microplastics, which have been detected in aquatic organisms, drinking water, and even human blood (Barboza et al., 2018). These microplastics now contaminate not only water and marine environments but also the atmosphere and terrestrial ecosystems.

Systematic reviews indicate the growing severity of this threat. Microplastics have been linked to oxidative stress, metabolic disorders, neurotoxicity, and reproductive harm in humans (Health Effects of Microplastic Exposures, 2024). Chronic exposure has been associated with gastrointestinal and respiratory inflammation, hormonal imbalances, and potential immune system dysfunction (Health-care Waste WHO, 2024; Health Risk of Human Exposure to Microplastics, 2024).

Taken together, these findings underscore the critical importance of effective plastic segregation and proper disposal practices—not only to enhance recycling efficiency but also to protect public health and safeguard the environment from long-term, potentially irreversible damage.

2.5. Practices for Managing Plastic in the Philippines

In the Philippines, plastic waste management is governed primarily by Republic Act No. 9003, also known as the Ecological Solid Waste Management Act

of 2000. This law mandates that individuals and organizations segregate their waste into recyclables, organic waste, residuals, and special wastes, including hazardous materials. The Act also prescribes the use of clearly labeled, color-coded bins—blue for recyclables, green for organics, black for residuals, and red for special wastes. Despite this legislation, enforcement has been inconsistent nationwide. In 2019, only 28% of major plastic resins consumed were recycled, while the remainder was either improperly managed or incinerated as refuse-derived fuel in cement production, raising concerns regarding air quality and public health (PEMSEA, 2022).

The Extended Producer Responsibility (EPR) law, enacted in 2022, shifts the responsibility for plastic recovery to manufacturers, targeting a 40% recovery rate by 2024 and 80% by 2028. Initial data indicate a 37% year-over-year increase in EPR registrations, though challenges remain with financing and logistics (World Bank, 2024). Informal waste collectors continue to handle the bulk of waste recovery, leading to collection coverage gaps of 80–90% in major urban areas and as low as 40% in rural regions (PEMSEA, 2022). Pilot deposit-return programs in select municipalities have shown over a 20% increase in bottle returns, highlighting the effectiveness of linking incentives to proper waste segregation.

Despite these regulatory frameworks, plastic waste management in the Philippines faces numerous structural and societal obstacles. According to the National Commission on Solid Waste Management (NCSWM, 2022), only 35% of the country's 1,600 local government units have material recovery facilities; the remainder rely on open dumpsites or uncontrolled landfills. Infrastructure limitations are particularly pronounced in rural and impoverished areas. Public education and engagement are inconsistent, leaving many households unaware of regulations or

lacking access to the proper bins and guidance for effective segregation (USAID Clean Cities Blue Ocean, 2021). Consequently, many communities continue to mix waste despite source-level segregation mandates. Those that comply often segregate based on perceived economic value, such as separating PET bottles or aluminum cans for sale to junk shops. In some cities, segregation occurs only at collection or transfer stations, undermining the intended benefits of source-level separation (DENR-EMB, 2023).

Collaborative initiatives between the private sector and communities offer promising solutions. Programs such as Plastic Bank and Basura Mo, Bayad Ko provide financial incentives to residents who collect and sort plastic waste, effectively transforming waste into a revenue source (Plastic Bank, 2023). These programs leverage digital monitoring and blockchain technology to improve traceability and accountability in take-back initiatives. While national-scale implementation remains challenging, local successes demonstrate that combining behavioral incentives with regulatory enforcement can significantly improve plastic waste management practices in the Philippines.

2.6. Artificial Intelligence and IoT in Plastic Waste Management

The Internet of Things (IoT) is transforming waste collection logistics through the use of “smart bins” equipped with ultrasonic and weight sensors, providing real-time fill-level data. This information enables dynamic route optimization, reducing collection costs by 15–25% (Infosys, 2024). Smart bins minimize unnecessary pickups, resulting in fewer miles traveled, lower fuel consumption, and reduced greenhouse gas emissions (Recycling Inside, 2024).

Modern prototypes incorporate RFID tags and basic computer vision units at the bin entrance, allowing first-time material identification and minimizing contamination before transport (IJFMR, 2024). IoT central platforms aggregate bin-status information into dashboards that forecast fill-level trends, schedule maintenance proactively, and pinpoint contamination hotspots for targeted public education campaigns (Infosys, 2024). This connectivity provides comprehensive visibility of material flows, facilitating informed policy-making and robust circular economy monitoring.

Advances in deep learning and computer vision now achieve 90–99% classification rates on conveyor belts, efficiently separating PET, HDPE, and PP streams in under 200 milliseconds per unit (ScienceDirect, 2024). Industrial systems, such as TOMRA, employ multispectral cameras and convolutional neural networks to reach over 95% purity in PET sorting, successfully distinguishing between food-grade and non-food-grade plastics (MDPI, 2024).

A study by Banaag Jr. et al. (2024) at La Salle University - Ozamiz during A.Y. 2024–2025 developed a plastic waste segregation system using an ESP32-CAM, achieving 90% classification accuracy. The system effectively identified recyclable, non-biodegradable, biodegradable, and unclassified materials, demonstrating resilience in handling complex or crumpled items despite dirt or reflective surfaces.

Recent Edge-AI platforms operating on microcontrollers have also achieved around 90% accuracy in laboratory tests, enabling intelligent sorting in environments with bandwidth limitations (Springer, 2023). Additionally, federated learning techniques allow models to be trained across multiple facilities without exposing raw image data, addressing privacy and bandwidth concerns (Springer, 2023). While

laboratory results are promising, further field testing under variable lighting, contamination, and conveyor throughput conditions is essential to validate system robustness before large-scale deployment.

2.7. The 80/20 Principle in Data Splitting and Model Evaluation

In machine learning and data science, selecting an appropriate data-splitting ratio is critical for developing robust, generalizable models. Data splitting involves dividing a dataset into separate portions: one for training the model and another for testing its performance on unseen data. This ensures that the model not only learns the patterns within the training set but also adapts effectively to new, unseen inputs.

A widely accepted standard for data splitting is the 80/20 rule, which allocates 80% of the data for training and 20% for testing (Joseph, 2022; Wijaya, 2023). Wijaya (2023) links this approach to the Pareto Principle, which suggests that roughly 80% of outcomes arise from just 20% of the contributing factors. In machine learning, this implies that a relatively small subset of highly informative data points often drives most of the model's predictive performance. By following the 80/20 split, model training focuses on the most impactful data, offering a resource-efficient approach without sacrificing evaluation reliability.

However, Joseph (2022) notes that the optimal data-splitting ratio can vary depending on factors such as dataset size, model complexity, and the bias–variance trade-off. For instance, a 70/30 split may be preferable for large datasets, allowing more data for testing and evaluation, whereas a 90/10 split might suit smaller datasets, maximizing training data at the cost of less robust testing. The 80/20 ratio provides a

practical compromise between training sufficiency and testing reliability, minimizing risks of both underfitting and overfitting.

In summary, the 80/20 split balances learning capacity and evaluation accuracy, providing sufficient data for pattern recognition while reserving a meaningful portion for independent validation. Its alignment with the Pareto Principle and its consistent statistical effectiveness make it a default benchmark in model development, widely valued for its simplicity, practicality, and proven reliability across diverse applications.

2.8. Key Hardware Elements for Plastic Bottle Segregation

The development of an automated plastic bottle segregation system relies on several critical hardware components that enable real-time detection, classification, and sorting. At the core is the Raspberry Pi 5, an affordable yet powerful single-board computer. Equipped with a 64-bit quad-core Arm Cortex-A76 CPU operating at 2.4 GHz and up to 8 GB of LPDDR4X RAM, it efficiently executes lightweight convolutional neural networks (CNNs) for image-based classification. Optimized CNN models running on the Raspberry Pi have demonstrated high performance in identifying various waste types—including plastic bottles, aluminum containers, and cardboard—making it suitable for embedded and edge computing applications (Ju et al., 2023). Fteiha et al. (2024) utilized the Raspberry Pi in a reverse vending machine, achieving an 82% accuracy for bottle classification, highlighting its suitability for real-time automated sorting.

The Raspberry Pi Camera is integral to image acquisition, capturing high-quality visuals required for object detection and classification. Mwangi and

Mokoena (2019) demonstrated its effectiveness by evaluating PET bottles for recycling, achieving a classification accuracy of 98%. Its small form factor, seamless compatibility with the Raspberry Pi, and capability for detailed image capture make it a practical choice for low-cost, embedded waste management systems.

Microwave sensors offer non-contact detection by analyzing material dielectric properties through reflected electromagnetic waves. Zhang et al. (2019) highlighted their ability to classify materials based on permittivity shifts, making them suitable for low-visibility conditions. Meanwhile, the HC-SR04 ultrasonic sensor is commonly used for monitoring bin fill levels. It calculates distance by emitting ultrasonic waves and timing the echo, providing accurate, real-time bin capacity data. Onyedinma and Onwumbiko (2025) reported its reliability in IoT-based waste monitoring systems, demonstrating efficiency in reducing manual inspections. With a range of 2 to 400 centimeters, the HC-SR04 ensures precise continuous observation in automated waste segregation setups.

The KY-003 Hall Magnetic Sensor detects the presence of a magnetic field and provides a digital on/off output. Its simplicity and affordability make it ideal for applications requiring position or speed sensing. Studies by Kutaj and Boroš (2017) have employed Hall-effect sensors for contactless switching and advanced sensing applications, including security systems.

For visual feedback, the Liquid I2C Crystal Display (LCD) plays a critical role in displaying real-time updates, such as bin fill levels and the classification of detected bottles. Pratama and Harijono (2024) demonstrated that pairing an I2C LCD with the HC-SR04 sensor enhances system readability and responsiveness.

Finally, servomotors translate digital classification outputs into precise mechanical movements, directing bottles into the appropriate bins. Kavitha et al. (2025) demonstrated the integration of deep learning with servo-operated sorting, achieving reliable separation of waste categories with minimal human involvement. In this study, servomotors ensure accurate, consistent sorting of PET, HDPE, LDPE, and PP bottles, supporting the goal of a fully automated and efficient waste management system.

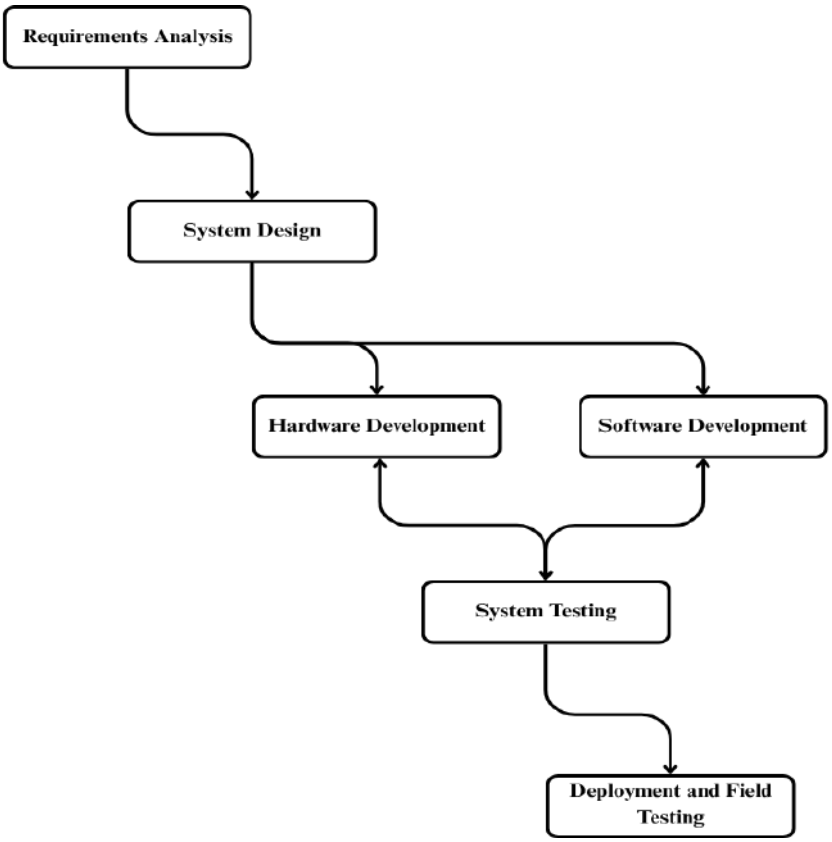
Chapter 3

METHODOLOGY

This study employs a modified Waterfall development approach, incorporating validation and verification at each phase to ensure the timely identification and correction of discrepancies. By integrating iterative feedback loops within the traditional linear Waterfall model, the researchers are able to refine system components progressively, enhance accuracy, and maintain alignment with project requirements throughout the development process.

Figure 2

Waterfall Development Approach for the Study



3.1. Research Process

The research process began with the requirements analysis phase, during which the problem was clearly identified, and the objectives of the study were formally established. This stage involved gathering background information, defining the purpose of the proposed system, and ensuring that all requirements were complete, measurable, and aligned with the overall research goals.

Next, the system design phase outlined both the structural and functional aspects of the intended solution. A comprehensive review of related literature was conducted to address existing knowledge gaps and strengthen design decisions. Design verification ensured that the proposed architecture satisfied the predefined requirements, while validation confirmed that the system was feasible and suitable for real-world applications.

Following this, the hardware and software development phases were carried out. The hardware components were assembled and tested according to the design specifications. At the same time, the software—comprising the system's logic, user interface, and control mechanisms—was developed to support the system's operational requirements. Each component underwent independent verification and validation to confirm accurate functionality and compliance with standards.

Once the hardware and software components were completed, the study proceeded to the integration and testing stage. Here, both components were combined and evaluated as a unified system. Comprehensive testing was conducted to ensure that the integrated system met all technical requirements and research objectives.

After successful integration, the implementation phase deployed the system in a real-world environment to assess its practical performance. This stage provided an

opportunity to evaluate system behavior under operational conditions and to collect relevant performance data.

Finally, the collected data were analyzed to derive insights, determine system effectiveness, and identify any limitations. The conclusions drawn from this analysis informed recommendations for system improvements and guided suggestions for future research. When necessary, earlier phases were revisited to refine and optimize the system, ensuring continuous improvement throughout the research process.

3.2. Requirement Analysis

The requirements analysis stage focused on identifying the technical, functional, and operational considerations necessary to develop the automated plastic bottle segregation system. This process began with understanding the existing issues in local waste segregation practices. Through observations and an interview with Engr. Esmael Larubis, the Environmental Sustainability Officer of La Salle University Ozamiz, noted that plastic bottles and cups are often disposed of together. This practice increases manual sorting time and labor demands for maintenance staff. This finding helped establish the system's primary goal—to automate the separation process, thereby reducing human effort and improving overall efficiency.

To support this goal, the researchers reviewed literature and explored the feasibility of using low-cost, readily available hardware components suitable for image-based classification and mechanical sorting. Components such as the Raspberry Pi 5, Pi Camera Module, microwave sensor, HC-SR04 ultrasonic sensor, hall-effect magnetic sensor, I2C LCD, and servomotors were identified as key elements. These

were selected due to their affordability, availability, and reliability in constrained environments.

The classification method was also defined based on the Resin Identification Code (RIC) system, focusing on physical characteristics that can be visually captured through image processing. This framework guided the preparation of the dataset and the training of the classification model in later stages. The system's core functions include capturing images, classifying bottle types, monitoring bin fill levels, providing real-time output display, and activating servomotors to sort plastic bottles into their designated containers.

3.3. System Design

Figure 3

System Design

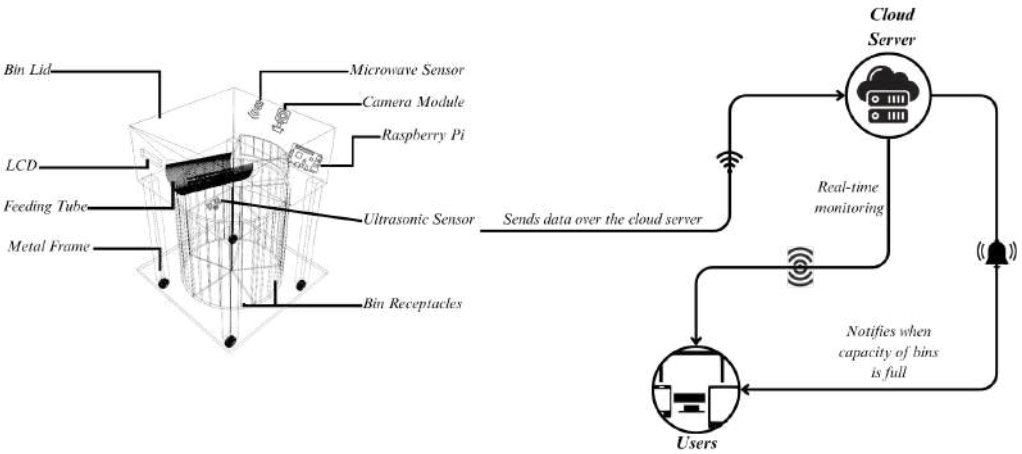


Figure 3 shows the conceptual design and system architecture of the automatic plastic bottle segregation system. The setup consists of a metal-framed structure housing five bin receptacles, each designated for a different category of plastic bottles. Key components include an LCD, a feeding tube, a bin lid, a microwave sensor, an ultrasonic sensor, a camera module, and a Raspberry Pi. Incoming bottles are

analyzed by sensors and a camera, enabling the system to identify their type and direct them into the corresponding bin. Data on bin status is processed by the Raspberry Pi and transmitted to a server for immediate monitoring. The system automatically notifies users when any bin reaches full capacity, ensuring efficient waste segregation and collection.

3.4. Hardware Development

The system is designed with a structured, multi-functional layout that enables accurate detection, classification, and sorting of plastic bottles. Users are guided to insert plastic bottles into the upper part of the system through a specific circular opening. This section contains the camera module and microwave sensor, which work together to detect the presence of the bottle and capture its image for classification.

The classification process identifies the plastic bottle as one of five types: PET, HDPE, LDPE, PP, or Unclassified. Once the type of plastic is determined, the bottle is directed to the appropriate compartment in the lower bin. The bin is divided into five separate sections, each specifically designated for one plastic type to ensure proper segregation.

Additionally, the system integrates a single-board computer (SBC) that manages the coordination between the sensors, camera, and sorting mechanism. The processed data from the AI model guides the servo and stepper motors in positioning the bottle accurately into its designated bin. An LCD display provides real-time feedback, showing the detected plastic type and the bin's current status. This automated method improves the accuracy of waste sorting while reducing the need for

human involvement, promoting efficiency and encouraging environmentally responsible disposal practices.

Figure 4

Hardware Design: Front View (top left), Rear View (top right), and Actual Design (bottom)

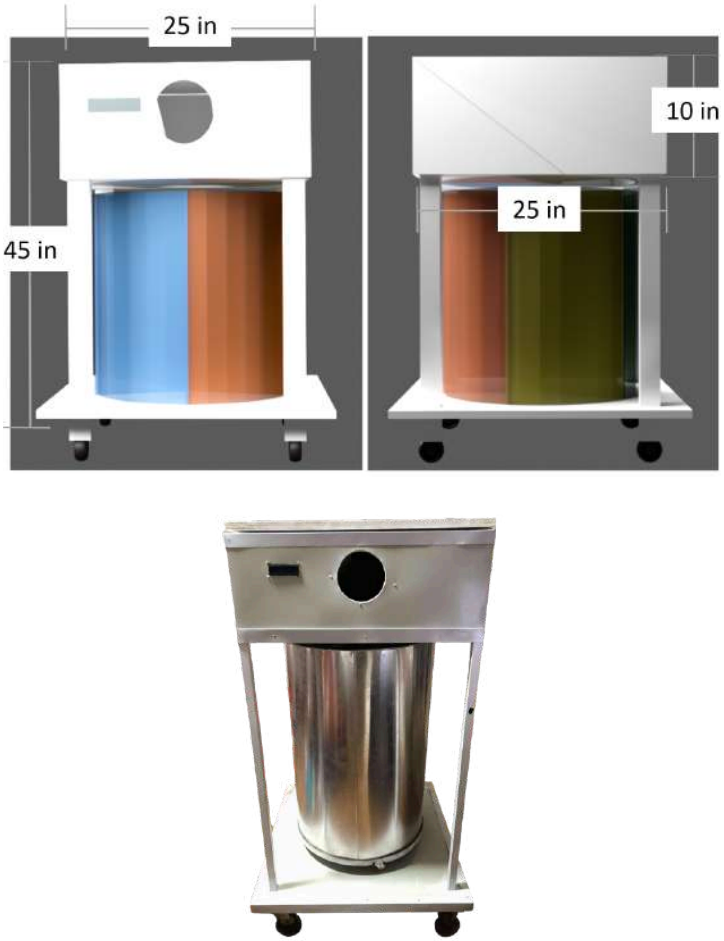


Figure 4 illustrates the hardware design of the automatic plastic bottle segregation system, which features a cylindrical structure with front (left) and rear (right) views. The upper front section contains a circular opening where users insert plastic bottles, designed to fit most standard sizes, along with an LCD for user feedback. Once a bottle is inserted, a microwave sensor detects its presence and activates a camera module to capture an image of it. The system then analyzes the

image to determine the bottle's plastic type—such as PET, HDPE, LDPE, or PP—or classifies it as unclassified if it cannot be recognized. The cylindrical design was selected to efficiently accommodate five sorting bins while enabling smooth, precise rotation via a stepper motor. This configuration offers better spatial efficiency, weight distribution, and stability compared to a box-type structure, reducing space requirements and mechanical complexity while improving operational reliability.

Figure 5

Perspective of the Hardware Design from the Side

Figure 5 presents a lateral perspective of the hardware configuration for the automatic plastic bottle segregation system. The structure features a sturdy steel frame

that supports both the bin and bin lid, with four vertical posts securely anchored to a flat base equipped with caster wheels for smooth mobility. The top section houses the bin lid, which includes the feed chute for inserting plastic bottles and the display panel for system feedback. Each bin receptacle or sector is designed to be removable, allowing for convenient cleaning and easy emptying when nearing full capacity.

Figure 6

Bin Receptacles: Side View (left) and Top View (right), and Actual Design (bottom)

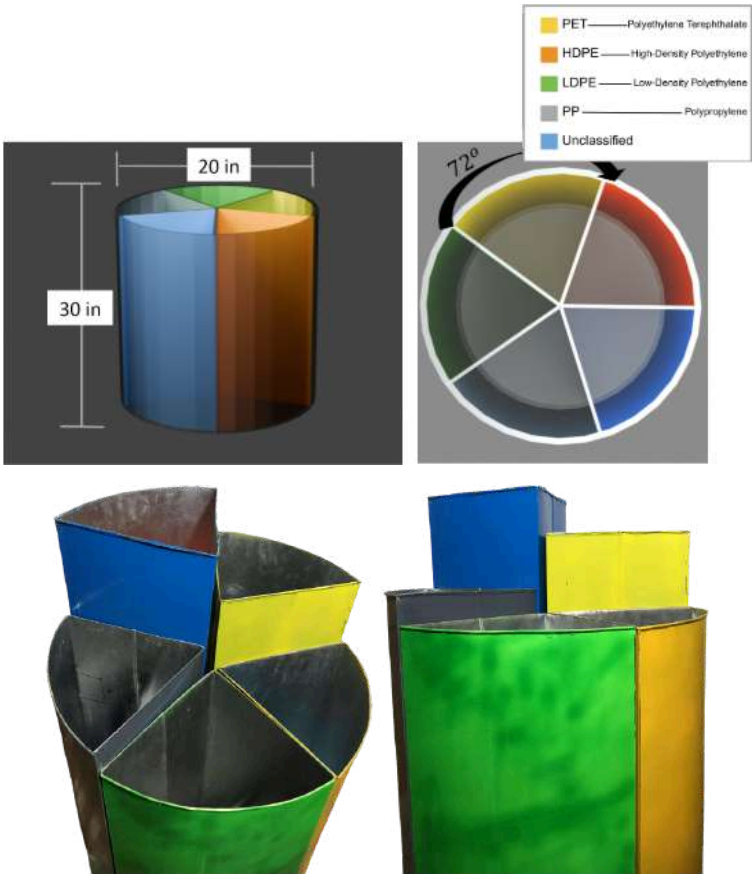


Figure 6 shows the bin receptacles of the automatic plastic bottle segregation system from the side (left) and top (right) views. The cylindrical bin, measuring 30 inches in height and 20 inches in diameter, is divided into five equal sectors, each spanning 72 degrees. Each sector is designated for a specific plastic type and is color-coded for easy identification: yellow for PET (Polyethylene Terephthalate),

orange for HDPE (High-Density Polyethylene), green for LDPE (Low-Density Polyethylene), gray for PP (Polypropylene), and blue for Unclassified. Bin receptacles are color-coded according to the Resin Identification Code (RIC) standard to ensure visual consistency with international plastic sorting conventions. This color-coding ensures clear visual guidance for sorting and facilitates efficient waste management.

Figure 7

Classification Area: Top (left), Cross Section (right) View, and Actual Design (bottom)

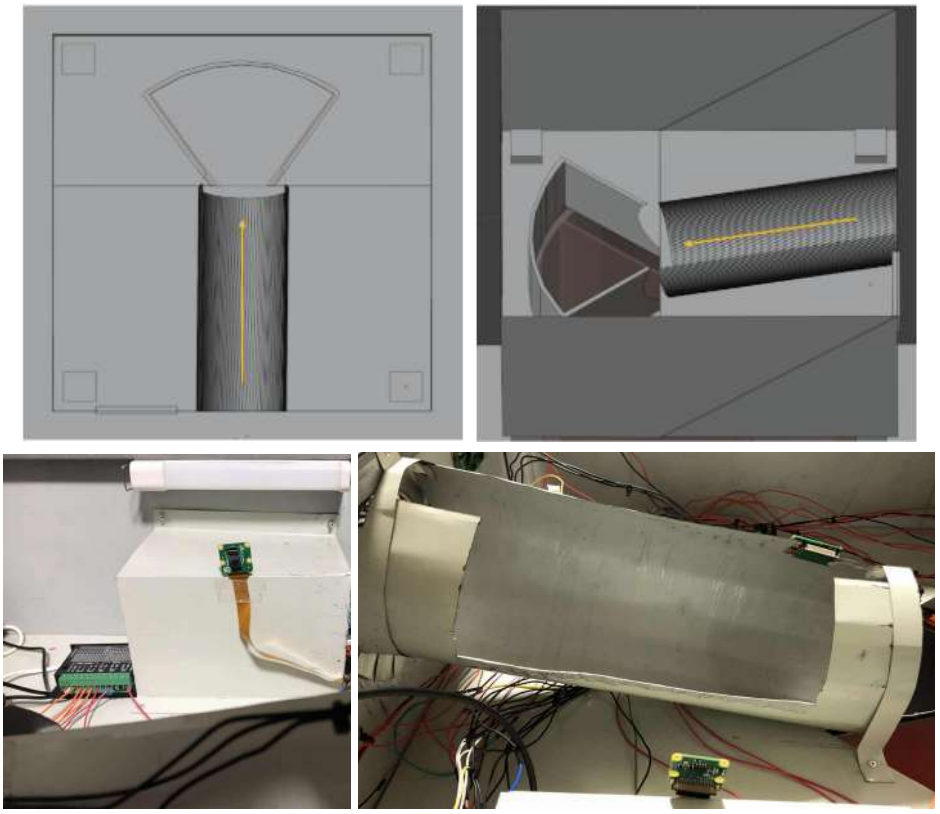


Figure 7 illustrates the classification area of the automatic plastic bottle segregation system. Located beneath the bin lid, the feeding tube serves as the entry point for plastic bottles, where integrated sensors determine the type of plastic used in the bottle. Once classification is complete, the bin receptacle rotates until the designated compartment is aligned with the mouth of the feeding tube. When properly

positioned, the bottle is released and dropped directly into the corresponding bin receptacle, ensuring accurate and efficient sorting.

3.5. Hardware Components

The hardware components are the foundation of the system. This section outlines the key electronic and mechanical components, including sensors, motors, and the single-board computer, which work together to ensure accurate detection, efficient sorting, and reliable system operation.

3.5.1. Single-Board Computer

Figure 8

Raspberry Pi 5

The Raspberry Pi 5 is a compact yet powerful single-board computer powered by a quad-core Arm Cortex-A76 processor running at 2.4 GHz, with up to 8 GB of LPDDR4X RAM. It supports dual 4K displays, high-speed USB 3.0 ports, Gigabit Ethernet, and includes a 40-pin GPIO header for peripheral interfacing. In this study, it serves as the main controller, handling image classification with deep learning models, processing sensor data, controlling motors, and displaying outputs via I2C

LCD. Its improved performance and expanded connectivity make it ideal for use in automated plastic bottle sorting systems.

3.5.2. Camera Module

Figure 9

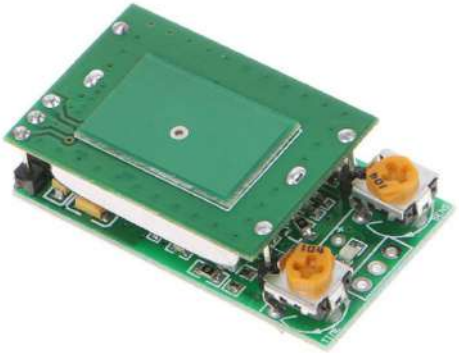
Raspberry Pi Camera Module V2

The Raspberry Pi Camera Module V2 is a versatile and reliable imaging component designed specifically for Raspberry Pi boards. It features an 8-megapixel Sony IMX219 sensor able to take high-resolution images and 1080p video resolution at 30 frames per second. Known for its balance of performance and affordability, the module is well-suited for various uses, including object tracking, image classification, robotics, and home automation. With its compact form factor and plug-and-play connectivity via the CSI-2 interface using a ribbon cable, it enables easy integration into embedded vision projects requiring clear and consistent visual input.

3.5.3. Sensors

Figure 10

HFS-DC06 Microwave Radar Sensor



The HFS-DC06 is a 5.8 GHz microwave radar motion sensor designed for non-contact motion detection. Unlike traditional PIR (passive infrared) sensors, it uses Doppler radar technology to detect movement by emitting microwave signals and analyzing the reflected waves. This allows it to sense motion through certain non-metallic materials, such as glass, plastic, or thin walls. The sensor offers high sensitivity, a wide detection range, and stable operation in various lighting and temperature conditions, rendering it appropriate for various applications such as automatic lighting control, security systems, and smart home devices. In this study, it will be used to detect the motion caused when a plastic bottle is thrown into the designated area, serving as part of the automated detection and classification process.

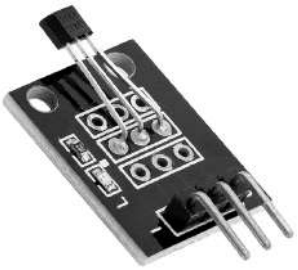
Figure 11

HC-SR04 Ultrasonic Sensor

The HC-SR04 is a cost-effective ultrasonic sensor that enables distance measurement without any physical contact by employing sonar technology. It emits sound pulses at a frequency of 40 kHz. By measuring the duration required for the echo to come back, it can determine distances ranging from 2 to 400 centimeters with an accuracy of approximately 3 millimeters. In this study, the sensor is vital to the overall process in monitoring the individual bin levels. Its instantaneous distance measurements help identify the optimal time for image capture, which aids in the precise classification and effective sorting of the bottles.

Figure 12

KY-003 Hall Magnetic Sensor



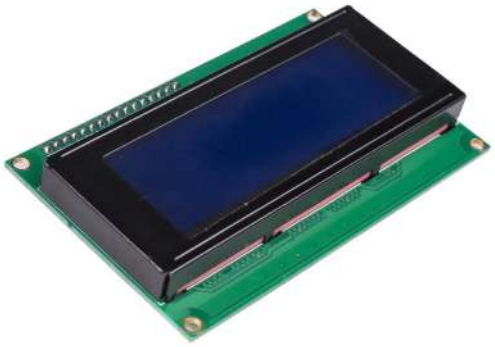
The KY-003 Hall Magnetic Sensor Module senses magnetic fields through an integrated Hall effect transducer, producing a digital output signal that alters its state when a magnetic field is detected. This module is specifically used to identify the

position of the PET bin by sensing a small magnet attached to it, ensuring accurate alignment. Proper detection is essential for the smooth operation of the automatic plastic bottle segregation process, as it prevents misplacement and sorting errors. The sensor's simplicity and reliability make it a key component in maintaining sorting efficiency.

3.5.4. Display Module

Figure 13

Liquid Crystal Display 12C



The LCD module, featuring an I2C interface, is a space-efficient and low-energy display that streamlines connections through the I2C communication protocol. Unlike traditional LCDs requiring multiple signal lines through a parallel interface, the I2C module employs just two data lines, namely data and clock, which makes it ideal for projects with limited microcontroller pins. This small display supports simple text and basic graphics and is commonly used in digital clocks, weather stations, and embedded systems. In this study, the I2C LCD module provides a straightforward, dependable visual interface that displays system status and real-time data, enhancing user interaction and monitoring.

3.5.5. Actuators

Figure 14

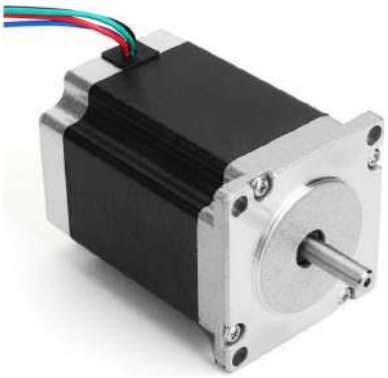
MG996R Servomotor

A servomotor is a precision rotary actuator designed to accurately control angular position, speed, and acceleration. It typically consists of a DC motor, a set of gears, a position sensor, and an integrated control circuit that enables closed-loop feedback for precise and repeatable motion. This high level of control makes servomotors essential in areas like robotics, remote-operated systems, industrial automation, and optical devices such as cameras.

In the context of this study, the servomotor plays a critical role in directing plastic bottles into their appropriate sorting bins after classification. Its fast and accurate response ensures that each bottle is deposited into the correct container, thereby maintaining the overall effectiveness and dependability of the automated sorting system.

Figure 15

Wantai Stepper Motor 57BYGH627



The Wantai Stepper Motor 57BYGH627 is a NEMA 23 stepper motor known for its reliable performance, consistent torque, and precise 1.8° step angle. With a holding torque of 19000 g·cm and a current rating of 3A per phase, it is well-suited for applications that demand accurate and repeatable motion, such as 3D printers, CNC machines, and other automated mechanisms. In this study, the motor is employed to rotate and align the sorting bins with the output chute, ensuring that each plastic bottle is accurately directed into its corresponding compartment. Coupled with a TB6600 driver, the motor provides smooth, controlled motion and stable operation, allowing for precise bin positioning and consistent performance. Its robust build and dependable response make it a crucial part of the system's hardware design, ensuring reliability and efficiency in the waste segregation process.

3.5.6. Buck Converter

Figure 16

12V-5V Buck Converter

The 12V–5V buck converter is a compact and efficient DC-DC converter that steps down a 12-volt input into a stable 5-volt output. It operates through high-frequency switching in combination with inductors, capacitors, and diodes, which together regulate the voltage with minimal energy loss. This switching-based design makes it significantly more efficient than traditional linear regulators, which tend to waste excess power as heat. In this study, a buck converter is employed to safely power low-voltage components, such as sensors, actuators, and display modules, from a single 12V supply. It ensures that each component receives the appropriate voltage level required for stable performance. Its efficiency, compact size, and thermal stability make it an ideal choice for embedded and automation systems, contributing to the overall reliability and energy efficiency of the waste segregation setup.

3.5.7. Power Adapter

Figure 17

Raspberry Pi 27W USB-C Power Supply

The Raspberry Pi 27W USB-C Power Supply is designed for the Raspberry Pi 5, providing a stable 5V/5A output through a USB-C connector. It includes built-in protections against overcurrent, overvoltage, and short circuits, ensuring durability and safety for continuous use. With support for a universal AC input of 100–240V, it operates effectively in a wide range of environments. In this study, it powers the Raspberry Pi exclusively, ensuring reliable operation under high processing loads and preventing issues like unexpected restarts or data corruption, which is essential for our automatic plastic bottle segregation system.

Figure 18

12V 5A Power Supply Adapter

The 12V 5A Power Supply Adapter is a reliable and efficient power source designed to provide a consistent 12-volt DC output with a maximum current capacity of 5 amperes. In this study, the adapter is utilized to power various hardware components of the system, including the stepper motor, motor driver, and sensors. Its sufficient current capacity allows multiple components to operate simultaneously without experiencing voltage drops, ensuring that the entire system functions effectively and reliably during operation.

3.6. Software Development

The applications employed in creating the program for the study are as follows:

3.6.1. Raspberry Pi OS

Figure 19

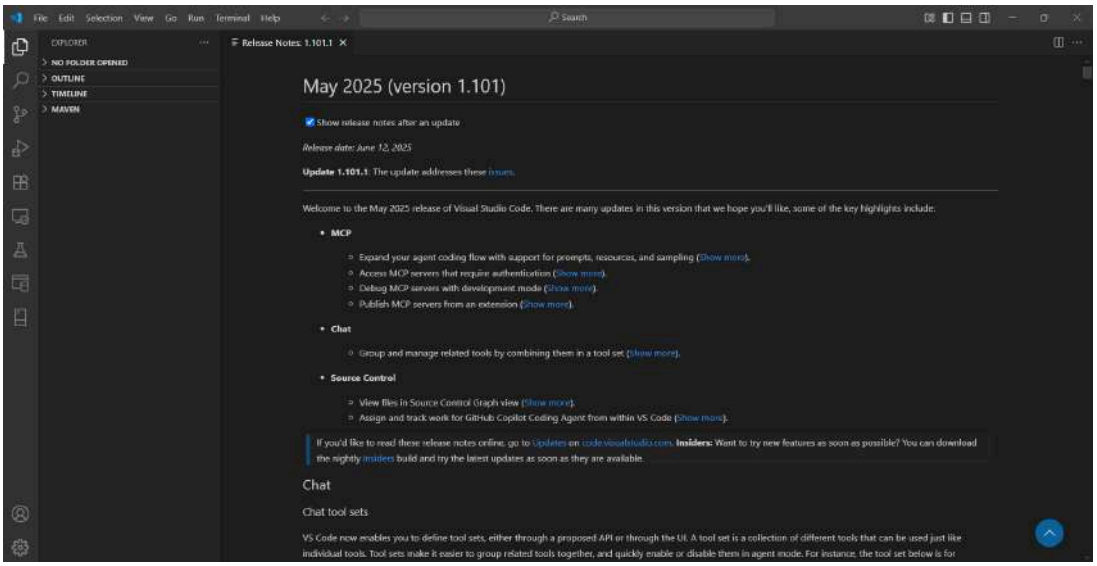
Raspberry Pi OS Imager

Raspberry Pi OS is the recognized operating system designed specifically for Raspberry Pi boards, providing a stable environment for both beginners and advanced users. It features a user-friendly desktop interface, supports programming languages such as Python and C++, and includes libraries for General Purpose Input/Output (GPIO) control. With pre-installed tools for coding, hardware interfacing, and internet connectivity, it is ideal for educational, research, and prototyping purposes. Its lightweight design makes it suitable for embedded systems, IoT projects, and AI applications.

3.6.2. Visual Studio Code

Figure 20

Visual Studio Code Editor Interface

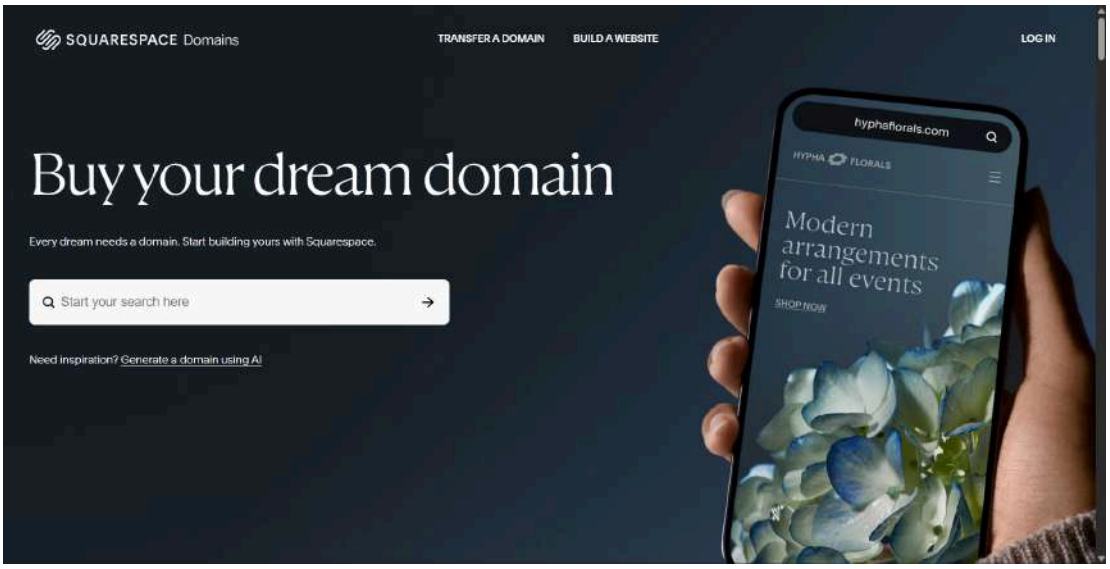


Visual Studio Code (VS Code) is a streamlined, open-source code editor created for effective software development. It supports various programming languages like Python, HTML, and JavaScript, and offers aspects like syntax highlighting, code auto-completion, debugging utilities, and Git integration. With a wide range of extensions, users can tailor their development environment to fit their unique requirements. In this study, VS Code is used to write, test, and manage Python scripts for controlling the Raspberry Pi, processing sensor data, and supporting an image classification system for the automatic segregation of plastic bottles.

3.6.3. Squarespace Domains

Figure 21

Squarespace Domains Homepage

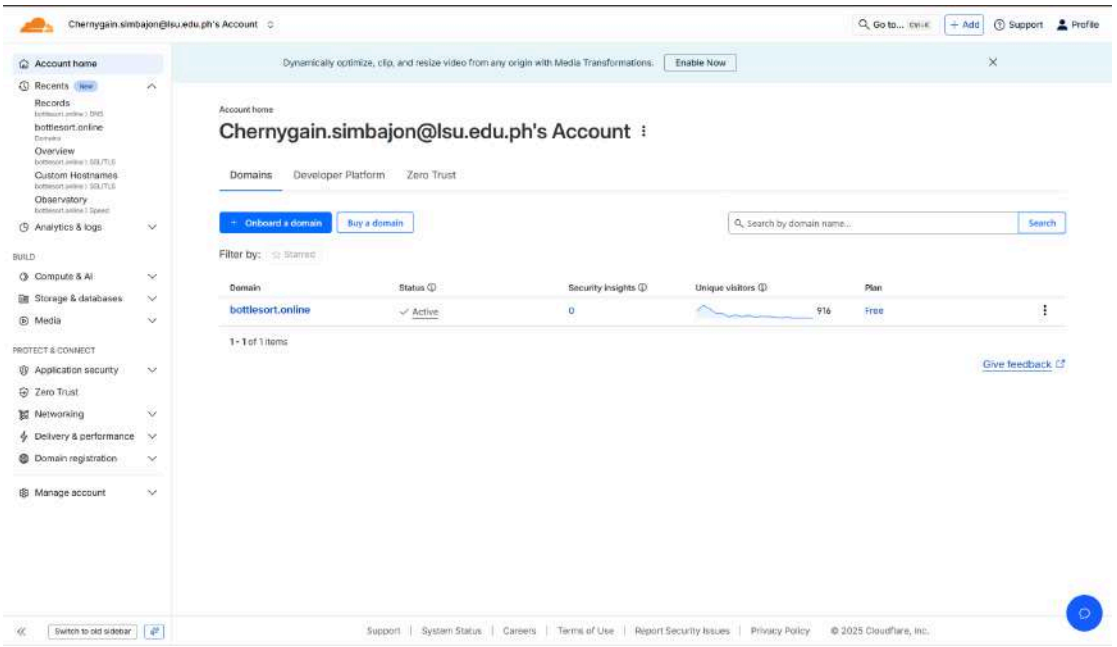


Squarespace Domains is a built-in service for registering and managing professional web addresses, offering transparent pricing, free privacy protection, and automatic SSL certificates. It connects seamlessly with Squarespace websites, eliminating the need for third-party providers while supporting DNS management, forwarding, and email setup. The custom domain bottlesort.online serves as the official entry point, strengthening credibility, branding, and accessibility, and providing visitors with a clear path to system information, documentation, progress updates, and results.

3.6.4. Cloudflare

Figure 22

Cloudflare Dashboard

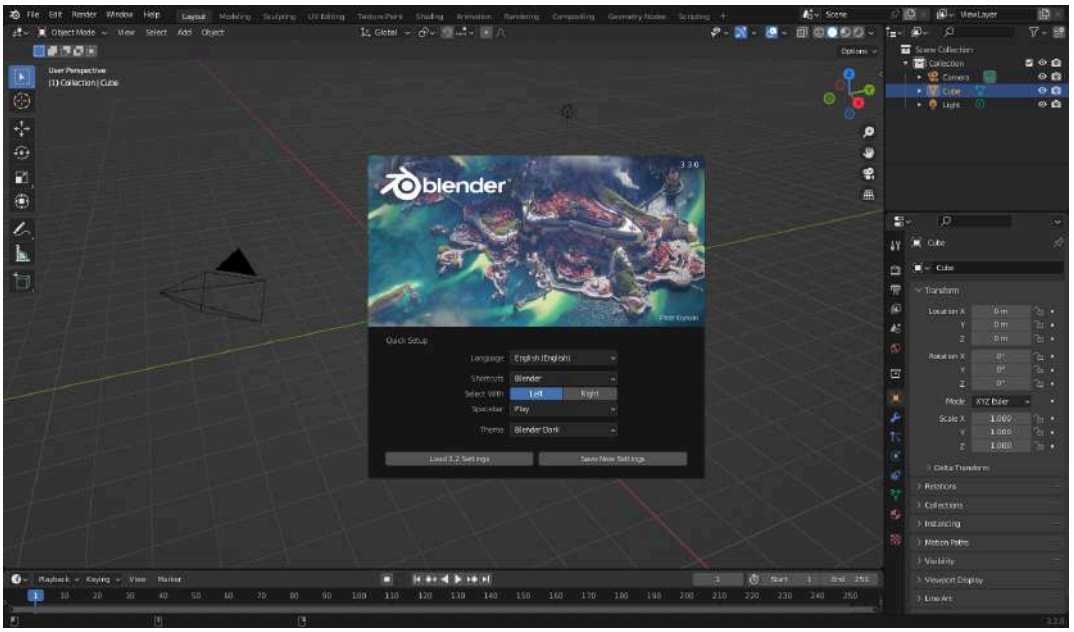


Cloudflare is a powerful platform for enhancing web performance and security, offering services such as content delivery, DDoS protection, and DNS management. It acts as a reverse proxy, enhancing website speed and security through its global data center network. In this study, Cloudflare is utilized to improve the accessibility and security of the system's web interface, protecting it from malicious traffic while ensuring seamless user access. Its performance optimization and threat mitigation features are crucial for maintaining a stable and secure system environment.

3.6.5. Blender

Figure 23

Blender 3D Modeling Workspace

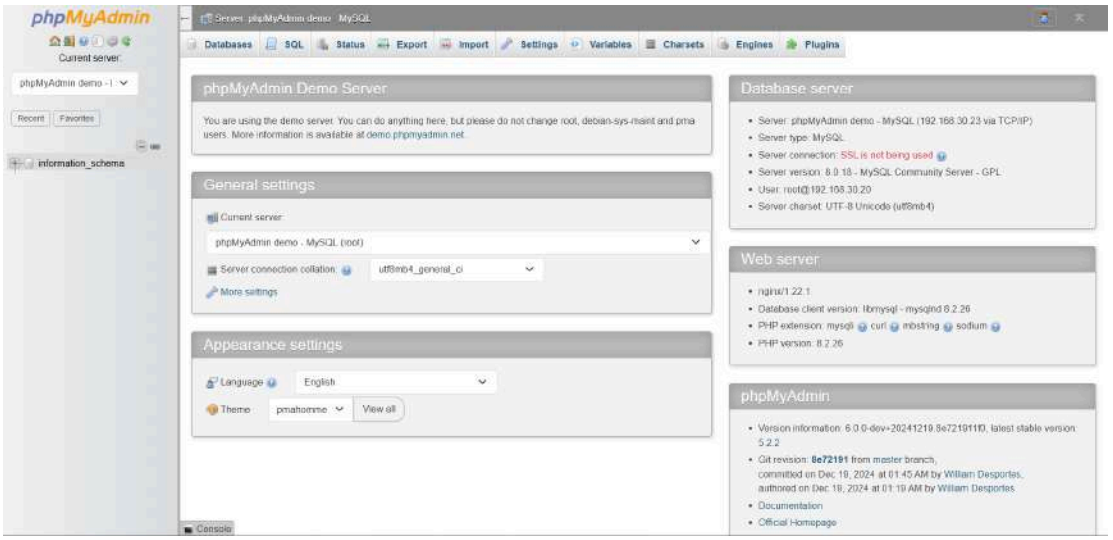


Blender is a robust, free, and open-source software for creating 3D content that covers modeling, sculpting, animation, rendering, simulation, compositing, and video editing. Its versatility makes it popular among artists, designers, and developers for creating high-quality visual content. With a customizable interface and an active community, Blender serves as a key visualization and design tool for accurately modeling system components and generating detailed 3D representations of the automated plastic bottle segregation setup, proving essential throughout the project's design and development stages.

3.6.6. phpMyAdmin

Figure 24

Database Management Using phpMyAdmin



phpMyAdmin is a freely available web-based tool that allows users to manage MySQL and MariaDB databases using an easy-to-navigate graphical interface. It offers features for creating, altering, and deleting databases and tables, running SQL queries, importing and exporting data, as well as managing user accounts and permissions effectively. In this study, phpMyAdmin is used to oversee the local database operations of an automated plastic bottle segregation system, ensuring organized data storage and easy retrieval of classification results and system logs. Its accessibility and user-friendliness make it a valuable tool for database management in both development and deployment environments.

Figure 25

Admins Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**admin_id**	int(11)			No	*None*		AUTO_INCREMENT
2	**admin_name**	varchar(100)	utf8mb4_unicode_ci		No	*None*		
3	**admin_email**	varchar(150)	utf8mb4_unicode_ci		No	*None*		
4	**admin_password**	varchar(255)	utf8mb4_unicode_ci		No	*None*		

The structure of the admins table contains essential information for managing system administrator accounts. It consists of four fields: admin_id, admin_name, admin_email, and admin_password. The admin_id serves as the primary key and is set to auto-increment, ensuring that each record is uniquely identified. The admin_name and admin_email fields store the administrator's name and email address, respectively, while the admin_password field securely holds the hashed password for authentication. This table ensures the secure and organized storage of administrator credentials within the system's database.

Figure 26

Admins Table

admin_id	admin_name	admin_email	admin_password
1	Jasper_BottleSort	jasperjohn.paitan@lsu.edu.ph	$2y$10$NlZlZj8OxPSfBSehW1Z4ne6cVlw1rdOpHnLbk6RXab9...
2	Hannah_BottleSort	hannahjoy.almendras@lsu.edu.ph	$2y$10$JCgWdiWJqwhkVuUgOualy.ZZ/4cJu4fb6KjJXfxcmHl...
3	Cherny_BottleSort	chernygain.simbajon@lsu.edu.ph	$2y$10$zAAbypkLjpVP2dLHAnc1k.X9csFUMyt8hsmVgJYHglg...

The admins table is designed to securely manage administrator accounts and authentication data. Each record contains essential information about the administrators, including their identification, contact details, and encrypted login credentials. This table ensures secure access control and effective management of administrative functions, enhancing the overall reliability and integrity of the system.

Figure 27

Plastic Bottle Logs Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**bin_id**	int(11)			No	*None*		
3	**record_time**	datetime			No	*None*		
4	**bottle_type**	enum('PET', 'HDPE', 'LDPE', 'PP', 'Unclassified')	utf8mb4_unicode_ci		No	*None*		
5	**confidence_level**	decimal(5,2)			Yes	*NULL*		

The plastic bottle logs table is designed to store data related to the classification and detection of plastic bottles processed by the system. It contains five fields: id, bin_id, record_time, bottle_type, and confidence_level. The id serves as the primary key and automatically increments with each new entry. The bin_id field functions as a foreign key that links each plastic bottle log entry to a specific bin in the system. The record_time field captures the exact date and time of the detection, while the bottle_type field specifies the identified category of plastic, such as PET, HDPE, LDPE, PP, or Unclassified. The confidence_level field indicates the system's confidence in classifying the bottle type. This table facilitates systematic data logging and analysis, supporting the evaluation of the system's accuracy and performance.

Figure 28

Plastic Bottle Logs Table

id	bin_id	record_time	bottle_type	confidence_level
1	1	2025-11-15 14:21:48	PET	0.98
2	1	2025-11-15 17:11:10	HDPE	0.94
3	1	2025-11-15 17:12:09	LDPE	0.94
4	1	2025-11-15 17:16:06	PET	0.96
5	1	2025-11-15 17:16:40	Unclassified	0.90
6	1	2025-11-15 17:17:21	PP	0.90

The plastic bottle logs table contains data related to the outcomes of the classification of plastic bottles identified by the system. Each entry records the specific

system, detection time, the identified bottle type, and the corresponding confidence level of the classification. This table provides a structured record of the system's performance, serving as a valuable reference for analyzing the accuracy and efficiency of the detection process.

Figure 29

Bin Level Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(11)			No	*None*		AUTO_INCREMENT
2	**bin_id**	int(11)			No	*None*		
3	**record_time**	datetime			No	*None*		
4	**BinLevel_PET**	int(11)			No	0		
5	**BinLevel_HDPE**	int(11)			No	0		
6	**BinLevel_LDPE**	int(11)			No	0		
7	**BinLevel_PP**	int(11)			No	0		
8	**BinLevel_Unclassified**	int(11)			No	0		

The plastic bottle bin level table is designed to monitor and record the fill levels of each plastic bottle bin in the system. It consists of eight fields: id, bin_id, record_time, BinLevel_PET, BinLevel_HDPE, BinLevel_LDPE, BinLevel_PP, and BinLevel_Unclassified. The id field serves as the primary key and has an auto-increment property to ensure that each entry is unique. The bin_id field functions as a foreign key that links each plastic bottle log entry to a specific system within a specific location. The record_time field logs the date and time of each update to the bin levels, while the remaining fields capture the fill levels of the categorized bins. This table enables real-time tracking and data management for the Automatic Plastic Bottle Segregation System, ensuring efficient monitoring of bin capacity and scheduling of collections.

Figure 30

Bin Level Table

id	bin_id	record_time	BinLevel_PET	BinLevel_HDPE	BinLevel_LDPE	BinLevel_PP	BinLevel_Unclassified
1	1	2025-11-15 14:41:12	5	10	20	30	90
2	1	2025-10-27 14:46:56	5	90	80	75	50
3	1	2025-10-27 14:47:47	75	0	0	0	0
4	1	2025-10-27 14:50:55	15	50	10	75	40
5	1	2025-10-27 14:50:55	15	50	10	75	40

The plastic bottle bin level table records the real-time fill levels of each categorized bin. Each entry includes the specific system, the date and time of measurement, along with the fill levels for PET, HDPE, LDPE, PP, and Unclassified materials. This table enables the systematic monitoring of bin capacity and provides valuable data for evaluating the system's efficiency and utilization.

Figure 31

Admin Activity Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**activity_id**	int(11)			No	*None*		AUTO_INCREMENT
2	**admin_id**	int(11)			No	*None*		
3	**login_time**	timestamp			No	current_timestamp()		
4	**report_generated**	varchar(255)	utf8mb4_unicode_ci		Yes	*NULL*		
5	**bottle_type**	varchar(255)	utf8mb4_unicode_ci		Yes	*NULL*		
6	**start_date**	date			Yes	*NULL*		
7	**end_date**	date			Yes	*NULL*		

The admin activity table logs administrative actions in the automated plastic bottle segregation system, ensuring accountability and monitoring. Each record is identified by the activity_id field, which serves as the primary key, and includes the admin_id as a foreign key to link activities to administrators. The login_time field records the exact timestamp of the admin's login, defaulting to the current time. Other fields, such as report_generated, bottle_type, start_date, and end_date, capture details

about reports created, the type of plastic bottle involved, and the date range for the reports.

Figure 32

Admin Activity Table

activity_id	admin_id	login_time	report_generated	bottle_type	start_date	end_date
1	1	2025-11-12 21:00:00	Plastic Bottle Collection Report	PET	2025-10-01	2025-10-31
2	3	2025-11-10 08:30:00	Plastic Bottle Collection Report	HDPE	2025-11-03	2025-11-09
3	1	2025-11-12 09:45:00	*NULL*	*NULL*	*NULL*	*NULL*
4	2	2025-11-12 21:41:20	Plastic Bottle Collection Report	*NULL*	2025-10-01	2025-11-12

The admin activity table logs administrative actions within the system, including a unique activity ID, the administrator's ID, and the timestamp of the activity. It also records details such as report names, types of plastic bottles related to the actions, and the report date ranges. This table facilitates tracking and auditing of administrative operations, ensuring transparency and accountability in the automated plastic bottle segregation process.

Figure 33

Bins Table Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**bin_id**	int(11)			No	*None*		AUTO_INCREMENT
2	**bin_code**	varchar(50)	utf8mb4_unicode_ci		No	*None*		
3	**bin_location**	varchar(255)	utf8mb4_unicode_ci		No	*None*		
4	**installation_date**	date			Yes	curdate()		
5	**status**	enum('active', 'maintenance', 'inactive')	utf8mb4_unicode_ci		Yes	active		
6	**created_at**	timestamp			No	current_timestamp()		
7	**updated_at**	timestamp			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP()

The bins table contains key data for each system. Each bin is uniquely identified by the bin_id primary key. The bin_code provides an assigned identifier, and bin_location specifies the bin's location. The installation_date indicates the

deployment date, while the status field denotes whether the bin is active, under maintenance, or inactive. The created_at and updated_at timestamps record when each entry was created and last modified, thereby facilitating precise monitoring and maintenance management.

Figure 34

Bins Table

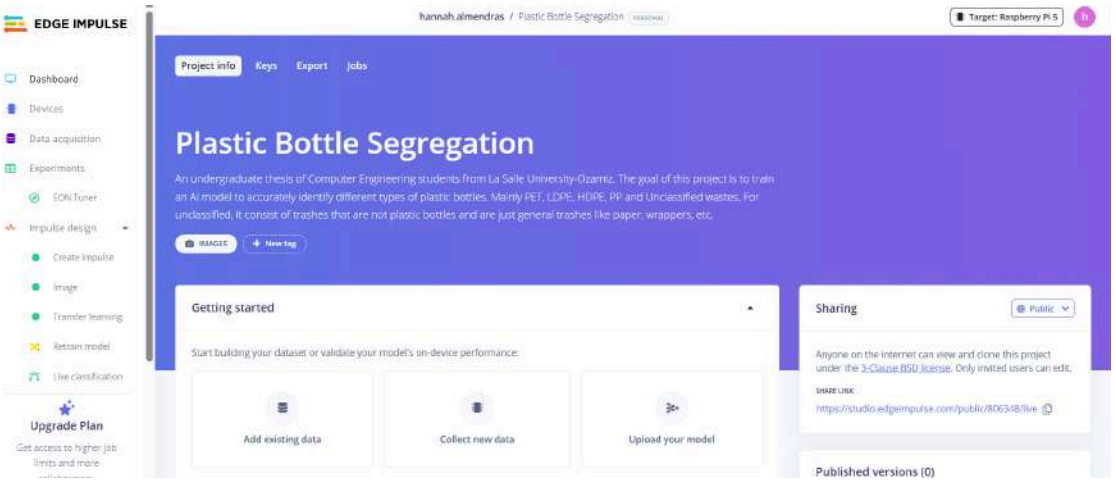
bin_id	bin_code	bin_location	installation_date	status	created_at	updated_at
1	BIN-001	SB-1F	2025-12-03	active	2025-12-03 22:21:58	2025-12-03 22:23:35

The bins table stores essential information for each waste collection bin in the system, including a unique identifier, assigned bin code, and recorded location. It also records the installation date and monitors the current status, such as active, inactive, or under maintenance. Furthermore, the table includes creation and update timestamps to document when each bin record was added or modified. This design facilitates effective management of bin placement and operational status within the automated plastic bottle segregation environment.

3.6.7. Edge Impulse

Figure 35

Online Platform for Building AI Models



Edge Impulse serves as a development platform designed for constructing, training, and deploying machine learning models tailored for embedded and edge devices. With its intuitive interface, the platform streamlines the entire machine learning process, from data collection to real-time classification. This capability enables the system to effectively recognize and categorize types of plastic bottles in real-time, establishing Edge Impulse as a crucial resource for integrating computer vision into efficient, low-power, and scalable waste management solutions.

Figure 36

Data Acquisition

The dataset for the plastic bottle classification model contains 11,712 labeled items across six classes: Polyethylene Terephthalate (PET), High-Density

Polyethylene (HDPE), Low-Density Polyethylene (LDPE), Polypropylene (PP), Unclassified, and Background. The Background class includes non-plastic areas, helping the model focus on relevant objects. Each plastic type has unique visual properties, which help in accurate classification.

Utilizing MobileNetV2 as its backbone through transfer learning, the model efficiently extracts visual features while remaining lightweight, making it suitable for deployment on systems like the Raspberry Pi. This combination enhances the model's classification accuracy and generalization, supporting the creation of a functional automated system for segregating plastic bottles.

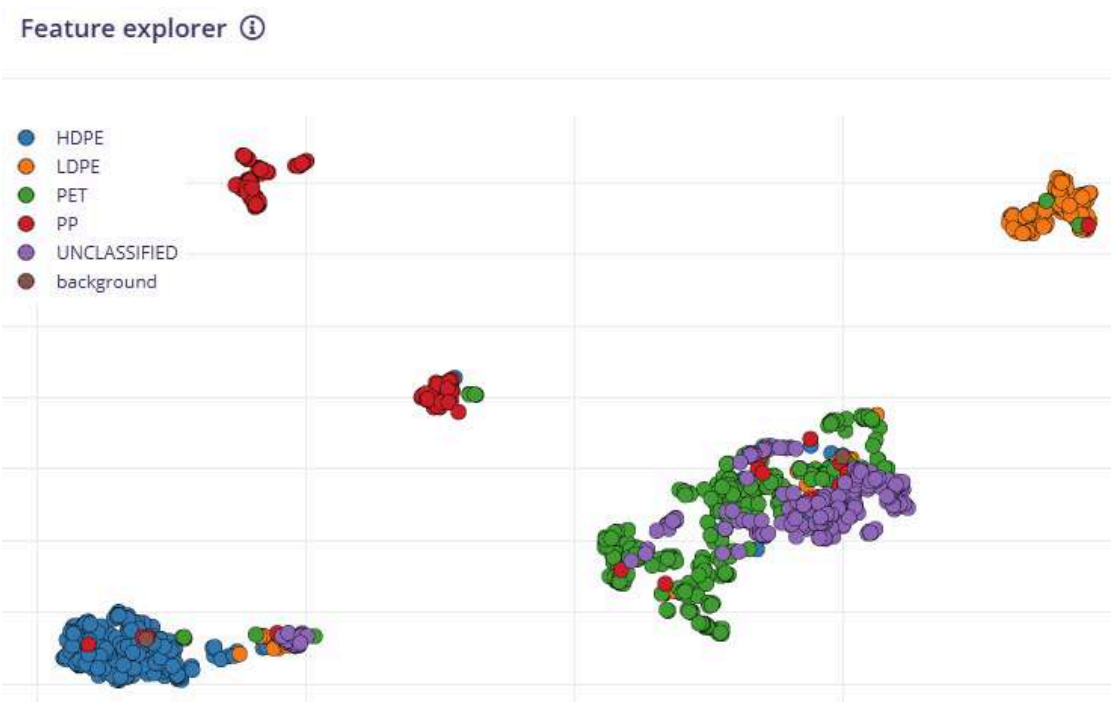
Figure 37

Train/Test Split of Data

The data is split into two parts: training and testing, with an 80/20 ratio. This indicates that 80% of the information is utilized to train the model, while the other 20% is set aside to assess its effectiveness. The training subset enables the model to learn and adjust its internal parameters based on the patterns and features found in the data. In contrast, the testing subset measures how effectively the model applies to new, unfamiliar samples. This division ensures that the model’s predictions are not merely customized to fit the training data but are adaptable to real-world variations. By implementing this data partitioning strategy, we can thoroughly validate the model’s reliability, accuracy, and robustness before deployment.

Figure 38

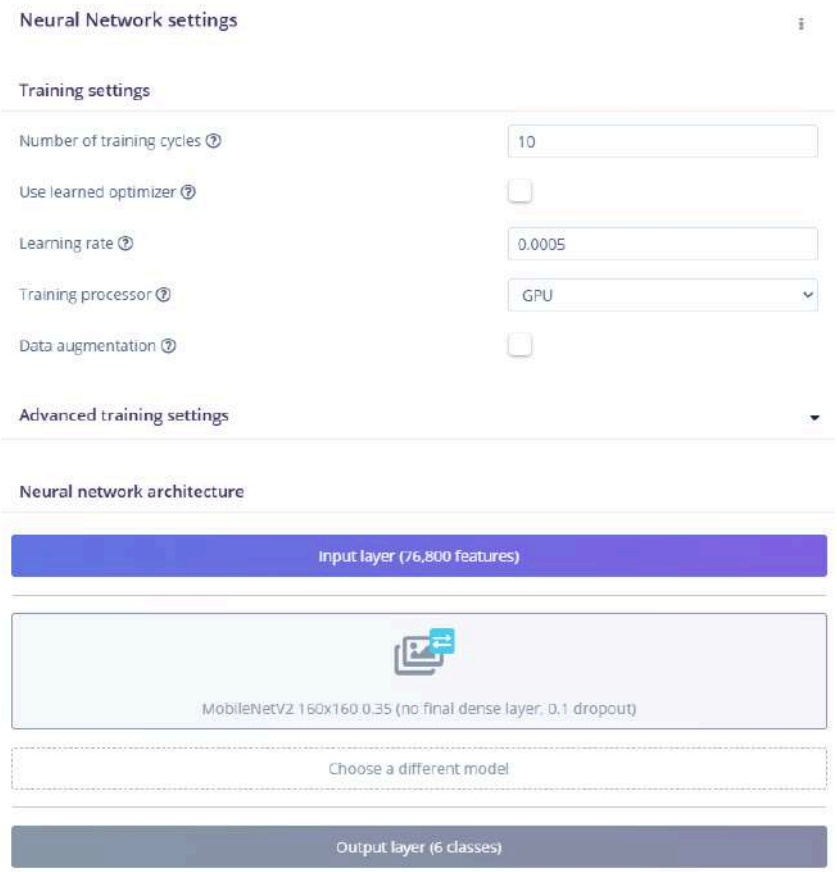
Feature Explorer



The Feature Explorer visualization illustrates the spatial distribution of data samples in a reduced-dimensional feature space, providing insight into how well the model distinguishes between plastic categories. Each point on the 2D plot represents an image sample and is color-coded: PET (green), HDPE (blue), LDPE (orange), PP (red), Unclassified (purple), and Background (brown). The distinct clustering of points shows that the features effectively separate most plastic types, though some overlap suggests shared visual characteristics that may complicate classification. Overall, the visualization confirms the model's capability in feature extraction and showcases the dataset's class separability.

Figure 39

Neural Network Settings

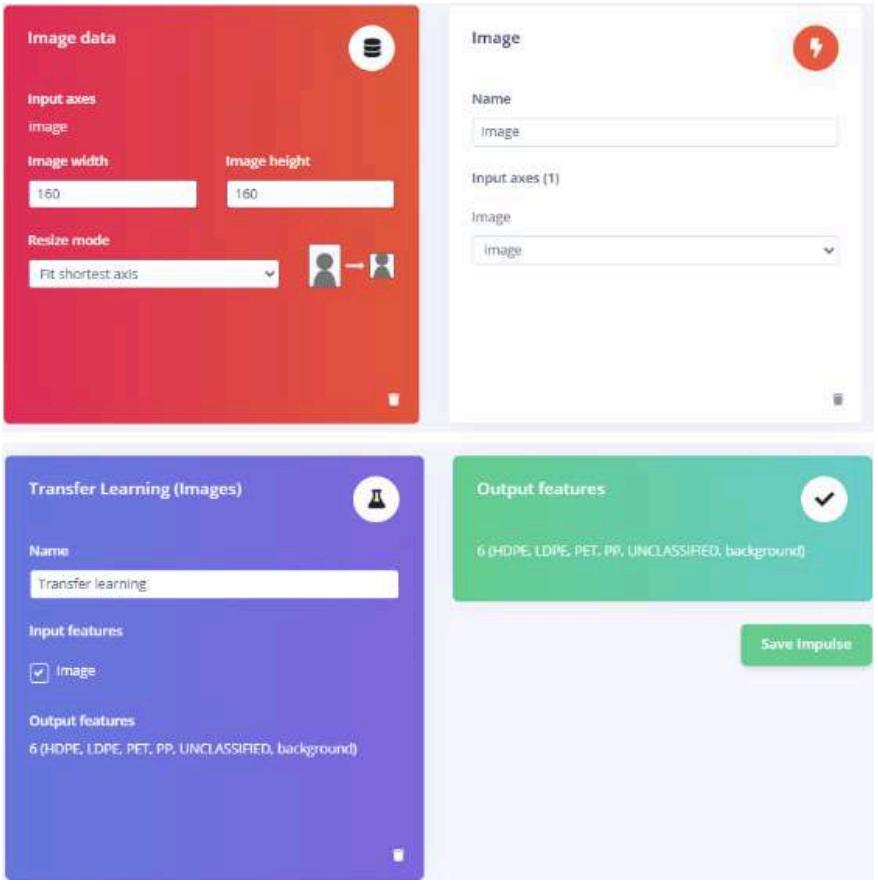


The neural network settings presented are designed to develop a classification model utilizing the MobileNetV2 architecture. The input resolution is set at 160×160 pixels, and a width multiplier of 0.35 is applied. To reduce overfitting and improve generalization performance, the model incorporates a dropout rate of 0.1.

The input layer consists of 76,800 features, while the output layer classifies data into six distinct categories: PET, HDPE, LDPE, PP, unclassified, and background. The training process runs for a duration of 10 epochs, using a learning rate of 0.0005, ensuring a steady and gradual optimization process. Training is conducted on a GPU, highlighting a focus on resource efficiency.

Figure 40

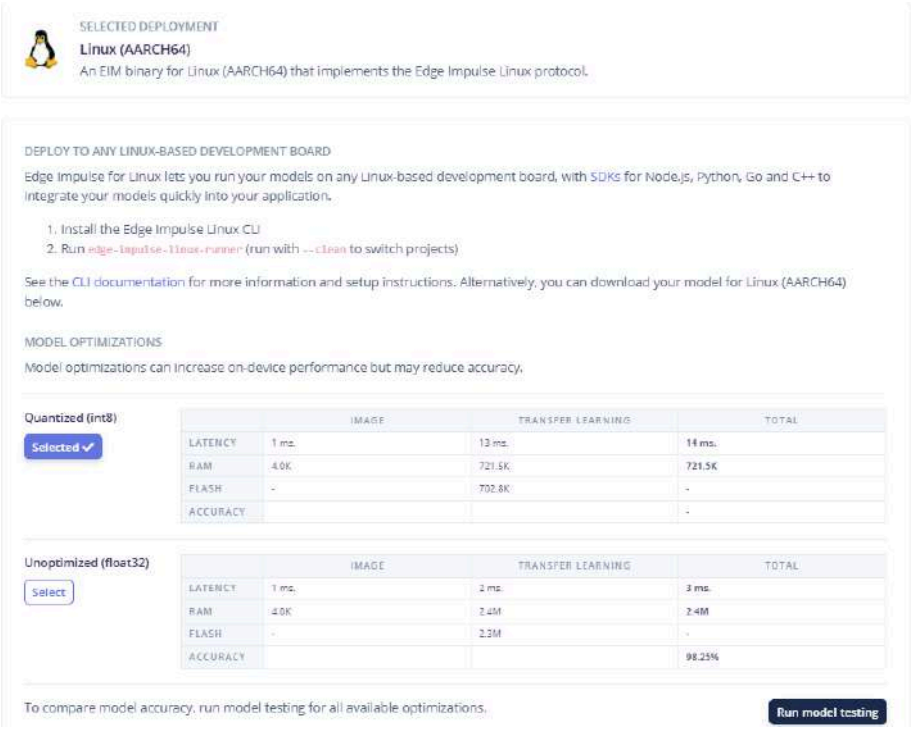
Creating the Impulse



The impulse design outlines the complete setup of the image classification system implemented in the project. The Image Data block defines the input parameters by setting the image resolution to 160 × 160 pixels and the resize mode to "Fit shortest axis." This ensures consistent image dimensions during preprocessing. The Transfer Learning (Images) block employs a pretrained MobileNetV2 model to extract high-level visual features, which are then fine-tuned to fit the project's dataset. The Output Features block specifies six classification categories: HDPE, LDPE, PET, PP, Unclassified, and Background. This design facilitates efficient feature extraction and accurate categorization, ensuring computational efficiency, which makes it appropriate for environments with limited resources, such as embedded or edge devices.

Figure 41

Deploying the Model



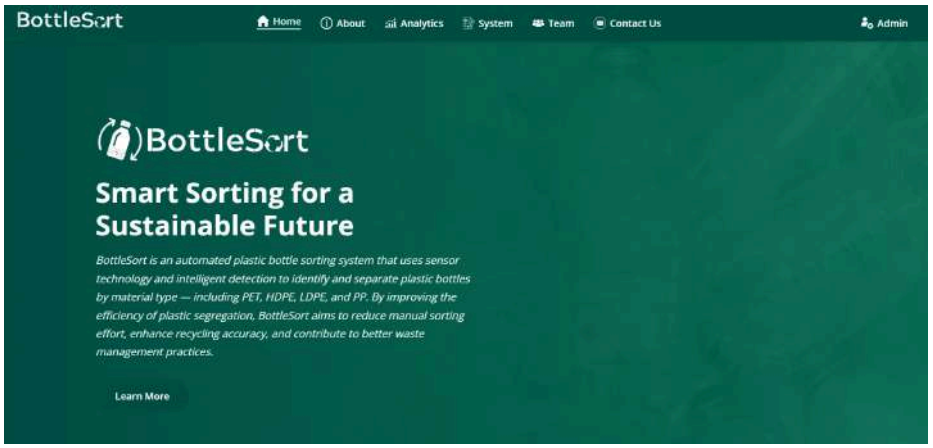
The deployment configuration utilizes the Linux (ARM64) library for efficient model operation on Linux-based boards, such as the Raspberry Pi. It integrates the Edge Impulse Linux CLI for real-time model testing. A quantized (int8) model was chosen to enhance performance, reducing memory use and processing time while ensuring accuracy. The model has a total latency of 14 ms, requires 721.5 KB of RAM and 702.8 KB of flash memory, making it ideal for lightweight edge AI systems focused on real-time waste classification.

3.7. Website Interface Design

The user interface provides a comprehensive and intuitive platform that enables both administrators and users to access the system, monitor real-time bin statuses, and manage operations efficiently. The following figures illustrate the key UI elements designed for all users.

Figure 42

Homepage



The homepage introduces BottleSort, an innovative system designed to streamline the categorization of plastic bottles. It emphasizes the platform's mission to enhance recycling efforts and minimize manual sorting through intelligent detection and real-time monitoring. This approach aims to promote a cleaner and more sustainable environment.

Figure 43

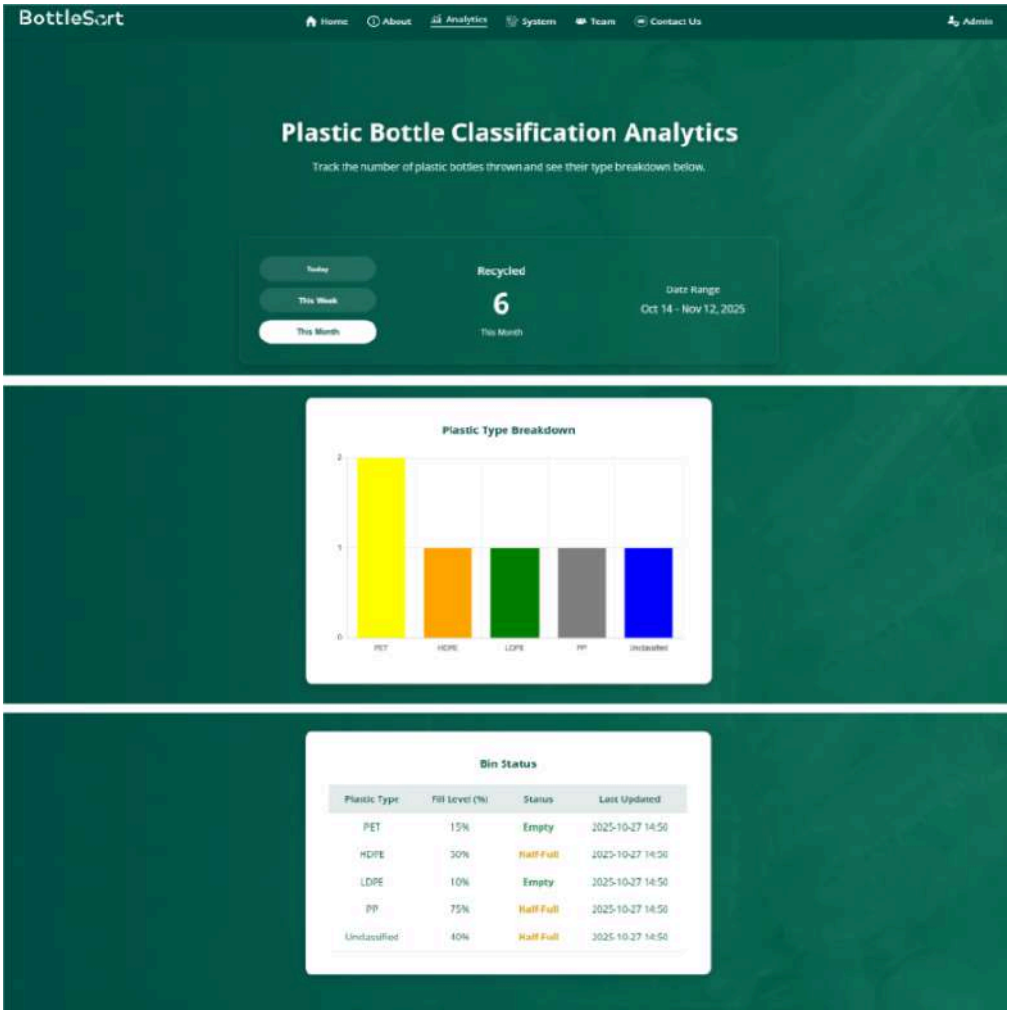
About Page

The About page of the website provides an overview of its automated plastic bottle sorting system, which utilizes sensor technology to identify and categorize

bottles into four main types of plastic: PET, HDPE, LDPE, and PP. Additionally, there is a designated bin for materials that cannot be classified. The page also includes brief descriptions of each type of plastic and its common uses.

Figure 44

Analytics Page



The Analytics Page of the website provides real-time information regarding the system's classification and sorting efficiency. It shows the total number of plastic bottles processed, categorized by specific time frames such as daily, weekly, and monthly. A bar graph illustrates the breakdown of plastic types—PET, HDPE, LDPE, PP, and Unclassified—displaying the quantity collected for each type. Additionally,

there is a table that offers detailed information on the current status of the bins, including the fill level percentage, condition (Empty, Half-Full, or Full), and the timestamp of the last update. This page enables users and administrators to monitor system activity, evaluate sorting efficiency, and track recycling progress over time.

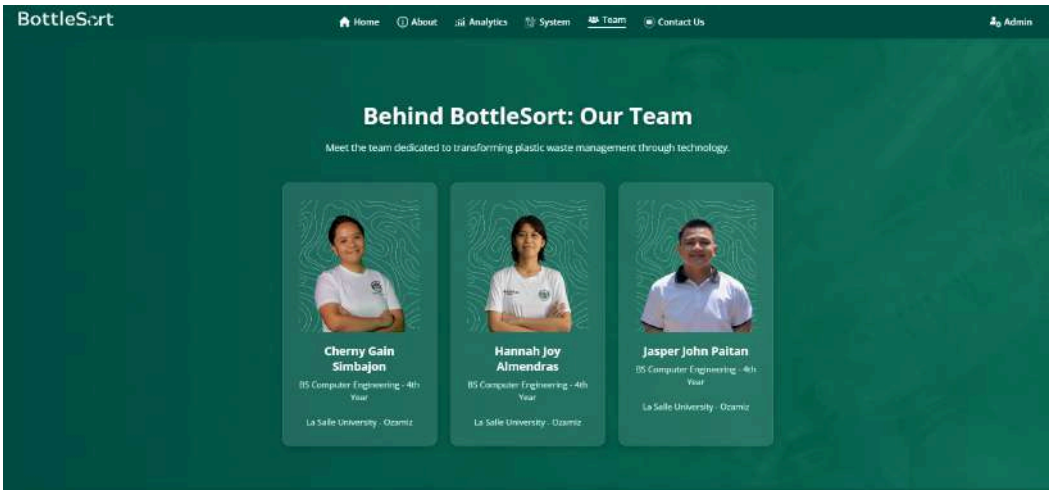
Figure 45

System Page

The System Page of the website provides a thorough summary of the system's structure and capabilities, highlighting the integration of hardware, software, and intelligent control systems that enable the automated classification and segregation of plastic materials. It focuses on three main components: the hardware design, which includes the physical assembly of sensors, actuators, and processing units; the classification area, where sensor data and machine learning models work together to recognize different categories of plastic bottles; and the bin receptacles featuring a color-coded collection system to ensure organized and efficient material segregation. This page offers users and administrators a detailed understanding of how each component contributes to the operation of the system.

Figure 46

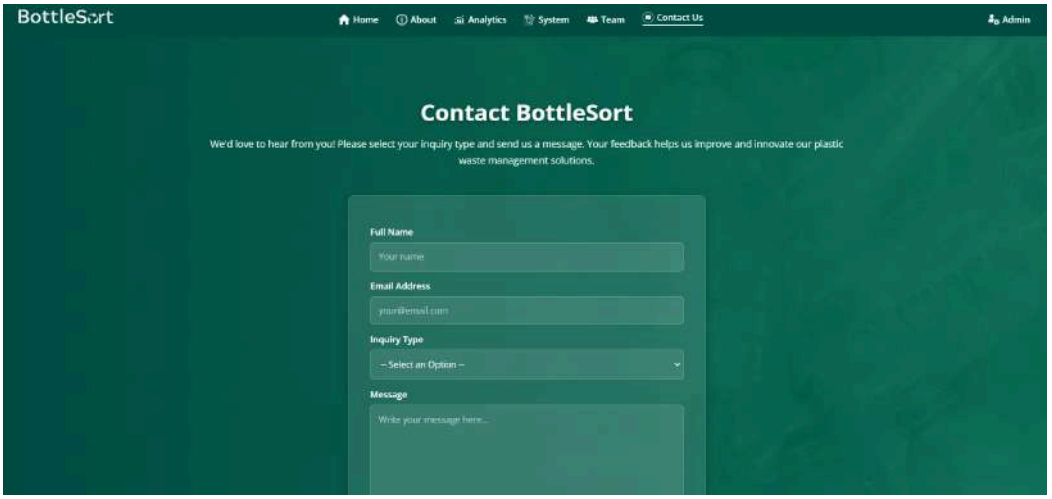
Team Page



The Team Page of the website introduces the group of researchers behind the project, emphasizing their shared commitment to improving plastic waste management through technology. It highlights the core members of the team—students from La Salle University - Ozamiz, who are pursuing degrees in Computer Engineering.

Figure 47

Contact Us Page



The Contact Us page on the website provides users with a straightforward way to connect with the team for inquiries, feedback, or to explore potential collaboration opportunities. Users can fill out a contact form that includes their name, email address, inquiry type, and message to get in touch. This page promotes communication between users and developers, enabling valuable feedback that supports the continuous improvement, innovation, and refinement of the system.

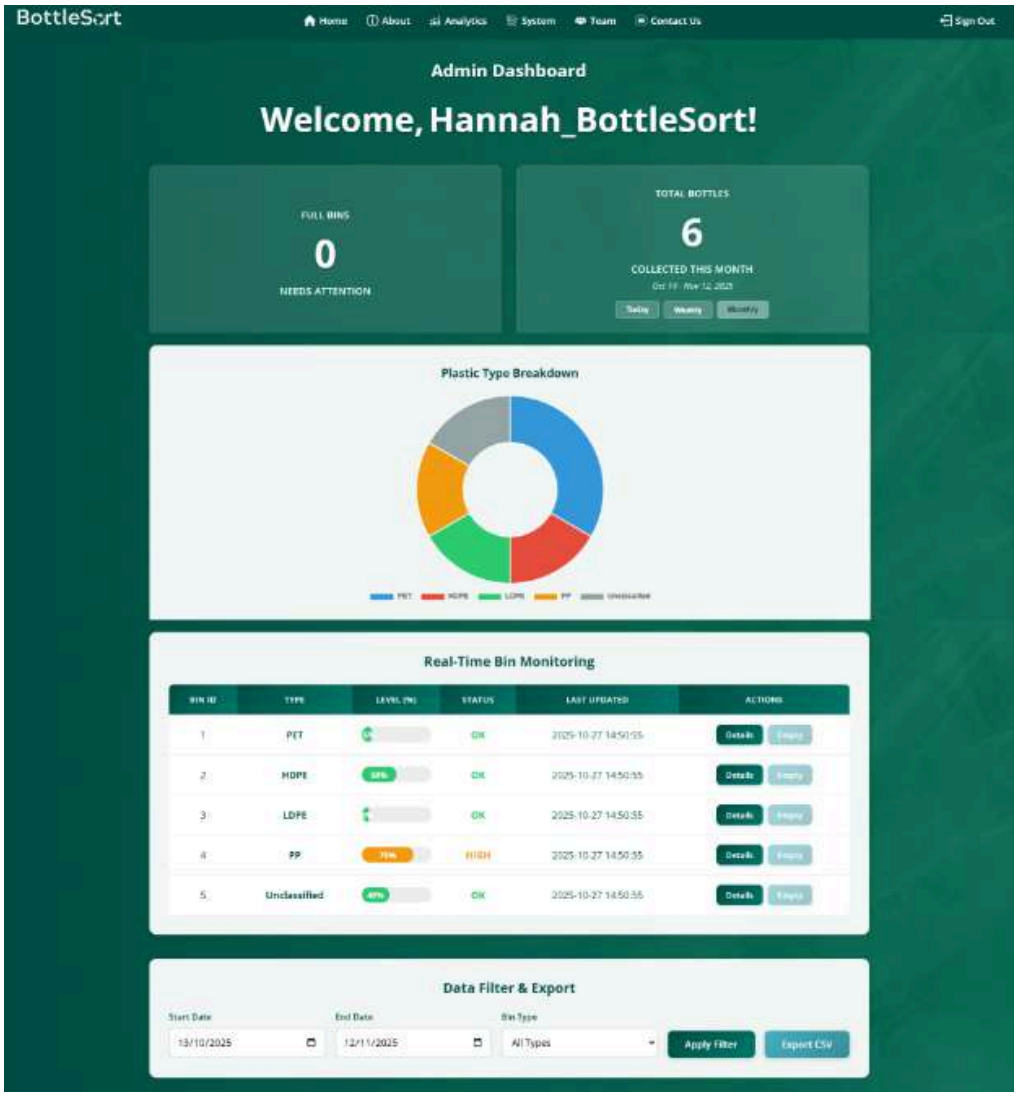
Figure 48

Administrator Login Page

The Administration Login Page of the website serves as a secure entry point for authorized administrators to manage and monitor system operations. To ensure data privacy and system integrity, valid login credentials are required. Through this portal, administrators can oversee classification data, update system configurations, and maintain overall functionality. This page is essential for safeguarding the system's performance and ensuring that only authorized personnel can perform administrative tasks within the platform.

Figure 49

Administrator Dashboard Page



The Administrator Dashboard Page of the website serves as a central hub for managing system data. It provides insights into various plastic types—PET, HDPE, LDPE, PP, and Unclassified—allowing effective monitoring of sorting trends and performance. Real-time bin monitoring shows current fill levels and statuses, facilitating efficient waste management. Additionally, data filtering and export tools help organize information and generate reports.

3.8. Data Acquisition

The data acquisition phase is a critical part of the system development process, as it provides the necessary image dataset for the purpose of training and evaluating the classification model. For this study, the focus is on collecting images of plastic bottles belonging to four target categories based on the Resin Identification Code (RIC): PET, HDPE, LDPE, and PP. These types were selected due to their prevalence in everyday packaging and their visual relevance for image-based sorting systems.

To ensure that the classification model can handle a variety of real-world conditions, each plastic bottle type is captured in four different physical states:

(1) with a cap and label,

(2) without a cap and label,

(3) without a cap but with a label, and

(4) with a cap but without a label.

These variations simulate common conditions in post-consumer plastic waste, helping the model learn to recognize each plastic category regardless of minor differences in appearance. An additional category labeled “Unclassified” will also be included to account for plastics or waste that do not fit any of the four main types, serving as a baseline for non-target materials.

The dataset consists of two parts. First, original images were captured using a camera in a top-view orientation, under consistent lighting conditions, and against a plain background. This setup reflects the planned configuration of the proposed system and helps reduce environmental noise during training. Second, additional bottle images were obtained from a dataset that is publicly accessible on Kaggle to enhance the dataset's diversity and size. By combining both sources, the dataset aims

to support more accurate training and testing of the model, ensuring its robustness in various sorting scenarios during future implementations.

Table 2

Image Count by Material Type

	PET	**HDPE**	**LDPE**	**PP**	**Unclassified**	**Total**
Number of Data Acquired						

Table 2 outlines the total number of images collected for each material type in the dataset. It provides a summary of the distribution among five categories: PET, HDPE, LDPE, PP, and Unclassified. This overview helps to assess data balance and representation. The first four categories include images of plastic bottles, while the Unclassified category features other waste items that do not fit into any of the main types. This information ensures that the dataset offers sufficient variety and sample size for effective model training and evaluation.

Table 3

Data Split for Training and Testing

Classification	**Total Acquired Images**	**Training (80%)**	**Validation (20%)**
PET			
HDPE			
LDPE			
PP			
Unclassified			

Table 3 displays the number of images collected for each material type in the dataset. It also shows how these images were divided into training and validation groups with an 80-20 ratio. This distribution ensures that each category—PET, HDPE, LDPE, PP, and Unclassified—has enough samples for model development while maintaining a balanced representation across all classes. The table provides a clear overview of the dataset preparation process, which supports reliable training, tuning, and performance evaluation of the classification model.

Table 4

Classification Matrix of Plastic Bottles by Material Type and the Target Categories

Physical State	**PET**	**HDPE**	**LDPE**	**PP**
with a cap and label				
without a cap and label				
without a cap, with a label				
with a cap, without a label				

Table 4 presents a classification matrix that organizes plastic bottle images based on their material type, labeled PET, HDPE, LDPE, PP, and Unclassified, and physical state, captured in four categories: with a cap and label, without a cap and label, without a cap but with a label, and with a cap but without a label.

The expected values to be entered into the cells are image counts per condition per plastic type, allowing researchers to assess balance and fair representation of real-world plastic bottle appearances.

These variations help ensure that the classification model can accurately recognize plastic bottle types despite minor differences in appearance. It suggests a comprehensive and balanced dataset designed to support the evaluation and assessment of the model for classifying plastic bottles. See Appendices D-G for images of the collected bottles.

Figure 50

Sample variations for PET plastic bottles: (top left) with cap and label, (top right) without cap and label, (bottom left) with label only, (bottom right) with cap only. See Appendix D.

Figure 51

Sample variations for HDPE plastic bottles: (top left) with cap and label, (top right) without cap and label, (bottom left) with label only, (bottom right) with cap only. See Appendix E.

Figure 52

Sample variations for LDPE plastic bottles: (top left) with cap and label, (top right) without cap and label, (bottom left) with label only, (bottom right) with cap only. See Appendix F.

Figure 53

Sample variations for PP plastic bottles: (top left) with cap and label, (top right) without cap and label, (bottom left) with label only, (bottom right) with cap only. See Appendix G.

Figure 54

Sample variations for unclassified wastes: (top left) crumpled paper, (top right) cardboard, (bottom left) chips wrapper, (bottom right) aluminum can. See Appendix H.

3.9. System Flowchart

Figure 55

System Flowchart (Part 1)

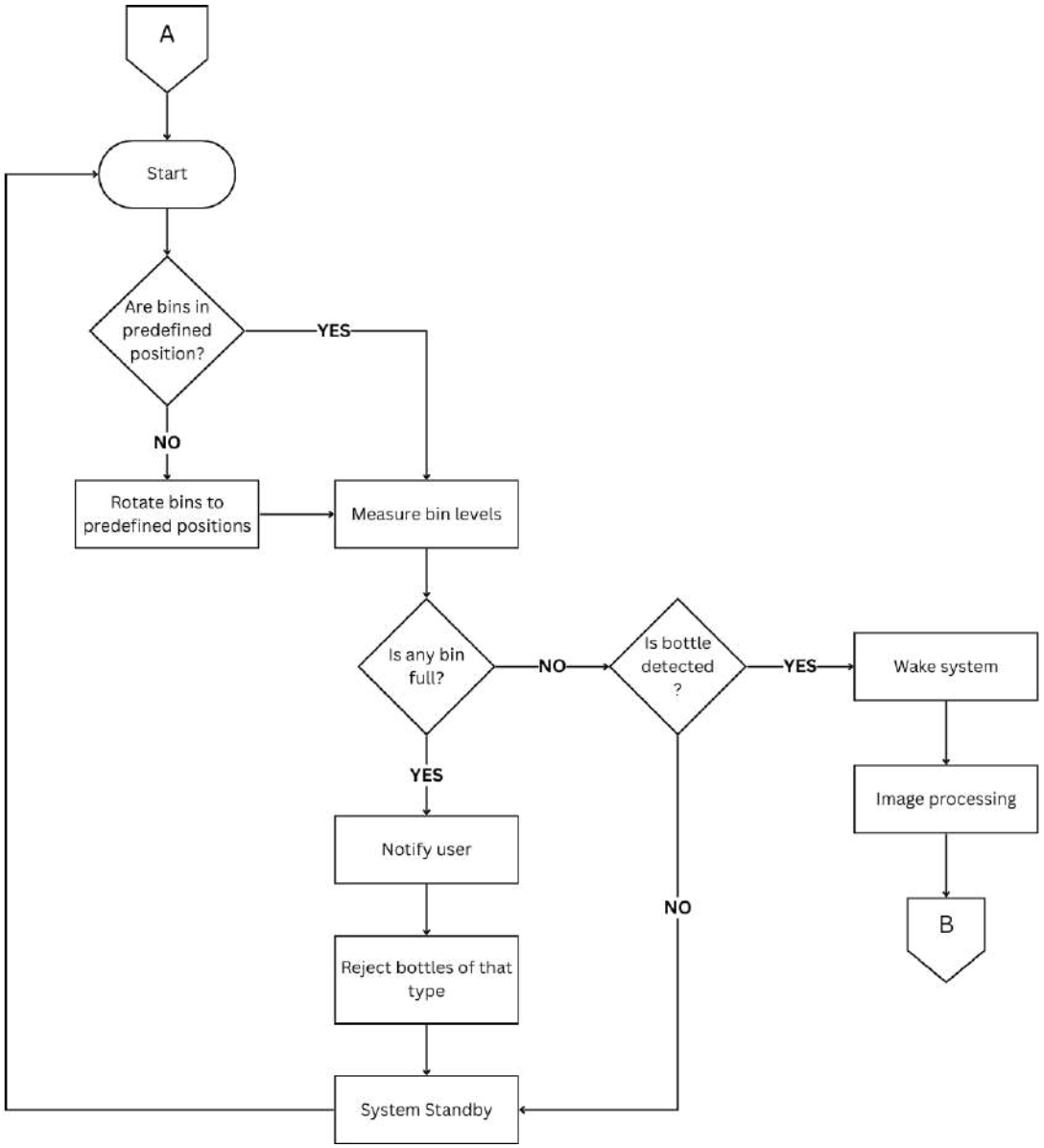


Figure 55 shows the initial operational sequence of the automated plastic bottle sorting system. The process starts by confirming all bins are in their predefined positions. If not, the system rotates the platform to correct its orientation. Once the bins are correctly positioned, the system measures their fill levels and checks if any bin is full. If a bin is full, the system notifies the user and pauses sorting by entering

standby mode. If all bins have capacity, the system checks for a plastic bottle. When a bottle is detected, the system exits standby mode and proceeds to image processing for sorting.

Figure 56

System Flowchart (Part 2)

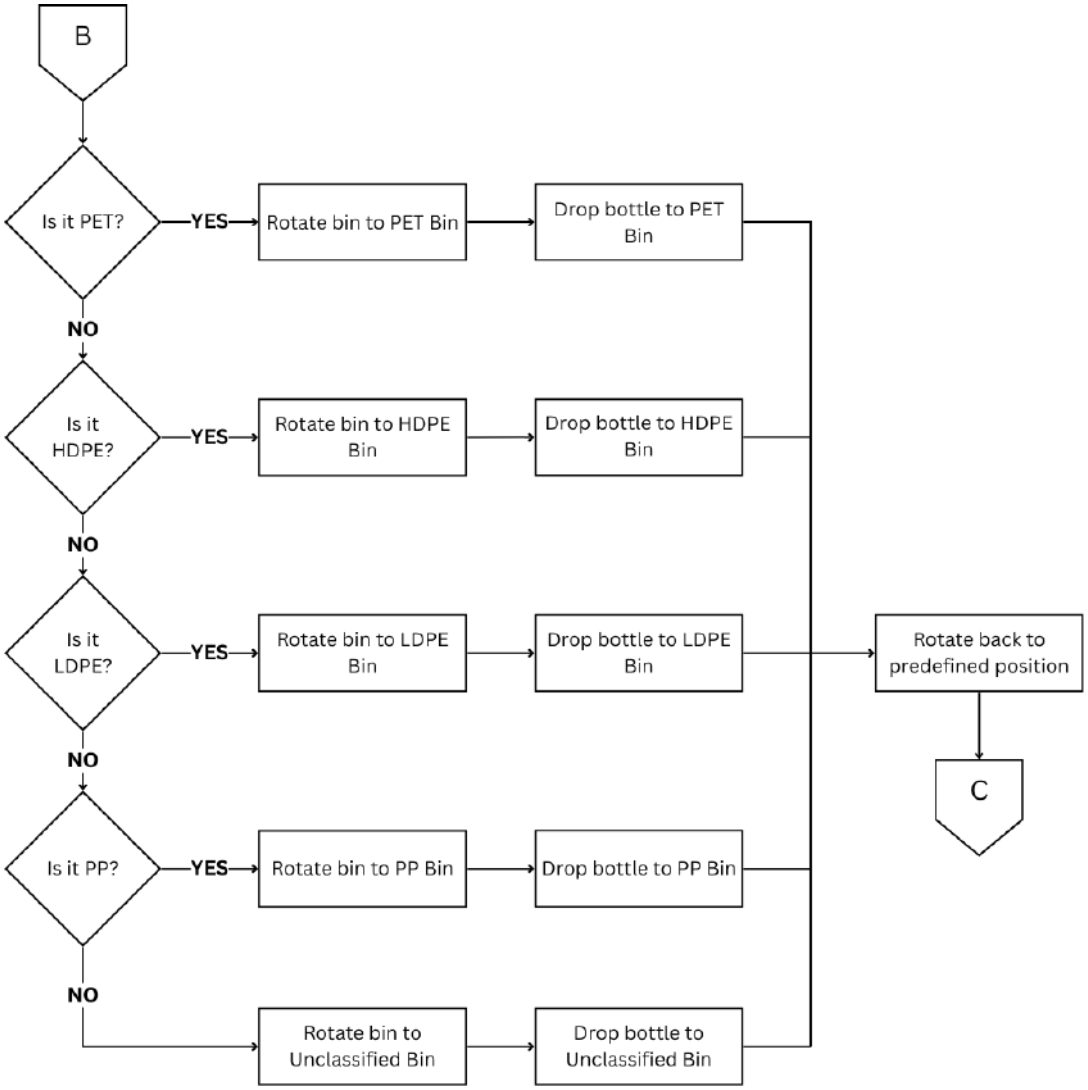


Figure 56 shows the second part of the system flowchart, which details the classification and sorting process after a bottle is detected. The system analyzes the captured image and classifies the bottle into a predefined plastic category using a trained model. Depending on the type (PET, HDPE, LDPE, PP, or Unclassified), the

system rotates the bin platform to position the correct bin under the disposal chute. The bottle is then released into the appropriate bin. Once sorting is complete, the platform returns to its starting position, ready for the next cycle.

Figure 57

System Flowchart (Part 3)

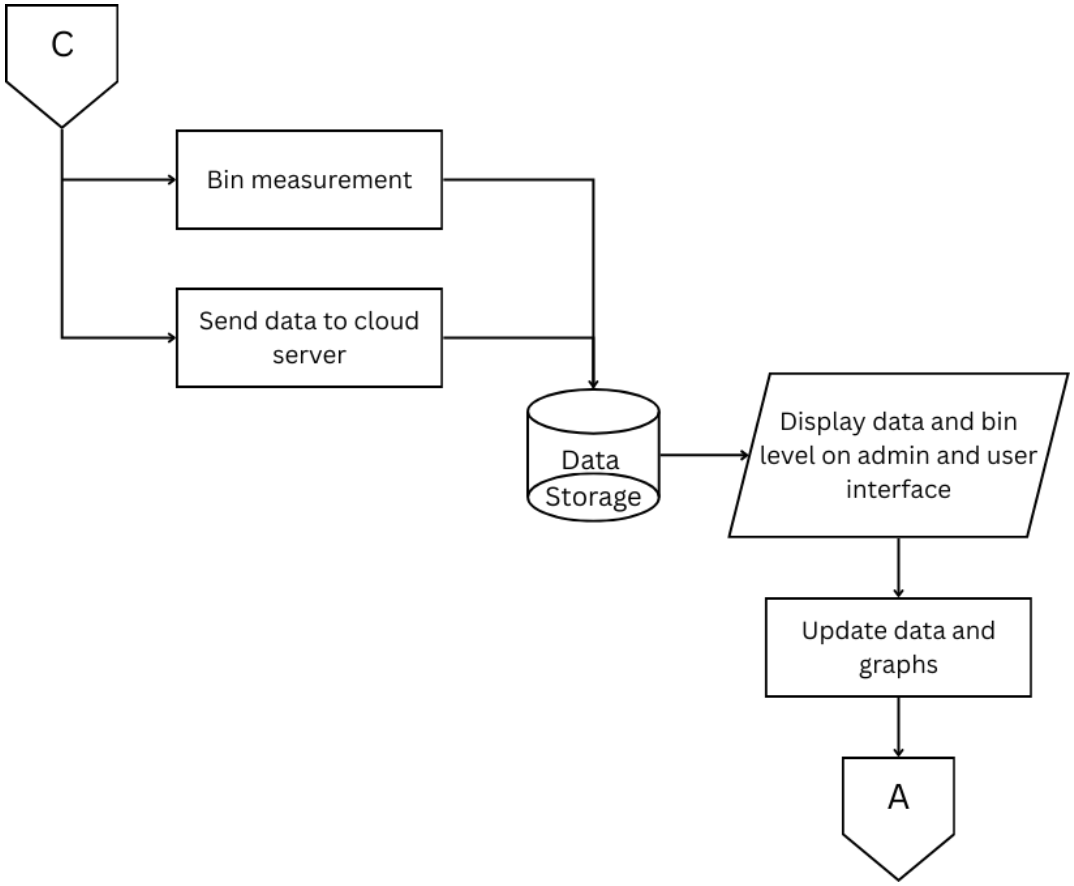


Figure 57 shows the final section of the system flowchart, which covers post-sorting monitoring and data management. After a bottle is deposited, the system measures the updated fill levels of all bins and sends this data to the server. The information is then displayed on both administrator and user interfaces, with visualizations, graphs, and status indicators updated in real time. Once data is synchronized and displayed, the process returns to the initial step, where the system checks bin positions and restarts the operational cycle.

3.10. Entity Relationship Diagram (ERD)

Figure 58

Entity Relationship Diagram of the System

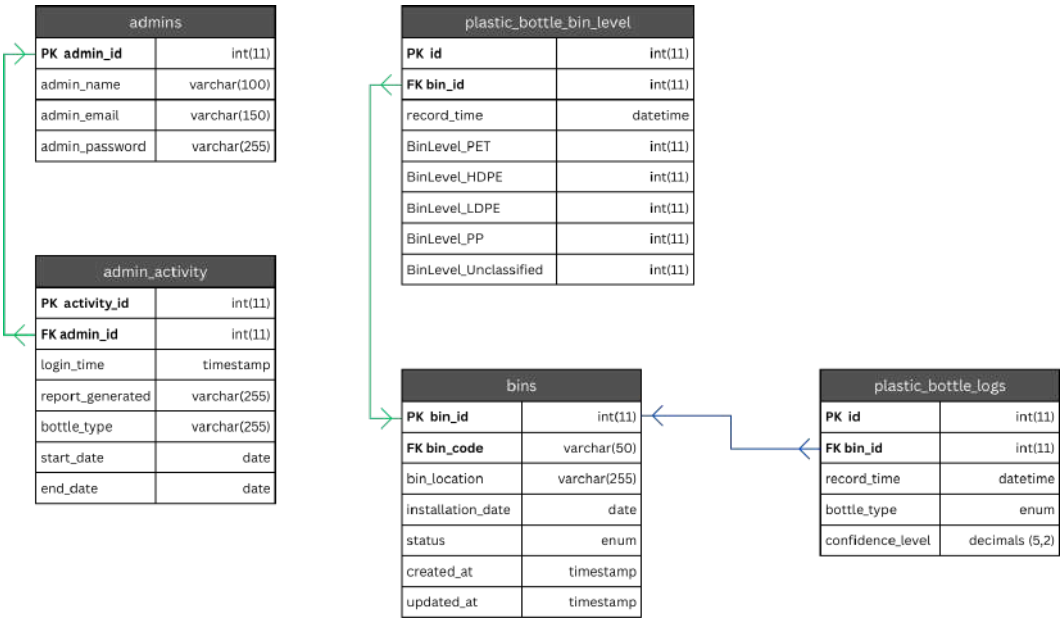
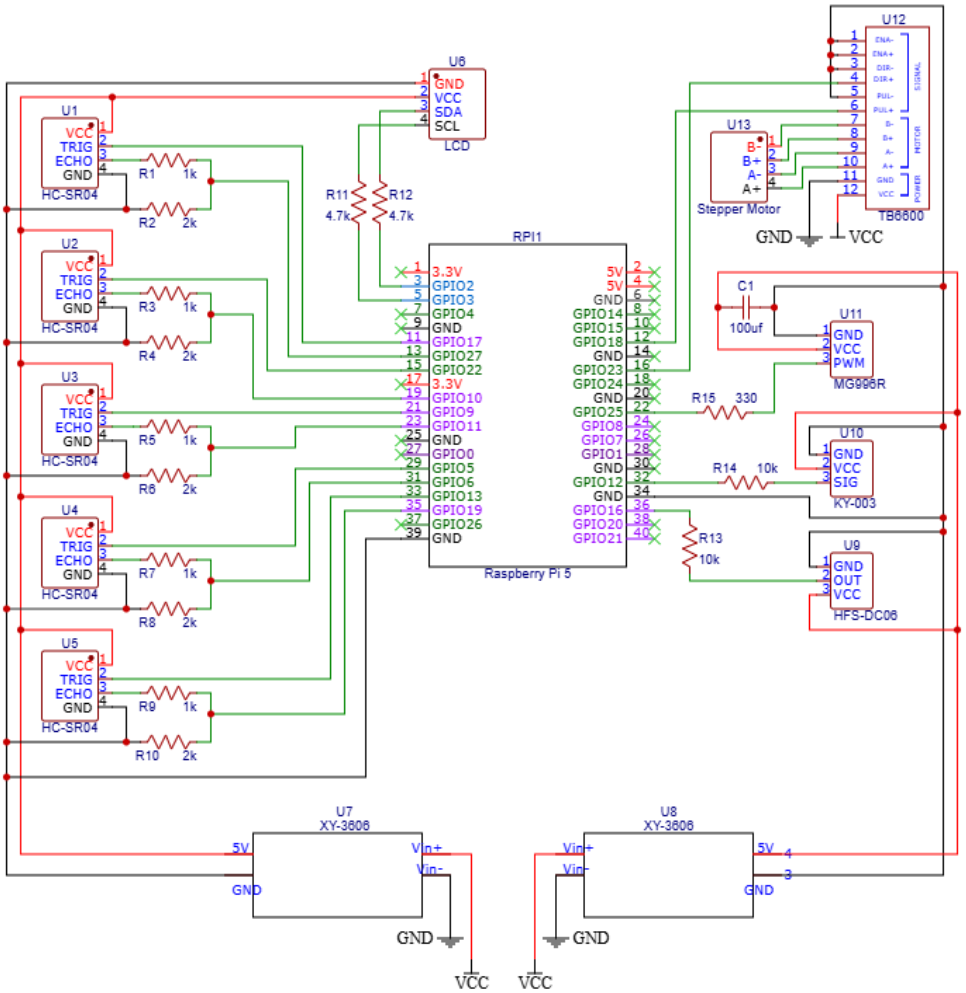


Figure 58 illustrates the Entity-Relationship Diagram (ERD) for the system, comprising five interconnected entities. The admins table establishes a one-to-many relationship with admin_activity, which logs administrative actions and reports. The bins table, representing physical collection bins at various locations, connects to plastic_bottle_logs, which records individual bottle detections with classification type and confidence level, and plastic_bottle_bin_level, which stores aggregated fill level data by plastic type. All relationships maintain referential integrity through foreign keys, ensuring data consistency and supporting efficient tracking of bottle detections and bin monitoring.

3.11. Circuit Diagram

Figure 59

Schematic Diagram



The Raspberry Pi 5 (RP1) acts as the central controller for the system, managing all sensor inputs, actuator outputs, and communication interfaces through its designated GPIO pins. Five HC-SR04 ultrasonic sensors (U1–U5) are connected to the Raspberry Pi, with each sensor's TRIG and ECHO lines interfaced through 1 kΩ and 2 kΩ resistors to ensure safe signal levels and minimize noise in distance measurements.

The 16×2 I2C LCD module (U6) communicates with the Raspberry Pi using the SDA and SCL lines, which are supported by two 4.7 kΩ pull-up resistors (R11,

R12) for stable I2C operation. Motor actuation is controlled via a dedicated driver and PWM lines. The TB6600 stepper motor driver (U12) receives direction and pulse signals from the Raspberry Pi and powers the NEMA 17 stepper motor (U13), providing the mechanical motion needed for bottle sorting.

Additionally, a high-torque MG996R servo motor (U11) is driven from a PWM pin through a series resistor (R15, 330 Ω) to regulate the control signal. Other sensing modules include the KY-003 Hall effect sensor (U10), connected through a 10 kΩ pull-down resistor (R14), and the HFS-DC06 microwave radar sensor (U9). Both of these sensors are supplied with a regulated 5V source.

Power distribution is managed through two XY-3606 buck converters (U7 and U8), which step down the system's main supply to a stable 5V line for the peripheral modules. A 100 μF capacitor (C1) is placed near the servo output to smooth out voltage fluctuations during motor operation. All VCC and GND rails are shared across components to maintain consistent reference levels and ensure reliable signal communication.

3.12. System Training and Validation Parameters

This section outlines the key settings and processes used in the computer vision and machine learning framework for classifying plastic bottles. Understanding the evaluation procedure and the model's capacity to generalize depends on the testing parameters outlined below. These parameters include the dataset configuration, performance metrics, and validation techniques used to assess classification accuracy and reliability.

Table 5

Criteria for Classifying Plastic Bottles Using Machine Learning

Parameter	Description
Input Data	*Categories of data inputs, such as images or written descriptions.*
Data Augmentation	*Method employed to enhance the variety of data used for training purposes.*
Training Epochs	*Number of iterations through the dataset throughout the training procedure.*
Learning Rate	*The rate at which the model modifies weights according to gradients.*
Batch Size	*Number of samples analyzed before adjusting the model's parameters.*
Validation	*The proportion of data reserved for assessing the model's performance while training.*
Performance Metrics	*Methods employed to assess how precisely the model performs.*

Table 5 describes the configuration employed to categorize plastic bottles into different groups, such as PET, HDPE, LDPE, PP, and Unclassified. It specifies the types of data inputs, mainly image-based, utilized for training the model. Data augmentation techniques are employed to enhance dataset diversity and boost generalization. The number of epochs indicates how many times the model reviews the complete training set, while the learning rate determines how responsively the model adjusts its parameters. Batch size influences the speed and efficiency of training by regulating the number of samples processed simultaneously. Validation confirms the model's effectiveness on new data, aiding in the prevention of overfitting. Metrics

like accuracy offer insights into the system's capability to identify and classify different plastic bottle materials.

Table 6

Training Parameters for Plastic Bottle Classification

Training Parameters	**Classifications**				
	PET	HDPE	LDPE	PP	Unclassified
Input Data					
Augmentation					
Epoch					
Learning Rate					
Batch Size					
Validation					
Accuracy					

This table outlines the training parameters employed to develop a machine learning model that categorizes plastic bottle images into five categories: PET, HDPE, LDPE, PP, and Unclassified. Each column corresponds to a specific plastic type, while each row outlines a key training component, such as input data volume, data augmentation techniques, number of epochs, learning rate, batch size, validation settings, and resulting classification accuracy. It provides a breakdown in comparison on how each classification type was trained and evaluated, emphasizing the consistency in model tuning across various plastic categories.

The “Input Data” row will contain the number of training images per class. “Augmentation” refers to techniques such as rotation, flipping, zooming, or brightness adjustments. “Epoch” will include an integer value representing the number of

training cycles. The “Learning Rate” will display a small decimal value, depending on the model tuning. The “Batch Size” parameter indicates the number of images processed together before the model updates during training. “Validation” refers to the percentage or number of images set aside per class for validation, typically 20% of the dataset. Finally, “Accuracy” will contain the expected classification performance per class, expressed as a percentage.

Table 7

Testing Parameters for Plastic Bottle Classification

Testing Parameters	**Classifications**				
	PET	HDPE	LDPE	PP	Unclassified
Input Data					
Augmentation					
Epoch					
Learning Rate					
Batch Size					
Validation					

This table outlines the testing parameters employed to evaluate the effectiveness of the machine learning model that categorizes plastic bottle images into five categories: PET, HDPE, LDPE, PP, and Unclassified. Each column corresponds to a specific plastic type, while each row outlines a key training component, such as input data volume, data augmentation techniques, number of epochs, learning rate, batch size, validation settings, and resulting classification accuracy. It provides a breakdown in comparison on how each classification type was trained and evaluated, emphasizing the consistency in model tuning across various plastic categories.

The “Input Data” row will contain the number of test images per class. “Augmentation” refers to any preprocessing applied to test images, such as resizing or normalization. However, minimal augmentation is expected during testing to preserve real-world accuracy. “Epoch” will reflect the final model state used for testing, typically the same value used during training. “Learning Rate” and “Batch Size” will match the configuration used during training. “Validation” during testing may refer to additional metrics or evaluation sets, often including a fixed 20% of the dataset or a separate unseen test set.

Table 8

Types of PET Plastic Bottles

	Result	**Remarks**
PET Bottle 1 - with cap and label		
PET Bottle 2 - without cap and label		
PET Bottle 3 - without cap but with label		
PET Bottle 4 - with cap but without label		

Table 8 presents the effectiveness of the system in categorizing various types of PET plastic bottles, each varying by the presence or absence of a label or cap. The “Result” column will display the classification accuracy of the model for each specific bottle type, expressed as a percentage. This reflects how accurately the system identifies each variant as PET. The “Remarks” column will contain short qualitative observations, such as whether the prediction was correct, misclassified, or uncertain.

Expected values in this column include notes like “Correctly classified as PET,” “Misclassified,” or “Low confidence due to missing label.” This format helps assess the model’s reliability and robustness when handling PET bottles with visual differences.

Table 9

Types of HDPE Plastic Bottles

	Result	**Remarks**
HDPE Bottle 1 - with cap and label		
HDPE Bottle 2 - without cap and label		
HDPE Bottle 3 - without cap but with label		
HDPE Bottle 4 - with cap but without label		

Table 9 presents the system's ability to categorize various types of HDPE plastic bottles, each varying by the presence or absence of a label or cap. The “Result” column will display the classification accuracy of the model for each specific bottle type, expressed as a percentage. This reflects how accurately the system identifies each variant as HDPE. The “Remarks” column will contain short qualitative observations, such as whether the prediction was correct, misclassified, or uncertain. Expected values in this column include notes like “Correctly classified as HDPE,” “Misclassified,” or “Low confidence due to missing label.” This format helps assess the model’s reliability and consistency when recognizing HDPE bottles with slight physical variations.

Table 10

Types of LDPE Plastic Bottles

	Result	**Remarks**
LDPE Bottle 1 - with cap and label		
LDPE Bottle 2 - without cap and label		
LDPE Bottle 3 - without cap but with label		
LDPE Bottle 4 - with cap but without label		

Table 10 presents the system's effectiveness in categorizing various kinds of LDPE plastic bottles, each varying by the presence or absence of a label or cap. The “Result” column will display the classification accuracy of the model for each specific bottle type, expressed as a percentage. This reflects how accurately the system identifies each variant as LDPE. The “Remarks” column will contain short qualitative observations, such as whether the prediction was correct, misclassified, or uncertain. Expected values in this column include notes like “Correctly classified as LDPE,” “Misclassified,” or “Low confidence due to missing label.” This format helps assess the model’s effectiveness and reliability in detecting LDPE bottles with subtle visual differences.

Table 11

Types of PP Plastic Bottles

	Result	**Remarks**
PP Bottle 1 - with cap and label		
PP Bottle 2 - without cap and label		
PP Bottle 3 - without cap but with label		
PP Bottle 4 - with cap but without label		

Table 11 presents the system's effectiveness in categorizing various kinds of PP plastic bottles, each varying by the presence or absence of a label or cap. The “Result” column will display the classification accuracy of the model for each specific bottle type, expressed as a percentage. This reflects how accurately the system identifies each variant as PP. The “Remarks” column will contain short qualitative observations, such as whether the prediction was correct, misclassified, or uncertain. Expected values in this column include notes like “Correctly classified as PP,” “Misclassified,” or “Low confidence due to missing label.” This format helps assess the model’s reliability and precision in recognizing PP bottles despite minor physical variations.

Table 12

Types of Unclassified Waste

	Result	**Remarks**
Unclassified Waste 1 - Crumpled Paper		
Unclassified Waste 2 - Plastic Wrapper		
Unclassified Waste 3 - Chips wrapper		
Unclassified Waste 4 - Aluminum Can		

Table 12 presents the effectiveness of the system in categorizing various kinds of unclassified waste, each varying by the presence or absence of a label or cap. The “Result” column will display the classification accuracy of the model for each specific bottle type, expressed as a percentage. This reflects the system's accuracy in identifying each variant of unclassified waste. The “Remarks” column will contain short qualitative observations, such as whether the prediction was correct, misclassified, or uncertain. Expected values in this column include notes like “Correctly classified,” “Misclassified,” or “Low confidence due to missing label.” This format helps assess the model’s reliability and precision in recognizing PP bottles despite minor physical variations.

Chapter 4

RESULTS AND DISCUSSION

This chapter presents a comprehensive analysis of the development, implementation, and evaluation of the automated plastic bottle segregation system. It details the system's data acquisition, training of the classification model, hardware and software integration, and the performance of real-time monitoring and analytics. Emphasis is placed on the model's ability to accurately identify different types of plastic materials, the effectiveness of sensor-based bin tracking, and the overall functionality of the web-based interface for administrators. The results are discussed in terms of classification accuracy, operational efficiency, and the system's potential to enhance sustainable waste management practices. Through these findings, this chapter demonstrates the practical application and effectiveness of integrating artificial intelligence, embedded systems, and IoT technologies in modern environmental solutions.

4.1. Training Result

Figure 60

Confusion Matrix and Metrics

Model

Model version: Quantized (int8)

Last training performance (validation set)

ACCURACY
98.6%

LOSS
0.10

Confusion matrix (validation set)

	HDPE	LDPE	PET	PP	UNCLASSIFIED	BACKGROUND
HDPE	98.7%	0%	1.0%	0.3%	0%	0%
LDPE	0%	97.9%	1.3%	0.9%	0%	0%
PET	0.3%	0.2%	99.1%	0.3%	0%	0%
PP	0.4%	2.5%	1.1%	96.1%	0%	0%
UNCLASSIFIED	0%	0%	0%	0%	100%	0%
BACKGROUND	0%	0%	0%	0%	0%	100%
F1 SCORE	0.99	0.97	0.99	0.97	1.00	1.00

Metrics (validation set)

METRIC	VALUE
Area under ROC Curve	1.00
Weighted average Precision	0.99
Weighted average Recall	0.99
Weighted average F1 score	0.99

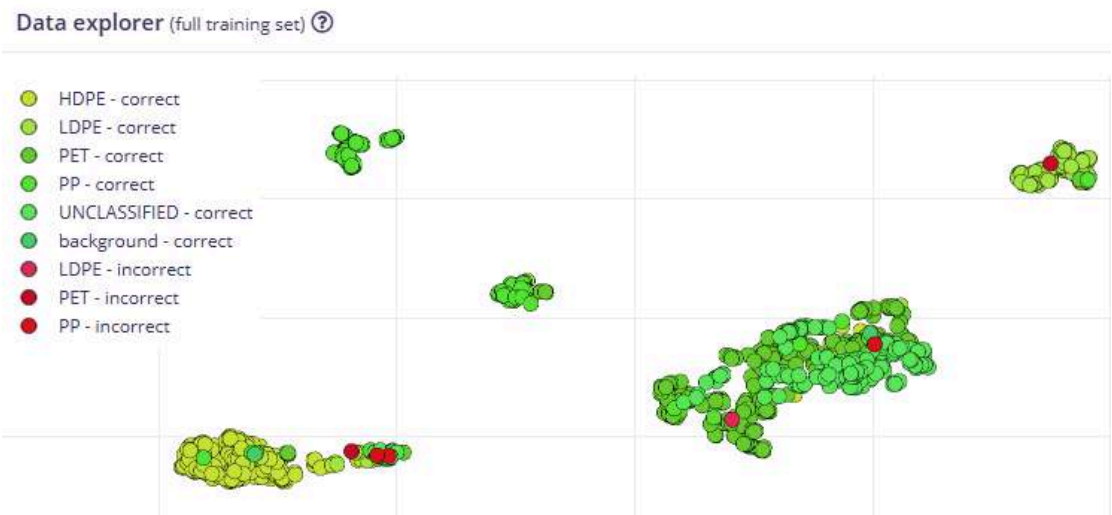
The training results indicate that the plastic bottle classification model performs exceptionally well. The validation set achieved an accuracy of 98.6% with a loss of 0.10, demonstrating the capability of the model to effectively learn and distinguish between plastic categories. The confusion matrix demonstrates a strong level of classification accuracy across all major classes: PET at 99.1%, HDPE at 98.7%, LDPE at 97.9%, PP at 96.1%, and Unclassified at 100%.

F1 scores across these categories are consistently high, highlighting an effective equilibrium between accuracy and recall. The weighted averages for

precision, recall, and F1 score are all at 0.99. Additionally, the Area Under the ROC Curve (AUC) is 1.00, which shows the model's excellent ability to separate the different classes. This AUC value reaffirms the strength and reliability of the model to accurately identify various types of plastic.

Figure 61

Data Exploration



The data explorer offers an understanding of the model's precision by visualizing the distribution of training samples in a reduced-dimensional space. Each point represents an image sample, color-coded according to its true material class and the accuracy of its classification. Clusters of green points indicate correctly classified samples for HDPE, LDPE, PET, PP, and unclassified categories, while red points highlight instances of misclassification.

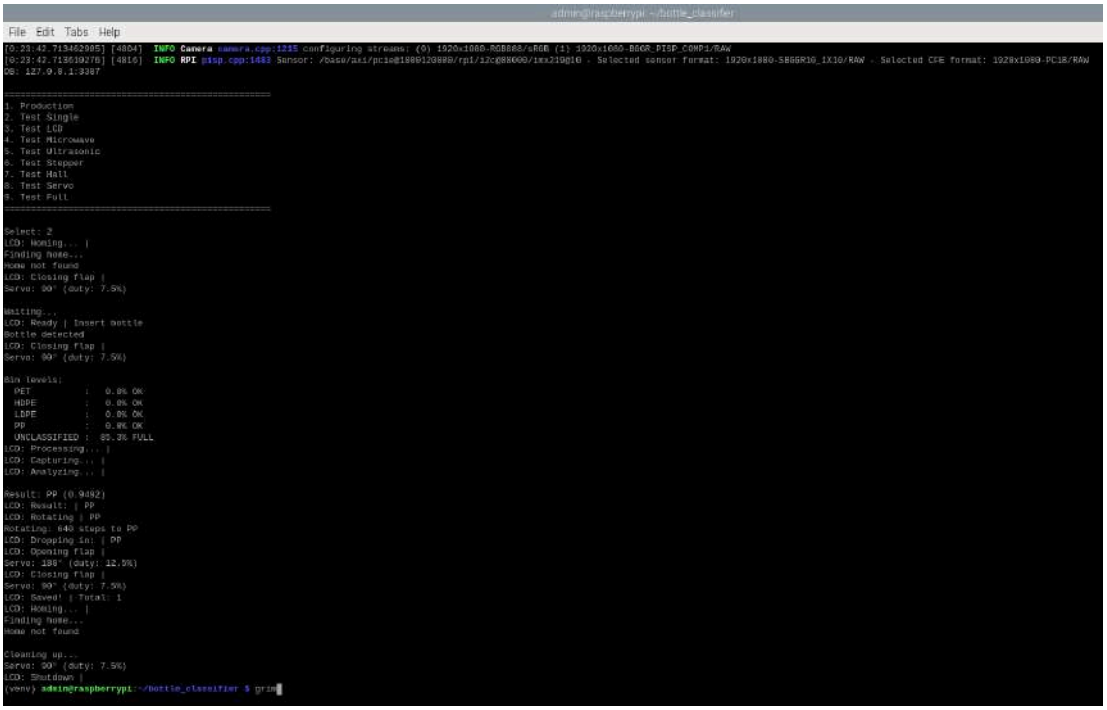
The spatial arrangement of these points demonstrates how visually similar samples tend to cluster together, revealing patterns within the dataset and identifying areas where the model faces challenges. This visualization is instrumental in

pinpointing overlapping classes, potential outliers, and regions that may require improved classification performance.

4.2. Raspberry Pi Command-Line Interface

Figure 62

Raspberry Pi CLI



The Raspberry Pi command-line interface (CLI) displays the real-time operation of the classification system. Upon initialization, the system first presents the current bin levels and the positions of the sorting bins, confirming that all components are in the correct state before detection and classification begin. The program then enters a waiting mode, monitoring the microwave motion sensor for activity.

When motion is detected, the system activates the camera to capture an image of the object. The captured image is subsequently processed by the classification model, which outputs the predicted material type along with confidence scores for all

classes. The CLI provides a detailed display of these probabilities, highlighting the class with the highest confidence and confirming the final classification.

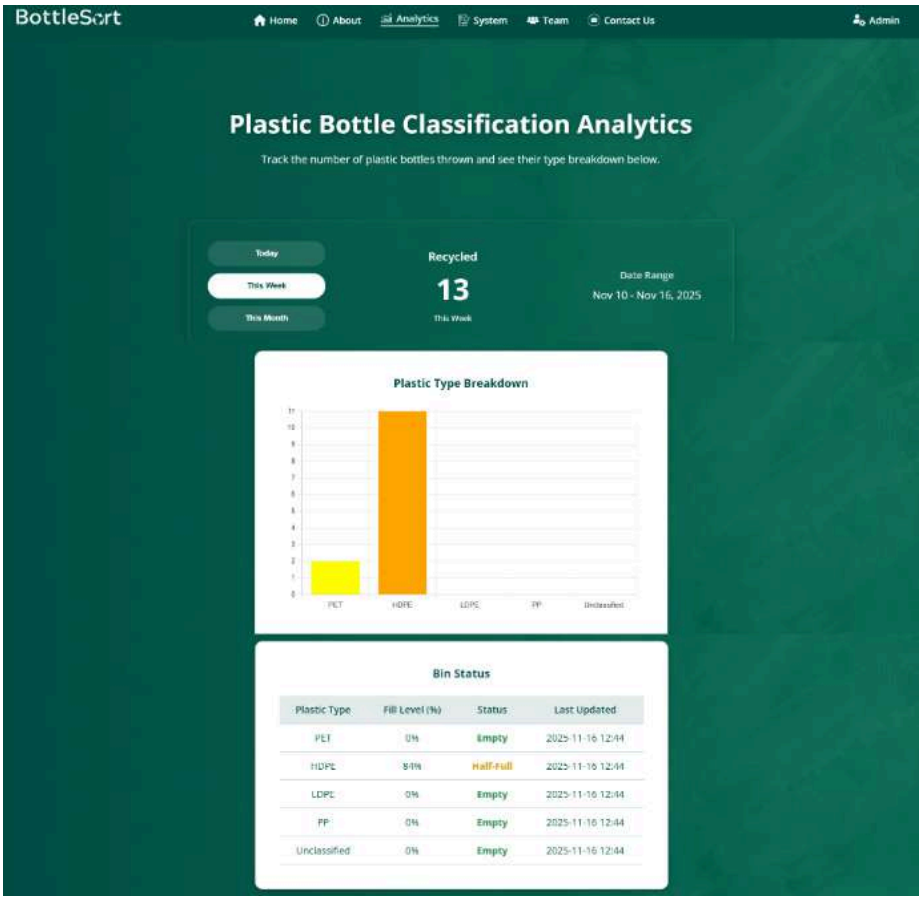
After classification, the system updates the bin status to reflect the outcome and any changes to bin levels. The results are then logged into the database, and the system executes its cleanup routine, which includes closing the LCD interface, stopping the camera, terminating the database connection, and resetting the GPIO pins.

4.3. Online Monitoring Interface

The online monitoring interface provides real-time outputs and logs from the system. Users and administrators can view bottle classification results, monitor bin fill levels, and analyze historical trends generated by the system. Information is retrieved from the database and displayed through visual charts, status indicators, and summarized statistics, offering a comprehensive view of system operations and performance.

Figure 63

Analytics Page (Monitoring Page)

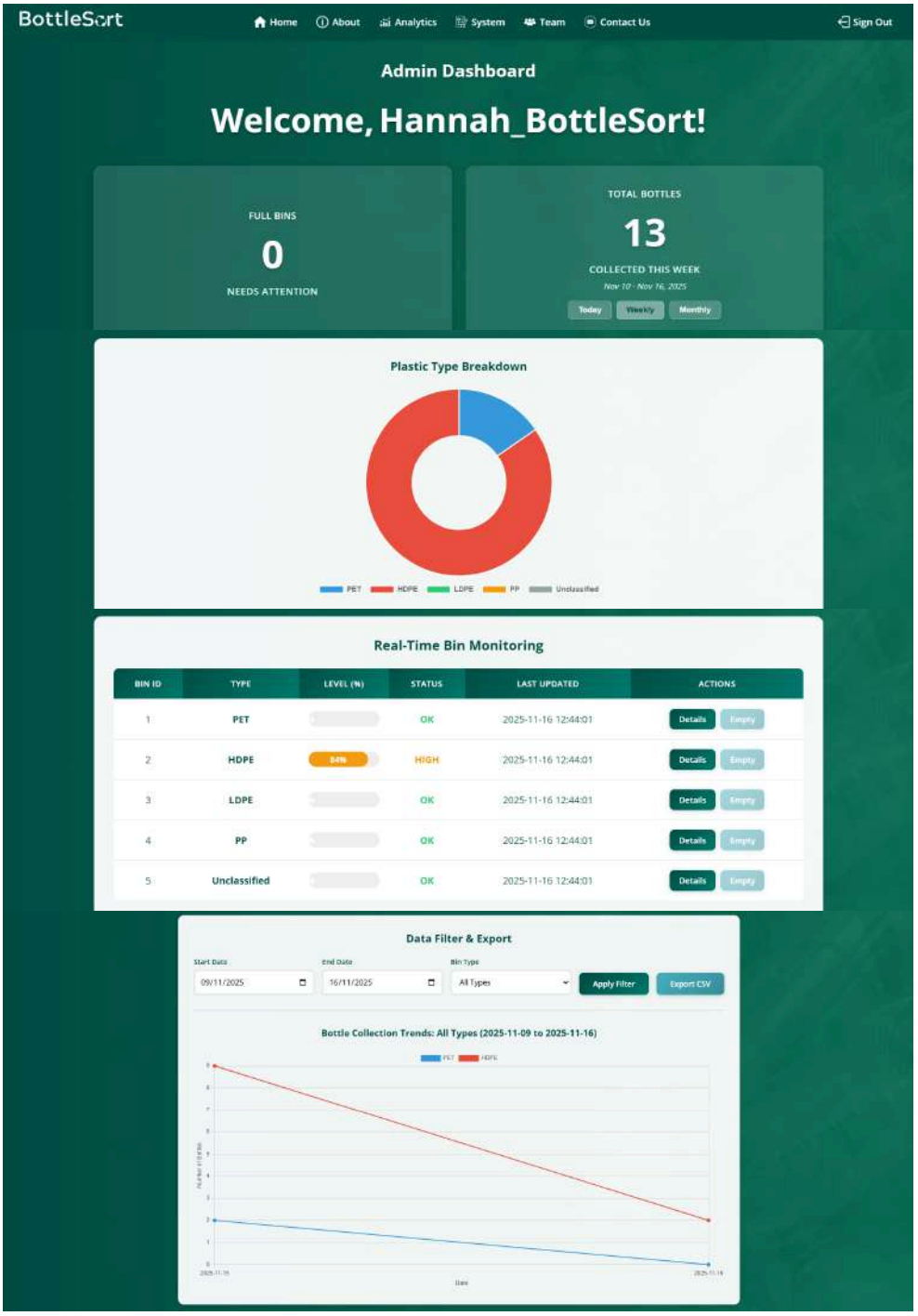


During testing, the Analytics Page effectively displayed real-time data generated by the system. The interface presented the total number of bottles processed within the selected date range, along with quick filters for daily, weekly, and monthly summaries. A bar chart illustrated the breakdown of detected plastic types, reflecting the model's classification outputs for PET, HDPE, LDPE, PP, and Unclassified items.

Additionally, the Bin Status table displayed the latest fill-level percentages for each bin, automatically updating as new classifications were recorded. Status indicators (e.g., Empty, Half-Full) were also included, confirming the system's capability to track and transmit bin capacity data in real time.

Figure 64

Admin Dashboard (Monitoring Page)



The Admin Dashboard provides an overview of the system's operational status. Testing results displayed counters for the total number of complete bins and the overall number of bottles collected, confirming successful synchronization of data

from the database. Additionally, a plastic-type breakdown chart visually summarized the proportions of each material detected during the testing period.

The Real-Time Bin Monitoring table accurately reflected the latest sensor readings, including fill percentages, status indicators, and timestamps. Administrators could view detailed information for each bin and initiate emptying actions directly from the dashboard. A data filter and export feature allowed filtering by date range and bin type, while a line graph illustrated trends in bottle collection over time, demonstrating the system's capability to generate historical analytics and support data-driven decision-making.

4.4. Database Logging

Figure 65

LCD Display for Data Logging

The data logging feature records all outputs generated by the system during operation. This includes every classified item, updates to bin fill levels, and administrative actions performed while the system is in use. These logs provide a reliable foundation for validating system performance, monitoring usage patterns, and tracking operational events throughout the testing period.

Figure 66

Plastic Bottle Logs Table

id	bin_id	record_time	bottle_type	confidence_level
1	1	2025-11-15 14:21:48	PET	0.98
2	1	2025-11-15 17:11:10	HDPE	0.94
3	1	2025-11-15 17:12:09	LDPE	0.94
4	1	2025-11-15 17:16:06	PET	0.96
5	1	2025-11-15 17:16:40	Unclassified	0.90
6	1	2025-11-15 17:17:21	PP	0.90

Figure 66 presents the classification events recorded by the system. The table includes the specific system the process was made in, detected material type, the timestamp of detection, and the confidence score assigned by the classification model for each processed item. This output highlights the system's ability to classify and log every detected plastic bottle in real time, providing detailed confidence metrics for each classification.

Figure 67

Plastic Bottle Bin Level Table

id	bin_id	record_time	BinLevel_PET	BinLevel_HDPE	BinLevel_LDPE	BinLevel_PP	BinLevel_Unclassified
1	1	2025-11-15 14:41:12	5	10	20	30	90
2	1	2025-10-27 14:46:56	5	90	80	75	50
3	1	2025-10-27 14:47:47	75	0	0	0	0
4	1	2025-10-27 14:50:55	15	50	10	75	40
5	1	2025-10-27 14:50:55	15	50	10	75	40

Figure 67 illustrates the current fill levels of each plastic bottle bin as recorded by the system. The table presents the percentage fill along with the timestamp of the most recent update. These results demonstrate the system's capability to continuously monitor and report bin capacities, facilitating timely maintenance and ensuring efficient operation.

Figure 68

Admin Activity Table

activity_id	admin_id	login_time	report_generated	bottle_type	start_date	end_date
1	1	2025-11-17 13:36:46	Plastic Bottle Collection Report	PET	2025-11-09	2025-11-16
2	3	2025-11-17 17:09:15	NULL	NULL	NULL	NULL

Figure 68 presents a log of administrative actions performed within the system. The table includes timestamps, administrator IDs, and descriptions of activities, such as viewing details and exporting reports. This output highlights the system's capability to monitor and document administrative interactions, promoting accountability and supporting effective system management.

4.5. Data Acquisition

The data acquisition process involved collecting image samples representing various types of plastic bottle materials processed by the system. This dataset served as the foundation for training and evaluating the classification model, ensuring the system could accurately identify different types of plastics. The collected images encompassed multiple categories, including PET, HDPE, LDPE, PP, and Unclassified materials.

Table 13

Total Number of Images by Material Type

	PET	HDPE	LDPE	PP	Unclassified	Total
Number of Data Acquired	3,628	2,429	1,631	1,589	2,420	11,697

Table 13 displays the total number of images collected for each type of plastic material included in the dataset. The dataset comprises 3,628 images of PET, 2,429 images of HDPE, 1,631 images of LDPE, 1,589 images of PP, and 4,527 unclassified images. In total, there are 13,804 samples.

Table 14

Dataset Train–Test Breakdown per Category

Classification	Total Acquired Images	Training (80%)	Testing (20%)
PET	3,628	2,904	724
HDPE	2,429	1,948	481
LDPE	1,631	1,310	321
PP	1,589	1,265	324
Unclassified	2,420	1,944	476

Table 14 illustrates how the acquired dataset is split into training and testing groups, adhering to an 80/20 ratio, a common practice in machine learning that helps balance model training and evaluation. Eighty percent of the dataset is allocated for training the classification model, while the other twenty percent is reserved for evaluating its performance. This separation guarantees that the model has sufficient data for learning meaningful patterns, while also offering an unbiased set of samples for the assessment of its accuracy and generalization.

Table 15

Physical State Distribution per Plastic Classification

Physical State	PET	HDPE	LDPE	PP
with cap and label	968	92	347	769
without cap and label	824	468	121	105
without cap but with label	867	49	23	409
with cap but without label	826	1,749	1056	121

Table 15 presents the number of samples categorized by their physical state, grouped into four categories: those with a cap and label, those without a cap and label, those without a cap but with a label, and those with a cap but no label. PET and HDPE show higher sample counts across most categories, indicating their greater availability and frequent occurrence during data collection. In contrast, LDPE and some PP states have fewer samples, as only a limited number of bottles of these types were collected during the data-gathering process.

4.6. System Training and Testing Results

This section presents the training process and evaluation of the classification model developed for identifying plastic bottle materials. It provides a detailed overview of the training setup and parameters, the performance metrics obtained from validation and testing, and an analysis of the model's effectiveness in categorizing

different types of plastics. The results demonstrate the system's ability to accurately distinguish among various plastic categories under diverse physical conditions.

Table 16

Training and Validation Setup for the Model

Training and Validation Parameters	
Input Data	9,383 training samples
Data Augmentation	Flipping, rotation, cropping, and scaling
Training Epochs	20
Learning Rate	0.0005
Batch Size	128
Validation	20% of Training Data = 1877
Performance Metrics	0.99 Weighted Average Precision
	0.99 Weighted Average Recall
	0.99 Weighted Average F1 Score
	0.10 Training and Validation Loss

Table 16 summarizes the training configuration and performance results of the classification model. The model was trained on 11,077 samples and employed data augmentation techniques—such as flipping, rotation, cropping, and scaling—to improve generalization. Training was conducted over 20 epochs with a learning rate of 0.0005 and a batch size of 128. For validation, 20% of the dataset—equivalent to 2,216 samples—was used.

The model achieved exceptional performance metrics, attaining weighted average scores of 0.99 for precision, recall, and F1-score, reflecting highly accurate

and balanced classification. Additionally, both training and validation losses stabilized at 0.10, indicating effective learning and minimal overfitting.

Table 17

Training Parameters per Classification

Training Parameters	**Classifications**				
	PET	**HDPE**	**LDPE**	**PP**	**Unclassified**
Input Data	2,904	1,948	1,310	1,265	3,638
Augmentation	Flipping, rotation, cropping, and scaling				
Epoch	20 Training Cycles				
Learning Rate	0.0005				
Batch Size	128				
Validation	20% of Data = 1,877				
Accuracy	98.6%				

The classification model was developed using a dataset comprising five categories of materials: PET, HDPE, LDPE, PP, and Unclassified. The number of samples varied across these categories. To enhance dataset diversity and improve the model's generalization capability, data augmentation techniques—including flipping, rotation, cropping, and scaling—were applied. Training was conducted over 20 epochs with a learning rate of 0.0005 and a batch size of 128. For validation, 20% of the dataset—equivalent to 2,739 samples—was set aside. As a result of this training, the model achieved an overall accuracy of 98.6%, demonstrating its effectiveness in accurately classifying different types of plastic bottles.

Table 18

Testing Parameters per Classification

Testing Parameters	**Classifications**				
	PET	HDPE	LDPE	PP	Unclassified
Input Data	724	481	321	324	476
Augmentation	Flipping, rotation, cropping, and scaling				
Epoch	20 Training Cycles				
Learning Rate	0.005				
Batch Size	128				
Validation	2,329 samples				

Table 18 presents the parameters used to evaluate the performance of the plastic bottle classification model. The test dataset consisted of 2,739 validation samples spanning all classes. To ensure a comprehensive assessment under diverse visual conditions, data augmentation techniques—such as flipping, rotation, cropping, and scaling—were applied. The evaluation employed 20 training cycles, a learning rate of 0.005, and a batch size of 128. These settings provided a consistent and rigorous framework to assess the model's ability to accurately classify and generalize across different types of plastic bottle materials during testing.

Table 19

PET Plastic Bottle Classification

	Result	Remarks
PET Bottle 1 - with cap and label	0.98	Classified consistently with high confidence.
PET Bottle 2 - without cap and label	0.94	Correctly identified even with missing features.
PET Bottle 3 - without cap but with label	0.96	Classified consistently with high confidence.
PET Bottle 4 - with cap but without label	0.95	Accurate classification despite partial features.

The results of the PET classification indicate consistently high confidence across all four physical states. The model successfully identifies PET bottles, even in cases where features like caps or labels are absent. Confidence values between 0.94 and 0.98 indicate that the system reliably recognizes PET material, regardless of variations in bottle appearance.

Figure 69

LCD Display for PET Plastic Bottle Classification

Figure 69 presents the LCD display showing the system's classification result after detecting and analyzing the input plastic bottle. In this instance, the system identifies the bottle as PET, and the LCD outputs this information in real time. This

display serves as the primary user interface for conveying the classification outcome during system operation.

Table 20

HDPE Plastic Bottle Classification

	Result	**Remarks**
HDPE Bottle 1 - with cap and label	0.98	Classified consistently with very high confidence.
HDPE Bottle 2 - without cap and label	0.90	Often misclassified as “LDPE”
HDPE Bottle 3 - without cap but with label	0.98	Accurate classification despite partial features.
HDPE Bottle 4 - with cap but without label	0.92	Often misclassified as “LDPE”

Here, the classification results for HDPE (High-Density Polyethylene) bottles indicate that the model performs exceptionally well when bottles are largely intact. Samples with complete or partially visible features, such as caps or labels, achieved high confidence scores, reflecting reliable detection. However, HDPE bottles lacking both caps and labels—or missing one of these features—were more prone to misclassification, often being mistaken for LDPE (Low-Density Polyethylene). These results suggest that the absence of key physical features reduces the model’s confidence and hinders its ability to accurately differentiate HDPE from visually similar materials.

Figure 70

LCD Display for HDPE Plastic Bottle Classification

Figure 70 illustrates the LCD display showing the system's classification result after detecting and analyzing the input plastic bottle. In this instance, the system identifies the bottle as HDPE and displays this information in real time on the LCD.

Table 21

LDPE Plastic Bottle Classification

	Result	**Remarks**
LDPE Bottle 1 - with cap and label	0.94	Classified consistently with high confidence.
LDPE Bottle 2 - without cap and label	0.90	Often misclassified as "HDPE"
LDPE Bottle 3 - without cap but with label	0.94	Accurate classification despite partial features.
LDPE Bottle 4 - with cap but without label	0.91	Often misclassified as "HDPE'

The results of the LDPE classification demonstrate strong performance across most physical states, with confidence scores consistently exceeding 0.90. The model accurately identified bottles with complete or partially visible features, highlighting its ability to recognize LDPE under varying conditions. However, samples lacking caps or labels were occasionally misclassified as HDPE, reflecting the visual similarities

between the two materials that can influence the model's predictions. Overall, the findings indicate that the system reliably detects LDPE while also identifying areas where additional training data could further enhance classification accuracy.

Figure 71

LCD Display for LDPE Plastic Bottle Classification

Figure 71 shows the LCD displaying the system's classification result after detecting and analyzing the plastic bottle. Here, the system identifies the bottle as LDPE and displays this in real time.

Table 22

PP Plastic Bottle Classification

	Result	**Remarks**
PP Bottle 1 - with cap and label	0.96	Classified consistently with strong confidence.
PP Bottle 2 - without cap and label	0.91	Often misclassified as "PET"
PP Bottle 3 - without cap but with label	0.93	Accurate classification despite partial features.
PP Bottle 4 - with cap but without label	0.90	Often misclassified as "PET"

The classification results for polypropylene (PP) plastic bottles demonstrate strong performance across different physical states. Bottles with caps and labels were consistently classified with high confidence, achieving an accuracy of 96%. In contrast, PP bottles without caps or labels showed a slightly lower accuracy, ranging from 90% to 91%, and were occasionally misclassified as polyethylene terephthalate (PET). Notably, PP bottles without caps but with labels still achieved accurate classification, with confidence scores exceeding 90%. These findings indicate that while the model performs reliably on fully intact PP bottles, it faces challenges in distinguishing PP from PET when key identifying features are absent.

Figure 72

LCD Display for PP Plastic Bottle Classification

Figure 72 displays the LCD showing the system's classification result for the detected plastic bottle. Here, the system identifies the bottle as PP and outputs this information in real time.

Table 23

Unclassified Waste Classification

	Result	Remarks
Unclassified Waste - Paper	0.92	Often misclassified due to a similar texture to other materials.
Unclassified Waste 2 - Cardboard	0.93	Generally identified accurately despite its close resemblance to paper.
Unclassified Waste 3 - Food Wrapper	0.96	Classified correctly despite varied shapes and reflections.
Unclassified Waste 4 - Aluminum Can	0.97	Occasionally confused with PET due to labels, but mostly accurate.

The classification accuracy for unclassified waste items varied across material types. Paper waste achieved an accuracy of 92%, though it was occasionally misclassified due to its texture, which closely resembles that of other materials. Cardboard recorded a slightly higher accuracy of 93% and was generally identified correctly despite its similarity to paper. Food wrappers presented classification challenges because of their irregular shapes and reflective surfaces; however, they were still classified correctly with an accuracy of 96%. Aluminum cans yielded the highest accuracy at 97%, with occasional misclassifications involving PET bottles, likely due to similarities in labeling. Overall, these results demonstrate the model's ability to classify a diverse range of unclassified waste types with strong and consistent accuracy.

Figure 73

LCD Display for Unclassified Plastic Bottle Classification

Figure 73 shows the LCD display presenting the system's classification result after detecting and analyzing the input plastic bottle. Here, the system classifies the waste as Unclassified, and the LCD outputs this information in real time.

Chapter 5

CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

This study successfully developed and implemented an automated plastic bottle segregation system that provides an efficient solution for classifying plastic bottles and monitoring bin levels. The system integrates machine learning, embedded hardware, and a web-based monitoring platform to automate critical functions such as material identification, bin-level tracking, and operational data recording for analysis and review.

The classification model demonstrated exceptional performance, achieving a validation accuracy of 98.6% with a loss of 0.10. It also achieved strong weighted averages of 0.99 for precision, recall, and F1-score. The confusion matrix and an AUC score of 1.00 further validate the model's robustness in distinguishing among PET, HDPE, LDPE, PP, and unclassified materials. Testing confirmed that the model performs consistently across various physical conditions of plastic bottles—whether capped, uncapped, labeled, or unlabeled—although minor misclassifications were observed between visually similar materials such as HDPE and LDPE, as well as PP and PET.

The Raspberry Pi–based classification pipeline operated reliably, providing real-time predictions, confidence scores, and bin-level updates through its command-line interface. Across all testing cycles, the system effectively executed sensor monitoring, camera image capture, classification, data logging, and cleanup routines.

The web-based monitoring interface displayed real-time classification results, bin fill percentages, historical analytics, and administrative logs. The Analytics Page and Admin Dashboard enabled users to review plastic type distributions, analyze collection trends, and monitor bin statuses. System data—stored in classification logs, bin-level tables, and administrative activity records—ensured transparency, traceability, and systematic documentation of all system events.

Additionally, the study produced a comprehensive dataset of 11,712 image samples representing different plastic categories in multiple physical states. This dataset played a crucial role in achieving the model's high recognition accuracy and strong generalization capability.

Overall, the study met all its objectives by demonstrating a high-performance classification model, a reliable embedded implementation, and an effective web-based monitoring interface. The findings confirm that automated plastic bottle classification is both feasible and practical for real-world waste management applications. The system enhances material sorting efficiency, supports data-driven recycling processes, and contributes to more sustainable waste management practices.

5.2. Recommendations

Based on the system's overall performance and testing results, the following recommendations are proposed to further enhance the system:

1. **Expand and Diversify the Dataset**

   To improve the model's accuracy, it is recommended to increase the number and diversity of images across all categories, especially for HDPE, LDPE, and PP. Including samples captured under different lighting conditions,

varying angles, deformations, occlusions, and levels of contamination will strengthen the model's ability to generalize and minimize misclassification among visually similar plastics.

2. **Upgrade Mechanical Components**

Implementing a higher-torque stepper motor or a geared motor is recommended to achieve smoother and more reliable bin rotation. This modification can prevent stalling, increase mechanical stability, and minimize rotational delays. As a result, the processing time for each plastic bottle is reduced, thereby improving overall system efficiency.

3. **Integrate Incentive-Driven User Features**

Incorporating user-centered functionalities such as account registration, reward points, or QR/RFID-based identification can increase user engagement. These incentive mechanisms will motivate users to consistently participate in proper plastic bottle segregation.

4. **Enhance Real-Time Monitoring Capabilities**

The monitoring interface can be strengthened by integrating push notifications, SMS alerts, or email updates. These features will improve real-time responsiveness, allow quicker decision-making, and provide more comprehensive oversight of system operations.

5. **Optimize Bin Capacity Based on Waste Volume**

Each bin's size should be evaluated and adjusted according to the anticipated volume for each plastic type. Since materials such as PET bottles tend to accumulate more rapidly, proportional bin sizing can prevent overflow,

decrease the frequency of manual emptying, and enhance the efficiency of the waste-handling process.

6. Include Additional Waste Categories

Expanding the system to classify other types of recyclable plastics will increase its versatility and make it more applicable to broader waste management processes.

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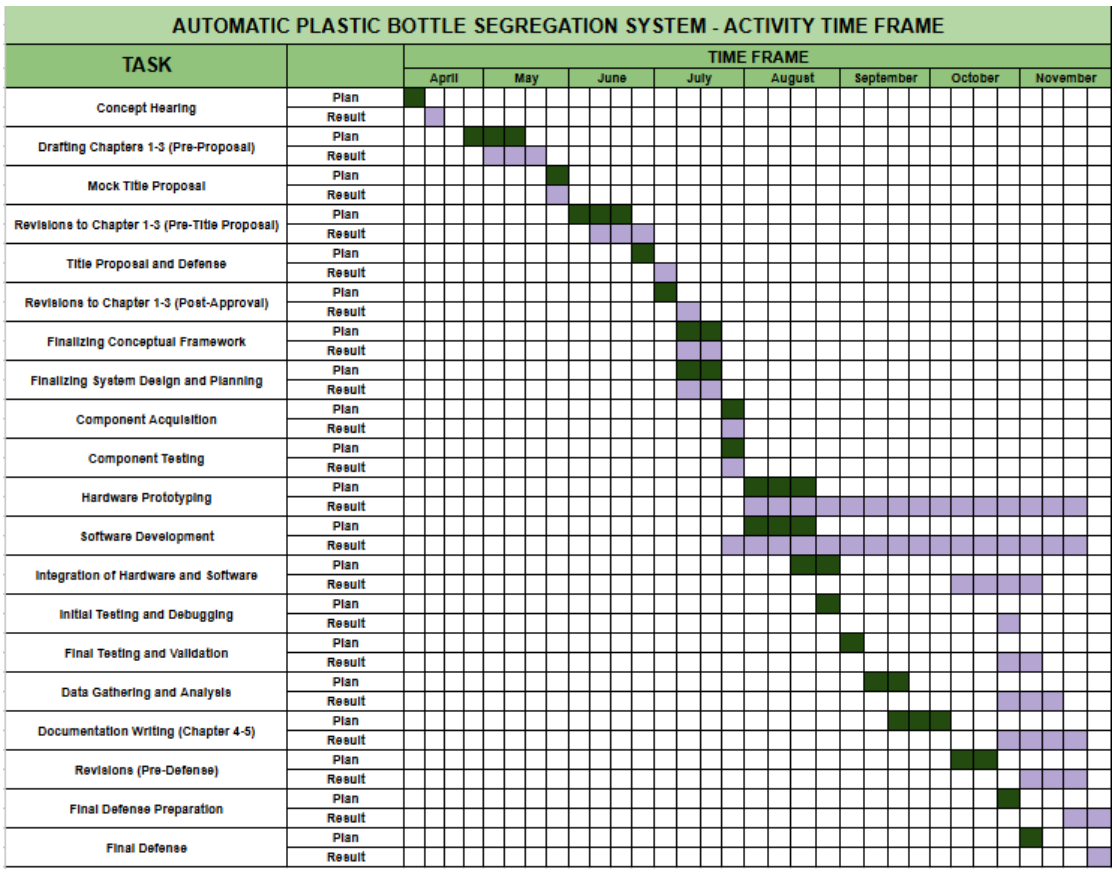
APPENDICES

Appendix A

Project Timeline and Activity Schedule

Table 24

Project Timeline and Activity Schedule



Appendix B

Project Expenses

Table 25

Project Expenses

Component	Quantity	Unit Price	Subtotal
Electronic Components			
Raspberry Pi 4 Model B 4GB RAM with Dual Fan Heat Sink, Power Supply, and 32GB SD Card	1	₱7,099.00	₱7,099.00
Raspberry Pi Camera V2	1	₱1,750.00	₱1,750.00
HC-SR04 Ultrasonic Sensor with Mounting Bracket	5	₱89.00	₱534.00
KY-003 Hall Magnetic Sensor Module	2	₱19.00	₱38.00
16x2 LCD Display I2C White on Blue	1	₱150.00	₱150.00
Tower Pro Digital Robot Servo Motor MG996R (180 Rotation)	1	₱250.00	₱250.00
Wantai Stepper Motor 57BYGH627	1	₱1,349.00	₱1,349.00
HW-688 12V To 5V DC-DC Step-Down buck converter	2	₱95.00	₱190.00
12V 5A Adapter Type Power Supply	1	₱300.00	₱300.00

True Rated			
Broadway 12V 5A Adapter Type Power Supply	1	₱680.00	₱680.00
TB6600 Stepper Motor Driver	1	₱350.00	₱350.00
Customized PCB	5	₱387.00	₱1,935.00
Surface Mount Capacitors	2	₱26.00	₱52.00
Surface Mount Resistors	16	₱14.00	₱224.00
Pinheaders	9	₱19.00	₱171.00
6 Pin Terminal Block	10	₱15.70	₱157.00
4 Pin Terminal Block	10	₱10.60	₱106.00
3 Pin Terminal Block	10	₱8.20	₱82.00
6 Pin Terminal Block	10	₱5.90	₱59.00
Solid Jumper Wires	30	₱5.00/meter	₱150.00
Stranded Jumper Wires	20	₱5.00/meter	₱100.00
Female-to-female Jumper Wires	20	₱5.00	₱100.00
Raspberry Pi Cam Flexible Flat Cable	5	₱47.8	₱239.00
L Standoff Nylon PCB Spacer	4	₱7.00	₱28.00
Stepper Motor Right Angle Mounting	1	₱150.00	₱150.00

Bracket			
Non-Electronic Materials			
Green Spray Paint	1	₱115.00	₱115.00
Orange Spray Paint	1	₱110.00	₱110.00
Yellow Spray Paint	1	₱110.00	₱110.00
Blue Spray Paint	1	₱115.00	₱115.00
Grey Spray Paint	1	₱115.00	₱115.00
Stainless Steel Hollow Square Bar	4	₱200.00	₱800.00
Push Cart Wheel	4	₱84.75	₱339.00
Push Cart Wheel (Small)	4	₱59.75	₱239.00
Zip Tie	1	₱30.00	₱30.00
Heat Shrink Tube	2	₱59.00	₱59.00
Extension Wire	1	₱109.00	₱109.00
Plywood Sheet 63cm x 63cm (15mm)	1	₱250.00	₱250.00
Aluminum Sheet 100x100cm (0.5mm)	2	₱1,682.00	₱3,364.00
Aluminum Sheet 80x80cm (1mm)	2	₱3,125.00	₱6,250.00
Drill Bits (Set)	1	₱219.00	₱219.00
Squarespace Domain (1 year)	1	₱249.00	₱240.00
Labor Cost	7 days	₱428.57/day	₱3,000.00
			TOTAL: **₱31,707.00**

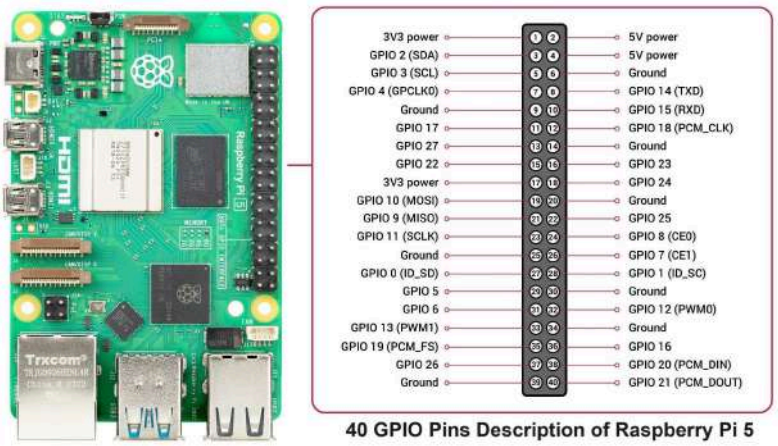
Appendix C

Components' Datasheets

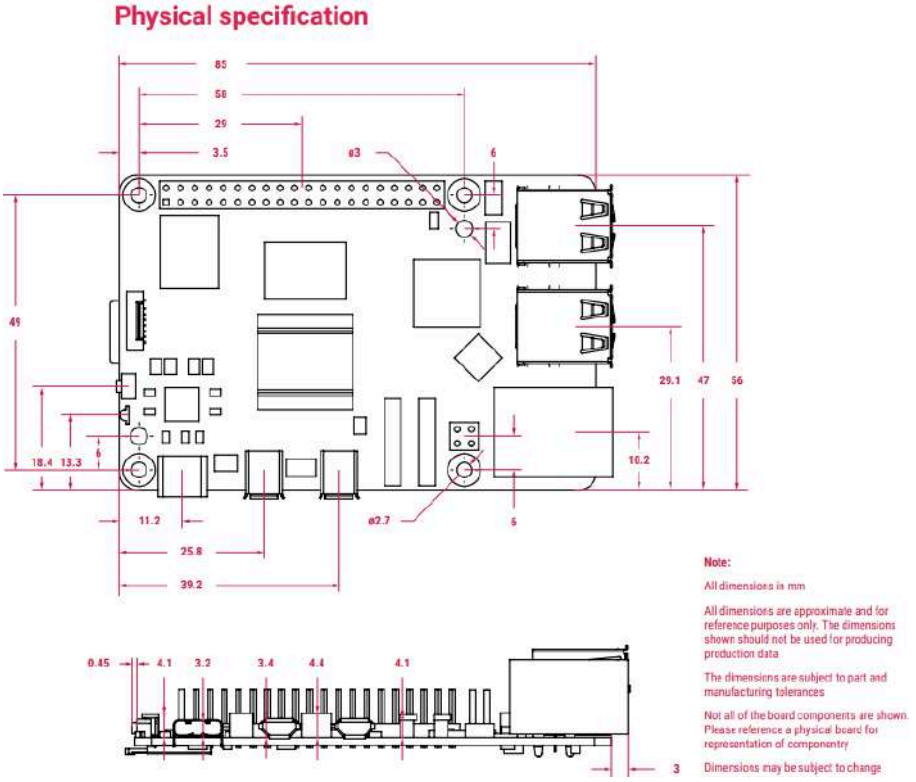
Raspberry Pi 5 Datasheet

Raspberry Pi 5 Module

Raspberry Pi 5 Pinout



Raspberry Pi 5 Physical Dimensions



Raspberry Pi 5 Specification and Features

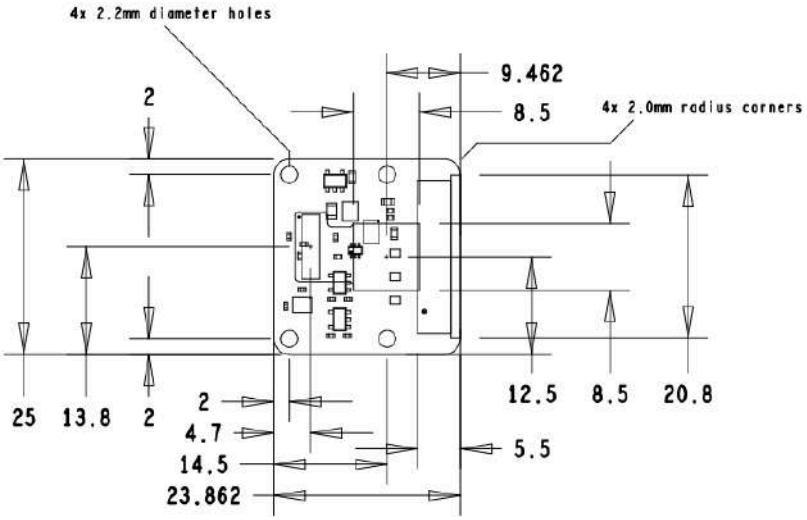
Specification

Processor	Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU, with Cryptographic Extension, 512KB per-core L2 caches, and a 2MB shared L3 cache
Features:	• VideoCore VII GPU, supporting OpenGL ES 3.1, Vulkan 1.2 • Dual 4Kp60 HDMI® display output with HDR support • 4Kp60 HEVC decoder • LPDDR4X-4267 SDRAM (options for 2GB, 4GB, 8GB and 16GB) • Dual-band 802.11ac Wi-Fi® • Bluetooth 5.0 / Bluetooth Low Energy (BLE) • microSD card slot, with support for high-speed SDR104 mode • 2 × USB 3.0 ports, supporting simultaneous 5Gbps operation • 2 × USB 2.0 ports • Gigabit Ethernet, with PoE+ support (requires separate PoE+ HAT) • 2 × 4-lane MIPI camera/display transceivers • PCIe 2.0 x1 interface for fast peripherals (requires separate M.2 HAT or other adapter) • 5V/5A DC power via USB-C, with Power Delivery support • Raspberry Pi standard 40-pin header • Real-time clock (RTC), powered from external battery • Power button

Raspberry Pi Camera Module V2 Datasheet

Raspberry Pi Camera Module V2

Raspberry Pi Camera Module V2 Physical Dimensions



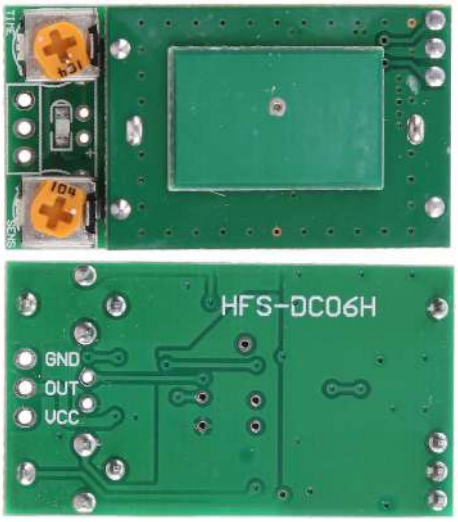
Raspberry Pi Camera Module V2 Specification and Features

Specifications

Image Sensor	Sony IMX 219 PQ CMOS image sensor in a fixed-focus module.
Resolution	8-megapixel
Still picture resolution	3280 x 2464
Max image transfer rate	1080p: 30fps (encode and decode) 720p: 60fps
Connection to Raspberry Pi	15-pin ribbon cable, to the dedicated 15-pin MIPI Camera Serial Interface (CSI-2).
Image control functions	Automatic exposure control Automatic white balance Automatic band filter Automatic 50/60 Hz luminance detection Automatic black level calibration
Temp range	Operating: -20° to 60° Stable image: -20° to 60°
Lens size	1/4"
Dimensions	23.86 x 25 x 9mm
Weight	3g

HFS-DC06 Microwave Radar Sensor Datasheet

HFS-DC06 Microwave Radar Sensor Module



HFS-DC06 Microwave Radar Sensor Module General Specifications

General specifications

- **Model:** HFS-DC06
- **Frequency:** 5.8GHz ISM band
- **Input Voltage:** DC5V
- **Minimum Input Current:** 30-40mA
- **Output Mode:** 5V level signal (high output when triggered)
- **Size:** 39 x 22 x 11mm
- **Working Principle:** Continuous Wave (CW) Doppler radar

HFS-DC06 Microwave Radar Sensor Module Installation and Wiring

Installation and wiring

- **Wiring:**
  - VCC connects to the power supply's positive pole.
  - GND connects to the power supply's negative pole.
  - The level signal output is connected to GND and OUT.
- **Installation Height:** 2-6 meters
- **Installation Precautions:** Do not install next to large metal bodies or within a fully enclosed metal shell to avoid performance issues.

HFS-DC06 Microwave Radar Sensor Module Performance and Adjustments

Performance and adjustments

- **Sensing Distance:** Adjustable from 0.5 to 12 meters.
- **Detection Angle:** 360 degrees from the front.
- **Adjustable Pulse (Time):** Adjustable from 2 to 120 seconds.
- **Adjustments:**
  - There are two onboard trim pots: one for adjusting the sensing range (SENS) and one for adjusting the pulse duration (TIME).
  - To increase the range/time, turn the preset clockwise; to decrease, turn it counterclockwise.

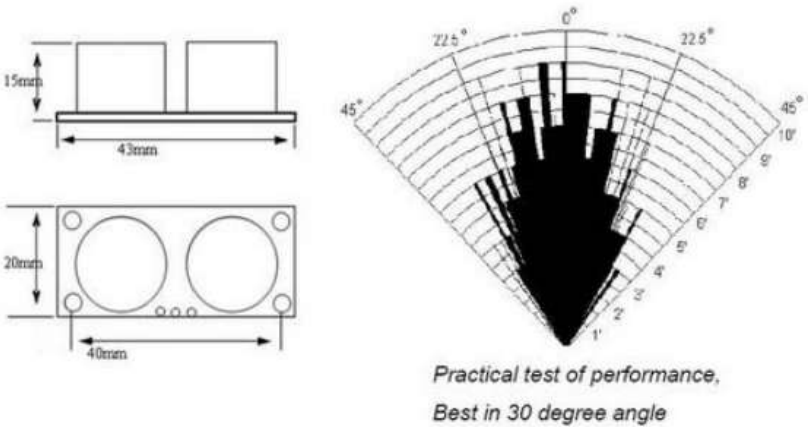
HC-SR04 Ultrasonic Sensor Datasheet

HC-SR04 Ultrasonic Sensor

HC-SR04 Ultrasonic Sensor Pinout

VCC	3.3v ~ 5V
TRIG	Triggering Input Pin. 10uS TTL Pulses
ECHO	TTL Logic Output Pin. Proportional to distance
GND	Ground Pin

HC-SR04 Ultrasonic Sensor Physical Dimensions



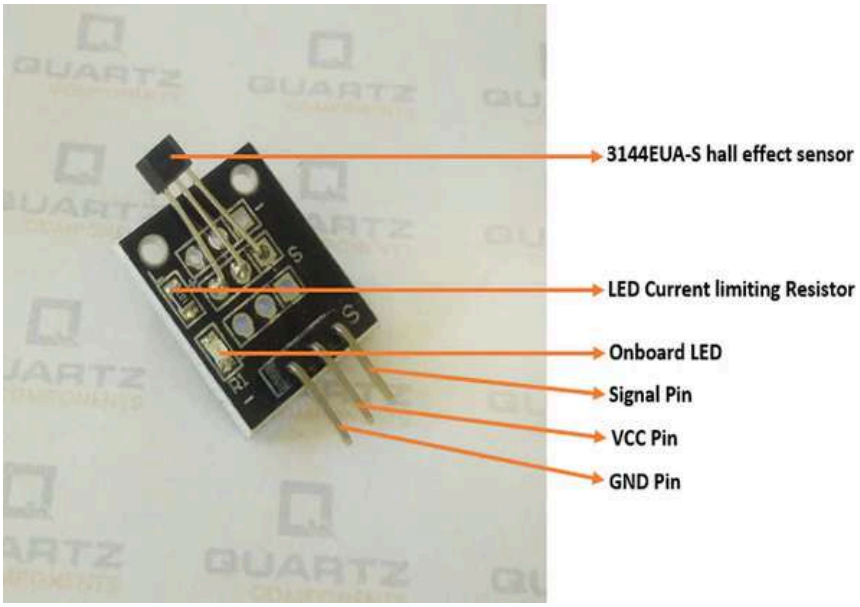
HC-SR04 Ultrasonic Sensor Specifications

Electrical Parameters	Value
Operating Voltage	3.3Vdc ~ 5Vdc
Quiescent Current	<2mA
Operating Current	15mA
Operating Frequency	40KHz
Operating Range & Accuracy	2cm ~ 400cm (1in ~ 13ft) ± 3mm
Sensitivity	-65dB min
Sound Pressure	112dB
Effective Angle	15°
Connector	4-pins header with 2.54mm pitch
Dimension	45mm x 20mm x 15mm
Weight	9g

KY-003 Hall Magnetic Sensor Datasheet

KY-003 Hall Magnetic Sensor

KY-003 Hall Magnetic Sensor Pinout



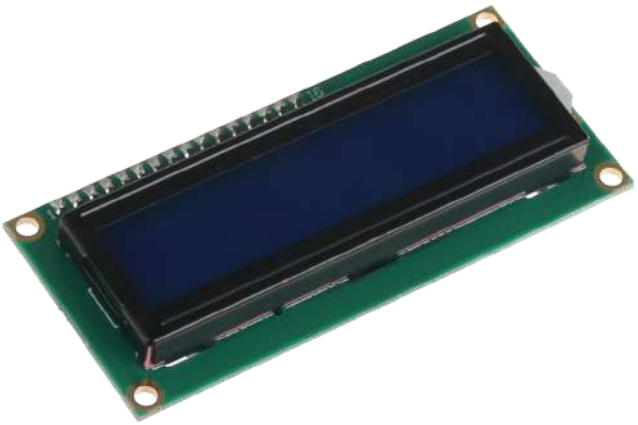
KY-003 Hall Magnetic Sensor Specifications

Specifications

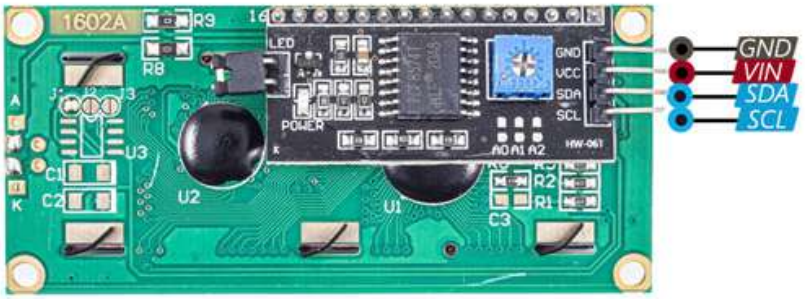
- Linear Output for Circuit Design Flexibility
- Low Noise Output Virtually Eliminates the Need for Filtering
- A Stable and Accurate Output
- Temperature Range of -40°C to +85°C
- Responds to Either Positive or Negative Gauss
- Onboard LED for Output detection

Liquid Crystal Display I2C Datasheet

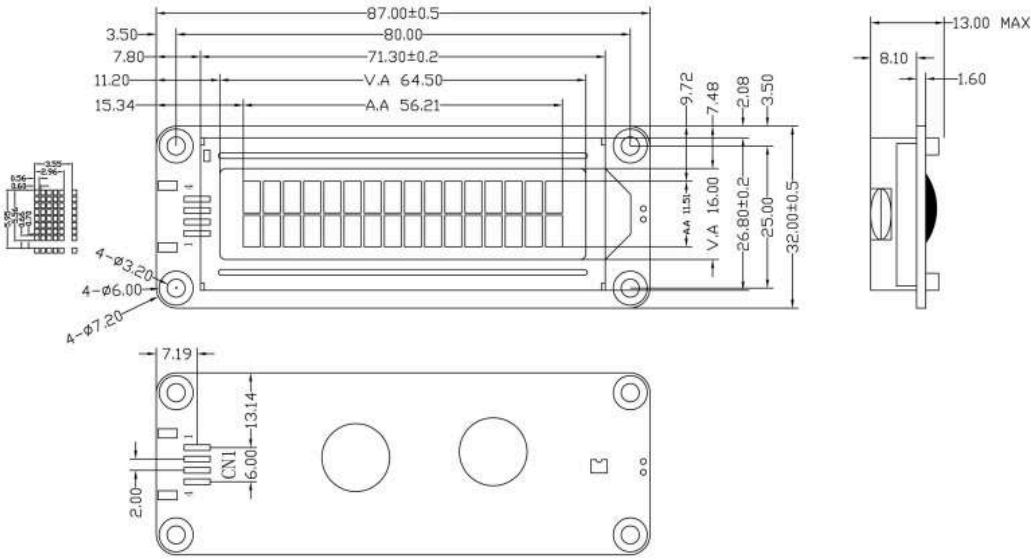
Liquid Crystal Display I2C Module



Liquid Crystal Display I2C Module Pinout



Liquid Crystal Display I2C Module Physical Dimensions



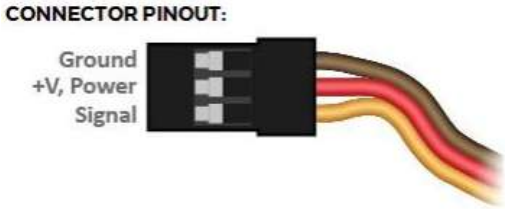
Liquid Crystal Display I2C Module Specifications

- Characters: 16×2 Lines
- LCD Mode: STN, Yellow –Green/Blue/Grayish;
- LCD Controller IC: AiP31068L or Equivalent
- Driving Method: 1/16 Duty; 1/5Bias
- Viewing Angie: 6 O'clock direction
- I^2C MPU Interface
- Backlight: White LED
- Operating Temperature Range: -20 to +70℃;
- Storage Temperature Range : -30 to +80℃;

MG996R Servomotor Datasheet

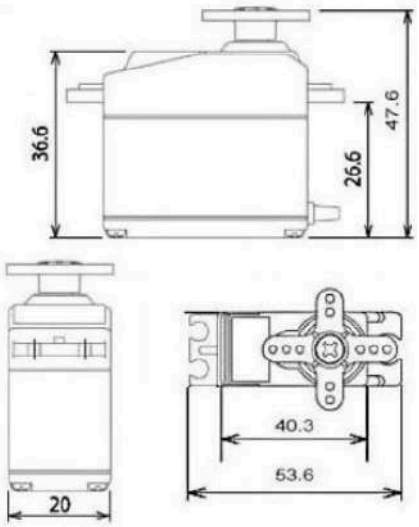
MG996R Servomotor

MG996R Servomotor Pinout



MG996R Servomotor Physical Dimensions

Unit: mm

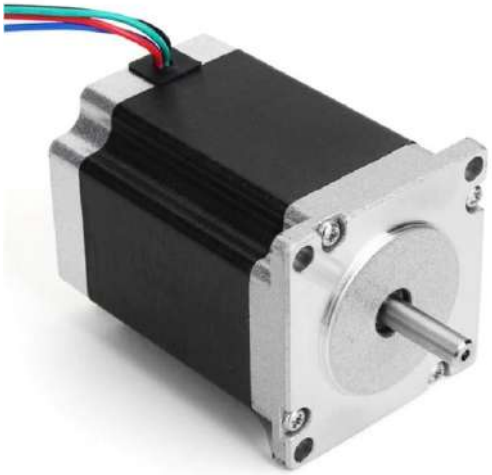


MG996R Servomotor Specifications

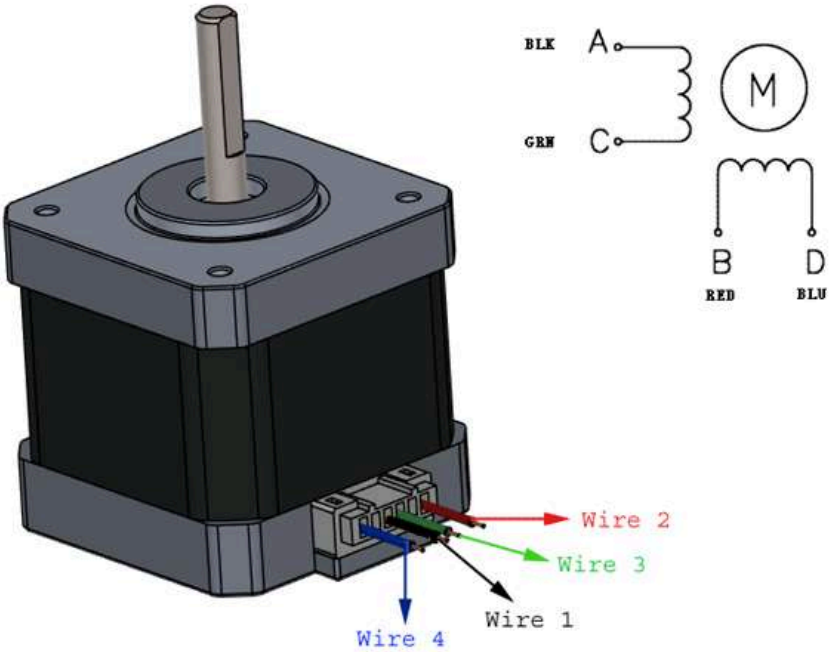
- Operating Voltage is +5V typically
- Current: 2.5A (6V)
- Stall Torque: 9.4 kg/cm (at 4.8V)
- Maximum Stall Torque: 11 kg/cm (6V)
- Operating speed is 0.17 s/60°
- Gear Type: Metal
- Rotation : 0°-180°
- Weight of motor : 55gm
- Package includes gear horns and screws

Wantai Stepper Motor Datasheet

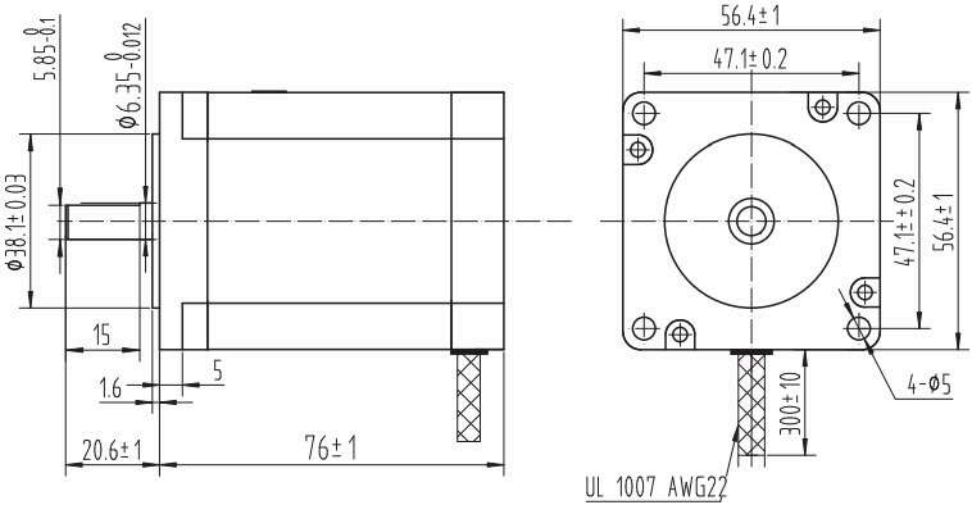
Wantai Stepper Motor



Wantai Stepper Motor Pinout



Wantai Stepper Motor Physical Dimensions



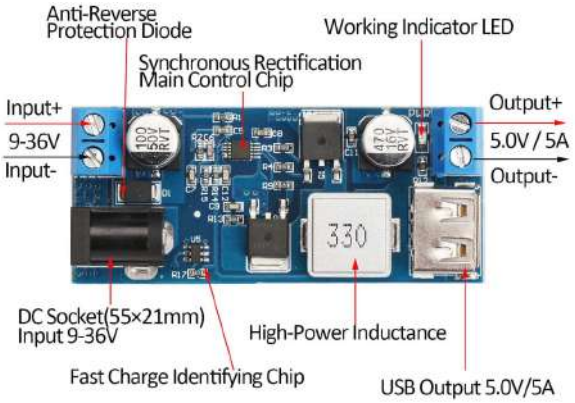
Wantai Stepper Motor Specifications

- Voltage Rating: 3.2V
- Current Rating: 2.8A
- Holding Torque: 270 oz. in
- Step Angle: 1.8 deg.
- Steps Per Revolution: 200
- No. of Phases: 4
- Motor Length: 3.1 inches
- No. of Leads: 4
- Inductance Per Phase: 3.6mH

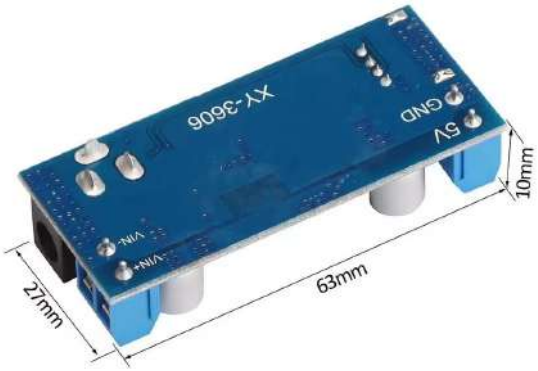
12V-5V Buck Converter Datasheet

12V-5V Buck Converter

12V-5V Buck Converter Pinout



12V-5V Buck Converter Physical Dimensions



12V-5V Buck Converter Specifications

Product Name	12V/24V to 5V 5A Power Module DC-DC XY-3606
Input Voltage Range	8V to 28V DC
Output Voltage	Fixed 5V DC
Output Current	Up to 5A
Output Power	Max 25W
Efficiency	Up to 95%
Regulation Method	Synchronous Rectification
Protection Features	Short-circuit, thermal shutdown
Operating Temperature	-20°C to +85°C
Board Dimensions	~45mm × 20mm × 15mm
Use Case	Automotive chargers, router power, Arduino

Appendix D

PET Bottles Dataset

With cap and label

Coca Cola (10 images)

Royal (160 images)

Mountain Dew (167 images)

Nature Spring (171 images)

Wilkins (320 images)

Sprite (16 images)

Cobra Energy Drink (12 images)

Nature Spring pH9 (18 images)

Mogu Mogu (9 images)

Kopiko Lucky Day (15 images)

C2 (12 images)

Kalamansi (10 images)

Sting Energy Drink (6 images)

Summit (15 images)

Sip (19 images)

Without cap and label

Coca Cola (29 images)

Royal (160 images)

Mountain Dew (160 images)

Nature Spring (160 images)

Wilkins (313 images)

Sprite (2 images)

With cap but without label

Coca Cola (7 images)

Royal (159 images)

Mountain Dew (166 images)

Nature Spring (166 images)

Wilkins (313 images)

Mogu mogu (3 images)

Kopiko Lucky Day (5 images)

C2 (3 images)

Kalamansi (2 images)

Sting Energy Drink (2 images)

Without cap but with label

Coca Cola (8 images)

Royal (160 images)

Mountain Dew (169 images)

Nature Spring (170 images)

Wilkins (316 images)

Sprite (7 images)

Mogu mogu (6 images)

Kopiko Lucky Day (7 images)

C2 (6 images)

Kalamansi (8 images)

Sting Energy Drink (4 images)

SM Bonus (8 Images

Amlan (6 images)

Appendix E

HDPE Bottles Dataset

With cap and label

Cleaning Products (92 images)

Food Products (49 images)

Hair Care Product (1 image)

Without cap and label

Skincare Products (225 images)

Haircare Products (401 images)

Cosmetic Products (500 images)

Food Products (235 images)

Cleaning Products (202 images)

Personal Hygiene Products (186 images)

With cap but without label

Skincare Products (9 images)

Haircare Products (262 images)

Food Products (199 images)

Without cap but with label

Cleaning Products (49 images)

Appendix F

LDPE Bottles Dataset

Without cap and label

Red Squeeze Bottle (19 images)

Yellow Squeeze Bottle (51 images)

Transparent Squeeze Bottle with red cap (34 images)

Transparent Squeeze Bottle with orange cap (40 images)

Transparent Squeeze Bottle with white cap (34 images)

With cap but without label

Red Squeeze Bottle (220 images)

Yellow Squeeze Bottle (240 images)

Transparent Squeeze Bottle with red cap (624 images)

Transparent Squeeze Bottle with orange cap (87 images)

Transparent Squeeze Bottle with white cap (232 images)

Appendix G

PP Bottles Dataset

With cap and label

Transparent Plastic Container (120 images)

Colored Plastic Bottle (324 images)

Without cap and label

Transparent Plastic Container (168 images)

Colored Plastic Bottle (86 images)

With cap but without label

Transparent Plastic Container (216 images)

Colored Plastic Bottle (215 images)

Without cap but with label

Transparent Plastic Container (86 images)

Colored Plastic Bottle (189 images)

Appendix H

Unclassified Waste Dataset

Crumpled Paper (800 images)

Nova (161 images)

Piattos (670 images)

Aluminum Can (640 images)

Oreo (149 images)

Appendix I

Source Code

Hardware Code

bottle_classifier.py

```
#!/usr/bin/env python3
"""
BottleSort Classification System
La Salle University-Ozamiz
Raspberry Pi 5 Compatible
"""

import os
import sys
import time
from datetime import datetime

import numpy as np
from PIL import Image
from picamera2 import Picamera2

# GPIO for sensors
try:
    import RPi.GPIO as GPIO
    GPIO_AVAILABLE = True
    print("Using RPi.GPIO for hardware control")
except ImportError:
    print("Warning: RPi.GPIO not available. Running in test mode.")
    GPIO_AVAILABLE = False

# LCD Display (I2C)
try:
```

```
    from RPLCD.i2c import CharLCD
    LCD_AVAILABLE = True
except ImportError:
    print("Warning: RPLCD not available. Install with: pip3 install RPLCD")
    LCD_AVAILABLE = False

# MySQL connector
try:
    import pymysql
except ImportError:
    print("Error: PyMySQL not found! Install with: sudo apt install python3-pymysql")
    sys.exit(1)

# Edge Impulse
try:
    from edge_impulse_linux.image import ImageImpulseRunner
except ImportError:
    print("Error: Edge Impulse Linux SDK not found! Install via: pip3 install
edge_impulse_linux")
    sys.exit(1)

# ==================== CONFIGURATION ====================

# Database configuration
DB_CONFIG = {
    'host': '127.0.0.1',
    'port': 3307,
    'user': 'root',
    'password': 'root',
    'database': 'bottlesort',
    'charset': 'utf8mb4',
    'connect_timeout': 10
```

```
}

# Edge Impulse model
MODEL_PATH = '/home/admin/bottle_classifier/modelfile.eim'

# Image capture directory
CAPTURE_DIR = '/home/admin/bottle_classifier/captures'

# GPIO Pin Configuration
MICROWAVE_SENSOR_PIN = 16  # U9 - Bottle detection
HALL_SENSOR_PIN = 12       # U10 - Home position
SERVO_PIN = 25             # U11 - Flap servo
STEPPER_DIR_PIN = 23       # Stepper direction
STEPPER_PUL_PIN = 18       # Stepper pulse

# Stepper motor config
STEPS_PER_REVOLUTION = 200
MICROSTEPS = 32
TOTAL_STEPS = STEPS_PER_REVOLUTION * MICROSTEPS
STEPS_PER_BIN = TOTAL_STEPS // 5  # 5 bins

# Servo motor config
SERVO_FREQUENCY = 50
SERVO_CLOSED_ANGLE = 0      # Flap closed
SERVO_OPEN_ANGLE = 45       # Flap open
SERVO_DROP_DELAY = 2.0      # Time for bottle to drop

# Ultrasonic Sensors (Trig, Echo)
ULTRASONIC_SENSORS = {
    'PET': {'trig': 17, 'echo': 27},
    'HDPE': {'trig': 22, 'echo': 10},
    'LDPE': {'trig': 9, 'echo': 11},
```

```
    'PP': {'trig': 5, 'echo': 6},
    'UNCLASSIFIED': {'trig': 13, 'echo': 19}
}

# Bin dimensions
BIN_HEIGHT = 50
BIN_FULL_THRESHOLD = 10  # cm from top

# Bin positions for stepper motor
BIN_POSITIONS = {
    'PET': 0,           # Home position
    'HDPE': 1,
    'LDPE': 2,
    'PP': 3,
    'UNCLASSIFIED': 4
}

# LCD Configuration
LCD_I2C_ADDRESS = 0x27
LCD_COLS = 16
LCD_ROWS = 2

# Valid bottle types
VALID_BOTTLE_TYPES = ['PET', 'HDPE', 'LDPE', 'PP']

# ==================== MAIN CLASS ====================

class BottleClassifierSystem:
    def __init__(self):
        self.camera = None
        self.db_connection = None
        self.lcd = None
```

```
        self.servo_pwm = None
        self.system_active = True
        self.total_bottles_processed = 0
        self.current_bin_position = 0  # 0=PET/home
        self.current_servo_angle = SERVO_CLOSED_ANGLE
        self.bottles_by_type = {b: 0 for b in VALID_BOTTLE_TYPES +
['UNCLASSIFIED']}
        self.bin_levels = {b: 0 for b in VALID_BOTTLE_TYPES +
['UNCLASSIFIED']}

        if not os.path.isfile(MODEL_PATH):
            print(f"Model file not found at path: {MODEL_PATH}")
            sys.exit(1)

    # ==================== HARDWARE SETUP ====================

    def setup_gpio(self):
        """Initialize GPIO for all sensors and actuators"""
        if not GPIO_AVAILABLE:
            print("Warning: Running without GPIO (test mode)")
            return False

        try:
            GPIO.cleanup()
        except:
            pass

        GPIO.setmode(GPIO.BCM)
        GPIO.setwarnings(False)

        # Sensors
        GPIO.setup(MICROWAVE_SENSOR_PIN, GPIO.IN)
```

```
        GPIO.setup(HALL_SENSOR_PIN, GPIO.IN, pull_up_down=GPIO.PUD_UP)

        # Stepper motor
        GPIO.setup(STEPPER_DIR_PIN, GPIO.OUT)
        GPIO.setup(STEPPER_PUL_PIN, GPIO.OUT)
        GPIO.output(STEPPER_DIR_PIN, GPIO.LOW)
        GPIO.output(STEPPER_PUL_PIN, GPIO.LOW)

        # Servo motor
        try:
            GPIO.setup(SERVO_PIN, GPIO.OUT)
            self.servo_pwm = GPIO.PWM(SERVO_PIN, SERVO_FREQUENCY)
            self.servo_pwm.start(0)
            time.sleep(0.1)
            self.set_servo_angle(SERVO_CLOSED_ANGLE)
            time.sleep(0.5)
        except Exception as e:
            print(f"Warning: Servo initialization warning: {e}")
            self.servo_pwm = None

        # Ultrasonic sensors
        for pins in ULTRASONIC_SENSORS.values():
            GPIO.setup(pins['trig'], GPIO.OUT)
            GPIO.setup(pins['echo'], GPIO.IN)
            GPIO.output(pins['trig'], GPIO.LOW)

        print("All GPIO initialized")
        time.sleep(0.5)
        return True

    def setup_lcd(self):
        """Initialize I2C LCD display"""
```

```
    if not LCD_AVAILABLE:
        print("Warning: LCD not available (test mode)")
        return False

    try:
        self.lcd = CharLCD(
            i2c_expander='PCF8574',
            address=LCD_I2C_ADDRESS,
            port=1,
            cols=LCD_COLS,
            rows=LCD_ROWS
        )
        self.lcd.clear()
        self.lcd.write_string("BottleSort v2\nStarting...")
        print("LCD display initialized")
        time.sleep(2)
        return True
    except Exception as e:
        print(f"Warning: LCD initialization error: {e}")
        return False

def setup_camera(self):
    """Initialize Raspberry Pi Camera"""
    print("Initializing camera...")
    try:
        self.camera = Picamera2()
        config = self.camera.create_still_configuration(
            main={"size": (1920, 1080), "format": "RGB888"},
            buffer_count=2
        )
        self.camera.configure(config)
        self.camera.start()
```

```
        time.sleep(2)
        print("Camera ready!")
    except Exception as e:
        print(f"Camera initialization error: {e}")
        sys.exit(1)

def setup_database(self):
    """Connect to MySQL database"""
    try:
        print(f"Connecting to {DB_CONFIG['host']}:{DB_CONFIG['port']}...")
        self.db_connection = pymysql.connect(**DB_CONFIG)
        print(f"Database connected!")
        self.create_tables()
    except Exception as err:
        print(f"\nDatabase connection failed: {err}")
        sys.exit(1)

def create_tables(self):
    """Verify database tables exist"""
    try:
        cursor = self.db_connection.cursor()
        cursor.execute("SHOW TABLES LIKE 'plastic_bottle_logs'")
        if not cursor.fetchone():
            print("Warning: 'plastic_bottle_logs' table not found!")
        cursor.execute("SHOW TABLES LIKE 'plastic_bottle_bin_level'")
        if not cursor.fetchone():
            print("Warning: 'plastic_bottle_bin_level' table not found!")
        cursor.close()
        print("Database tables verified")
    except Exception as e:
        print(f"Warning: Table verification warning: {e}")
```

```
    # ==================== SERVO MOTOR CONTROL
====================

    def angle_to_duty_cycle(self, angle):
        """Convert angle (0-180°) to PWM duty cycle (2.5-12.5%)"""
        return 2.5 + (angle / 180.0) * 10.0

    def set_servo_angle(self, angle):
        """Set servo to specific angle"""
        if not self.servo_pwm:
            print(f"  TEST MODE: Would set servo to {angle}°")
            self.current_servo_angle = angle
            return

        try:
            angle = max(0, min(180, angle))
            duty_cycle = self.angle_to_duty_cycle(angle)
            self.servo_pwm.ChangeDutyCycle(duty_cycle)
            time.sleep(0.5)
            self.servo_pwm.ChangeDutyCycle(0) # Stop signal to prevent jitter
            self.current_servo_angle = angle
            print(f"Servo set to {angle}°")
        except Exception as e:
            print(f"Servo error: {e}")

    def open_flap(self):
        """Open servo flap to drop bottle"""
        print(f"Opening flap ({SERVO_OPEN_ANGLE}°)...")
        self.display_lcd("Opening flap", "Dropping...")
        self.set_servo_angle(SERVO_OPEN_ANGLE)

    def close_flap(self):
```

```
    """Close servo flap"""
    print(f"Closing flap ({SERVO_CLOSED_ANGLE}°)...")
    self.display_lcd("Closing flap", "")
    self.set_servo_angle(SERVO_CLOSED_ANGLE)

# ==================== SENSOR FUNCTIONS ====================

def read_ultrasonic(self, trig_pin, echo_pin, timeout=0.5):
    """Read distance from ultrasonic sensor in cm"""
    if not GPIO_AVAILABLE:
        import random
        return random.uniform(15, 45)

    try:
        GPIO.output(trig_pin, GPIO.LOW)
        time.sleep(0.00001)

        GPIO.output(trig_pin, GPIO.HIGH)
        time.sleep(0.00001)
        GPIO.output(trig_pin, GPIO.LOW)

        pulse_start = time.time()
        timeout_start = time.time()
        while GPIO.input(echo_pin) == GPIO.LOW:
            pulse_start = time.time()
            if pulse_start - timeout_start > timeout:
                return -1

        pulse_end = time.time()
        timeout_start = time.time()
        while GPIO.input(echo_pin) == GPIO.HIGH:
            pulse_end = time.time()
```

```
            if pulse_end - timeout_start > timeout:
                return -1

        distance = (pulse_end - pulse_start) * 17150

        return distance if 2 <= distance <= 400 else -1
    except Exception:
        return -1

def check_bin_levels(self):
    """Check all bin levels and save to database"""
    print("\nChecking bin levels...")

    for bin_name, pins in ULTRASONIC_SENSORS.items():
        distance = self.read_ultrasonic(pins['trig'], pins['echo'])

        if distance > 0:
            fill_percentage = ((BIN_HEIGHT - distance) / BIN_HEIGHT) * 100
            fill_percentage = max(0, min(100, fill_percentage))
            self.bin_levels[bin_name] = fill_percentage
        else:
            print(f"  Warning: {bin_name} sensor error")

    self.save_bin_levels()
    print("Bin levels updated.")

def is_bin_full(self, bin_name):
    """Check if a specific bin is full"""
    return self.bin_levels.get(bin_name, 0) >= 95

def wait_for_bottle(self):
    """Wait for microwave sensor to detect a bottle"""
```

```
        if not GPIO_AVAILABLE:
            input("\nTEST MODE: Press Enter to simulate bottle detection...")
            print("Bottle detected!")
            return True

        print("\nWaiting for bottle...")
        self.display_lcd("Ready", "Insert bottle")

        try:
            while self.system_active:
                if GPIO.input(MICROWAVE_SENSOR_PIN) == GPIO.HIGH:
                    print("Bottle detected!")
                    time.sleep(0.5)
                    return True
                time.sleep(0.1)
        except KeyboardInterrupt:
            return False

    # ==================== STEPPER MOTOR CONTROL
====================

    def step_motor(self, steps, delay=0.001):
        """Step the motor a specific number of steps"""
        if not GPIO_AVAILABLE:
            print(f"  TEST MODE: Would step {steps} steps")
            time.sleep(0.5)
            return

        for _ in range(abs(steps)):
            GPIO.output(STEPPER_PUL_PIN, GPIO.HIGH)
            time.sleep(delay)
            GPIO.output(STEPPER_PUL_PIN, GPIO.LOW)
```

```
        time.sleep(delay)
    def rotate_to_bin(self, target_bin_name):
        """Rotate stepper motor to the correct bin"""
        target_position = BIN_POSITIONS.get(target_bin_name)
        if target_position is None:
            print(f"Error: Unknown bin: {target_bin_name}")
            return False

        if not GPIO_AVAILABLE:
            print(f"TEST MODE: Would rotate to {target_bin_name}")
            self.current_bin_position = target_position
            return True

        position_diff = target_position - self.current_bin_position

        # Calculate shortest path (5 bins total)
        if position_diff > 2:
            position_diff -= 5  # Go backwards
        elif position_diff < -2:
            position_diff += 5  # Go forwards

        steps_to_move = position_diff * STEPS_PER_BIN

        if steps_to_move == 0:
            print(f"Already at {target_bin_name}")
            return True

        print(f"Rotating to {target_bin_name}...")
        self.display_lcd(f"Rotating to:", target_bin_name)

        # Set direction
```

```
        GPIO.output(STEPPER_DIR_PIN, GPIO.HIGH if steps_to_move > 0 else
GPIO.LOW)
        time.sleep(0.001)

        self.step_motor(abs(steps_to_move), delay=0.001)
        self.current_bin_position = target_position
        print(f"Positioned at {target_bin_name}")
        return True

    def home_stepper(self):
        """Return stepper to home position using hall sensor"""
        print("Homing stepper motor...")
        self.display_lcd("Homing", "Please wait...")

        if not GPIO_AVAILABLE:
            print("  TEST MODE: Simulating home")
            self.current_bin_position = 0
            return True

        # Check if already at home
        if GPIO.input(HALL_SENSOR_PIN) == GPIO.LOW:
            print("  Already at home position")
            self.current_bin_position = 0
            return True

        # Rotate slowly (CCW) to find home
        GPIO.output(STEPPER_DIR_PIN, GPIO.LOW)
        time.sleep(0.001)

        max_steps = TOTAL_STEPS + STEPS_PER_BIN # Max 1.2 rotations
        for _ in range(max_steps):
            if GPIO.input(HALL_SENSOR_PIN) == GPIO.LOW:
```

```
            print("  Home position found!")
            self.current_bin_position = 0
            return True

        # Take one step
        GPIO.output(STEPPER_PUL_PIN, GPIO.HIGH)
        time.sleep(0.002) # Slower for homing
        GPIO.output(STEPPER_PUL_PIN, GPIO.LOW)
        time.sleep(0.002)

    print("  Warning: Home position not found after full rotation!")
    self.current_bin_position = 0
    return False

# ==================== LCD DISPLAY ====================

def display_lcd(self, line1, line2=""):
    """Display message on LCD"""
    if self.lcd:
        try:
            self.lcd.clear()
            self.lcd.write_string(line1[:16] + "\n" + line2[:16])
        except Exception as e:
            print(f"Warning: LCD error: {e}")
    else:
        print(f"LCD: {line1[:16]} | {line2[:16]}")

# ==================== CLASSIFICATION ====================

def normalize_bottle_type(self, bottle_type):
    """Normalize bottle type to standard format"""
    if not bottle_type:
```

```
        return None
    normalized = bottle_type.upper().strip()
    return normalized if normalized in VALID_BOTTLE_TYPES else None

def capture_and_classify(self):
    """Capture image and classify bottle type"""
    os.makedirs(CAPTURE_DIR, exist_ok=True)
    timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
    filename = f"bottle_{timestamp}.jpg"
    full_path = os.path.join(CAPTURE_DIR, filename)
    db_path = f"captures/{filename}"

    print(f"\nCapturing image...")
    self.display_lcd("Capturing...", "")

    self.camera.capture_file(full_path)
    print(f"Image saved: {filename}")

    bottle_type, confidence = self.classify_image(full_path)
    return bottle_type, confidence, db_path

def classify_image(self, image_path):
    """Classify bottle using Edge Impulse model"""
    self.display_lcd("Analyzing", "Please wait...")

    img = Image.open(image_path).convert('RGB')
    img_arr = np.array(img)

    with ImageImpulseRunner(MODEL_PATH) as runner:
        runner.init()
        features, _ = runner.get_features_from_image(img_arr)
        res = runner.classify(features)
```

```
        predictions = res['result']['classification']
        filtered = {k: v for k, v in predictions.items() if k.lower() != 'background'}

        if not filtered:
            print("Warning: No valid classification (only background detected)")
            return None, 0

        # Get best prediction
        best_label = max(filtered, key=filtered.get)
        best_confidence = min(filtered[best_label], 0.98) # Cap at 0.98

        print(f"Result: {best_label} (Confidence: {best_confidence:.4f})")
        return best_label, best_confidence

    # ==================== DATABASE OPERATIONS
====================

    def save_classification(self, bottle_type, confidence, db_path):
        """Save classification result to database"""
        try:
            cursor = self.db_connection.cursor()
            confidence_level = int(confidence * 100) # Convert 0.98 to 98

            query = """
                INSERT INTO plastic_bottle_logs (record_time, bottle_type,
confidence_level)
                VALUES (%s, %s, %s)
            """
            values = (datetime.now(), bottle_type, confidence_level)
            cursor.execute(query, values)
            self.db_connection.commit()
            cursor.close()
```

```
            print(f"Database saved: {bottle_type} at {confidence_level}%")
        except Exception as err:
            print(f"Error: Database save error: {err}")

    def save_bin_levels(self):
        """Save all bin levels to database"""
        try:
            cursor = self.db_connection.cursor()

            query = """
                INSERT INTO plastic_bottle_bin_level
                (record_time, BinLevel_PET, BinLevel_HDPE, BinLevel_LDPE,
BinLevel_PP, BinLevel_Unclassified)
                VALUES (%s, %s, %s, %s, %s, %s)
            """
            values = (
                datetime.now(),
                int(self.bin_levels.get('PET', 0)),
                int(self.bin_levels.get('HDPE', 0)),
                int(self.bin_levels.get('LDPE', 0)),
                int(self.bin_levels.get('PP', 0)),
                int(self.bin_levels.get('UNCLASSIFIED', 0))
            )
            cursor.execute(query, values)
            self.db_connection.commit()
            cursor.close()
        except Exception as err:
            print(f"Warning: Bin level save error: {err}")

    # ==================== MAIN WORKFLOW ====================

    def process_bottle(self):
```

```
"""Main workflow for processing a single bottle"""
try:
    self.close_flap()
    self.check_bin_levels()

    self.display_lcd("Processing", "Bottle...")
    raw_type, confidence, db_path = self.capture_and_classify()

    if not raw_type:
        print("Error: Classification failed")
        self.display_lcd("Error:", "No bottle found")
        time.sleep(3)
        return False

    normalized_type = self.normalize_bottle_type(raw_type)

    # Default to UNCLASSIFIED
    final_type = 'UNCLASSIFIED'

    if normalized_type:
        if self.is_bin_full(normalized_type):
            print(f"Warning: {normalized_type} bin is FULL!")
            self.display_lcd(f"{normalized_type} Bin", "FULL!")
            time.sleep(2)
        else:
            final_type = normalized_type # Use the correct bin

    # Check if UNCLASSIFIED bin is also full
    if final_type == 'UNCLASSIFIED' and self.is_bin_full('UNCLASSIFIED'):
        print(f"Error: All relevant bins are FULL!")
        self.display_lcd("All Bins", "FULL!")
        time.sleep(3)
```

```
            return False

        self.display_lcd(final_type, f"Conf: {confidence:.4f}")
        print(f"Final result: {final_type}")
        time.sleep(3)

        if not self.rotate_to_bin(final_type):
            print("Error: Failed to rotate to bin")
            return False

        time.sleep(1)

        print(f"\nDropping bottle into {final_type} bin...")
        self.open_flap()
        time.sleep(SERVO_DROP_DELAY)
        self.close_flap()
        time.sleep(0.5)
        self.save_classification(final_type, confidence, db_path)

        # Update local statistics
        self.total_bottles_processed += 1
        self.bottles_by_type[final_type] += 1

        self.display_lcd("Success!", f"Total: {self.total_bottles_processed}")
        print(f"Total bottles: {self.total_bottles_processed}")
        time.sleep(2)

        self.home_stepper()
        return True

    except Exception as e:
        print(f"Error: Processing error: {e}")
```

```
        self.display_lcd("System Error", "Check logs")
        return False

    def continuous_monitoring(self):
        """Main production loop"""
        print("\n" + "="*50)
        print("BottleSort System - ACTIVE")
        print("Press Ctrl+C to stop")
        print("="*50 + "\n")
        self.home_stepper()
        self.close_flap()
        self.check_bin_levels()
        self.display_lcd("System Ready", "Waiting...")

        try:
            while self.system_active:
                if self.wait_for_bottle():
                    print(f"\n--- Processing bottle #{self.total_bottles_processed + 1} ---")

                    if not self.process_bottle():
                        print(f"Warning: Bottle processing failed")
                        # Return home and close flap even on failure
                        self.home_stepper()
                        self.close_flap()

                    print("\nReady for next bottle...")
                    time.sleep(1)
                else:
                    break
        except KeyboardInterrupt:
            print("\n\nStopping system...")
            self.system_active = False
```

```
    finally:
        self.show_final_statistics()

def show_final_statistics(self):
    """Display final statistics on exit"""
    print("\n" + "="*50)
    print("FINAL STATISTICS")
    print("="*50)

    if self.total_bottles_processed == 0:
        print("No bottles processed this session.")
        print("="*50 + "\n")
        return

    print(f"Total Bottles: {self.total_bottles_processed}")
    print("\nBreakdown:")
    for bottle_type, count in sorted(self.bottles_by_type.items()):
        if count > 0:
            pct = (count / self.total_bottles_processed * 100)
            print(f"  {bottle_type:15s}: {count:3d} ({pct:5.1f}%)")
    print("="*50 + "\n")

    self.display_lcd(f"Total: {self.total_bottles_processed}", "Session ended")
    time.sleep(3)

# ==================== CLEANUP ====================

def cleanup(self):
    """Clean up all resources"""
    print("\nCleaning up...")

    if self.servo_pwm:
```

```
            try:
                self.close_flap()
                time.sleep(0.5)
                self.servo_pwm.stop()
                print("Servo stopped")
            except Exception as e:
                print(f"Warning: Servo cleanup warning: {e}")

        if self.lcd:
            try:
                self.display_lcd("System", "Shutting down...")
                time.sleep(1)
                self.lcd.close()
                print("LCD closed")
            except Exception as e:
                print(f"Warning: LCD cleanup warning: {e}")

        if self.camera:
            self.camera.stop()
            print("Camera stopped")

        if self.db_connection:
            self.db_connection.close()
            print("Database closed")

        if GPIO_AVAILABLE:
            GPIO.cleanup()
            print("GPIO cleaned up")

        print("Cleanup complete!")

# ==================== MAIN ====================
```

```
if __name__ == "__main__":
    print("="*50)
    print("BottleSort System v2.0")
    print("   La Salle University-Ozamiz")
    print("="*50 + "\n")

    system = BottleClassifierSystem()

    try:
        # Initialize all hardware
        system.setup_gpio()
        system.setup_lcd()
        system.setup_camera()
        system.setup_database()

        # Run the main production loop
        system.continuous_monitoring()

    except Exception as e:
        print(f"\nCritical Error: {e}")
    finally:
        # Ensure cleanup runs on exit or crash
        system.cleanup()
```

Webpage Code

about.php

```
<?php
// about.php
?>
<!DOCTYPE html>
<html lang="en">
<head>
```

```
  <meta charset="UTF-8">
  <title>About | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <!-- Fonts -->
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">

  <!-- AOS -->
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">

  <!-- Main CSS -->
  <link rel="stylesheet" href="css/style.css">

  <!-- About-specific styles -->
  <style>
    #about {
      position: relative;
      min-height: 40vh;
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      text-align: center;
      padding: 7rem 2rem 5rem;
      color: #004e44;
    }

    #about::before { display: none; }
    #about > * { position: relative; z-index: 1; }
```

```
#about h1 {
  font-size: 2.6rem;
  margin-bottom: 1rem;
  color: #ffffff;
  text-shadow: 0 2px 10px rgba(0,0,0,0.3);
}

#about p {
  max-width: 850px;
  line-height: 1.7;
  font-size: 1.1rem;
  margin-bottom: 2rem;
  color: #eafaf4;
}

.types { display: flex; gap: 25px; justify-content: center; flex-wrap: wrap; width:
100%; max-width: 1200px; }

.type-card {
  background: rgba(255,255,255,0.12);
  color: #ffffff;
  padding: 20px;
  border-radius: 12px;
  text-align: center;
  backdrop-filter: blur(10px);
  transition: all 0.3s ease;
  width: 220px;
  cursor: pointer;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  border: 1px solid rgba(255,255,255,0.15);
}
```

```
.type-card:hover { transform: translateY(-5px); background:
rgba(255,255,255,0.22); box-shadow: 0 6px 20px rgba(0,0,0,0.3); }

.type-card img { width: 90px; height: 90px; margin-bottom: 10px; object-fit:
contain; }
.type-card h3 { color: #ffffff; margin-bottom: 8px; }
.type-card p { color: #d9fff2; font-size: 0.95rem; margin: 0; }
/* ---------------------- MODALS ---------------------- */
.modal {
  display: none;
  position: fixed;
  z-index: 2000;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: rgba(0,0,0,0.75);
  justify-content: center;
  align-items: center;
}

.modal-content {
  background: #ffffff;
  color: #004e44;
  text-align: center;
  padding: 25px;
  border-radius: 14px;
  box-shadow: 0 0 30px rgba(0,0,0,0.5);
  width: 90%;
  max-width: 400px;
  position: relative;
}
```

```
    .modal-content h2 { margin-bottom: 15px; }
    .modal-content img { width: 100%; border-radius: 10px; margin-top: 10px; }

    .modal-close {
      position: absolute;
      top: 20px;
      right: 25px;
      font-size: 2.5rem;
      color: white;
      cursor: pointer;
      transition: color 0.2s ease;
      z-index: 2001;
    }

    .modal-close:hover { color: #00bfa6; }

    @media (max-width: 900px) {
      #about h1 { font-size: 2.2rem; }
      #about p { font-size: 1rem; }
      .type-card { width: 180px; }
    }
  </style>
</head>
<body>

<?php include 'navbar.php'; ?>

<!-- MAIN CONTENT WRAPPER -->
<main>
  <section id="about">
    <h1 data-aos="fade-down">Automated Plastic Bottle Sorting System</h1>
    <p data-aos="fade" data-aos-delay="300">
```

```
    <em>BottleSort is an innovative waste management solution designed to improve
plastic bottle segregation efficiency. Using sensor-based detection and automated
sorting mechanisms, the system identifies and separates plastic bottles into four main
plastic types — PET, HDPE, LDPE, and PP — with a separate bin for unidentified
materials. This approach aims to streamline the recycling process, improve sorting
accuracy, and support more effective waste management practices.</em>
  </p>

  <div class="types" data-aos="fade-up" data-aos-delay="600">

    <div class="type-card" id="pet-card">
      <img src="img/pet.png" alt="PET Bottles">
      <h3>PET (Polyethylene Terephthalate)</h3>
      <p><em>Transparent bottles commonly used for beverages and
water.</em></p>
    </div>

    <div class="type-card" id="hdpe-card">
      <img src="img/hdpe.png" alt="HDPE Bottles">
      <h3>HDPE (High-Density Polyethylene)</h3>
      <p><em>Opaque and rigid bottles used in milk jugs and detergents.</em></p>
    </div>

    <div class="type-card" id="ldpe-card">
      <img src="img/ldpe.png" alt="LDPE Bottles">
      <h3>LDPE (Low-Density Polyethylene)</h3>
      <p><em>Flexible plastic bottles used in squeezable containers.</em></p>
    </div>

    <div class="type-card" id="pp-card">
      <img src="img/pp.png" alt="PP Bottles">
      <h3>PP (Polypropylene)</h3>
```

```
        <p><em>Lightweight and heat-resistant bottles used in food
packaging.</em></p>
      </div>

      <div class="type-card" id="others-card">
        <img src="img/other.png" alt="Unclassified Bottles">
        <h3>Unidentified Materials</h3>
        <p><em>Items that cannot be classified into the four main plastic
types.</em></p>
      </div>
    </div>
  </section>
</main>

<!-- MODALS -->
<div id="petModal" class="modal">
  <div class="modal-content" data-aos="zoom-in">
    <h2>PET Bottles - Visual Examples</h2>
    <img src="img/pet-about.png" alt="PET Bottle Example">
  </div>
  <span class="modal-close" id="closePet">&times;</span>
</div>

<div id="hdpeModal" class="modal">
  <div class="modal-content" data-aos="zoom-in">
    <h2>HDPE Bottles - Visual Examples</h2>
    <img src="img/hdpe-about.png" alt="HDPE Bottle Example">
  </div>
  <span class="modal-close" id="closeHdpe">&times;</span>
</div>

<div id="ldpeModal" class="modal">
```

```
  <div class="modal-content" data-aos="zoom-in">
    <h2>LDPE Bottles - Visual Examples</h2>
    <img src="img/ldpe-about.png" alt="LDPE Bottle Example">
  </div>
  <span class="modal-close" id="closeLdpe">&times;</span>
</div>

<div id="ppModal" class="modal">
  <div class="modal-content" data-aos="zoom-in">
    <h2>PP Bottles - Visual Examples</h2>
    <img src="img/pp-about.png" alt="PP Bottle Example">
  </div>
  <span class="modal-close" id="closePp">&times;</span>
</div>

<div id="othersModal" class="modal">
  <div class="modal-content" data-aos="zoom-in">
    <h2>Unidentified Materials - Visual Examples</h2>
    <img src="img/others-about.png" alt="Other Bottle Example">
  </div>
  <span class="modal-close" id="closeOthers">&times;</span>
</div>

<?php include 'footer.php'; ?>

<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });

  const modalMap = [
    { card: 'pet-card', modal: 'petModal', close: 'closePet' },
    { card: 'hdpe-card', modal: 'hdpeModal', close: 'closeHdpe' },
```

```
    { card: 'ldpe-card', modal: 'ldpeModal', close: 'closeLdpe' },
    { card: 'pp-card', modal: 'ppModal', close: 'closePp' },
    { card: 'others-card', modal: 'othersModal', close: 'closeOthers' },
  ];

  modalMap.forEach(({card, modal, close}) => {
    const cardEl = document.getElementById(card);
    const modalEl = document.getElementById(modal);
    const closeEl = document.getElementById(close);

    cardEl.addEventListener('click', () => modalEl.style.display = 'flex');
    closeEl.addEventListener('click', () => modalEl.style.display = 'none');
    window.addEventListener('click', e => { if (e.target === modalEl)
modalEl.style.display = 'none'; });
  });
</script>

</body>
</html>
```

team.php

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Team | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">
```

```
<link rel="stylesheet" href="css/style.css">

<style>
  #team {
    min-height: 40vh;
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    text-align: center;
    padding: 7rem 2rem 5rem;
  }

  #team h1 {
    font-size: 2.6rem;
    margin-bottom: 1rem;
    color: #ffffff;
    text-shadow: 0 2px 10px rgba(0,0,0,0.3);
  }

  #team .section-desc {
    max-width: 850px;
    line-height: 1.7;
    font-size: 1.1rem;
    margin-bottom: 3rem;
    color: #eafaf4;
  }

  .team-grid {
    display: flex;
    gap: 25px;
    justify-content: center;
```

```
  flex-wrap: wrap;
  width: 100%;
  max-width: 1200px;
}

.team-card {
  background: rgba(255,255,255,0.12);
  color: #ffffff;
  padding: 30px;
  border-radius: 12px;
  backdrop-filter: blur(10px);
  transition: transform 0.3s ease;
  width: 280px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  border: 1px solid rgba(255,255,255,0.15);
}

.team-card:hover {
  transform: translateY(-5px);
}
.profile-img {
  width: 100%;
  height: 240px;
  object-fit: cover;
  border-radius: 8px;
  margin-bottom: 15px;
}

.team-card h3 {
  color: #ffffff;
  margin-bottom: 8px;
  font-size: 1.3rem;
```

```
    }

    .team-card p {
      color: #d9fff2;
      font-size: 0.95rem;
      line-height: 1.5;
    }

    @media (max-width: 900px) {
      #team h1 { font-size: 2.2rem; }
      #team .section-desc { font-size: 1rem; }
      .team-card { width: 260px; }
      .profile-img { height: 200px; }
    }

    @media (max-width: 480px) {
      #team h1 { font-size: 1.9rem; }
      .team-card { width: 100%; max-width: 300px; }
    }
  </style>
</head>
<body>

<?php include 'navbar.php'; ?>

<section id="team">
  <h1 data-aos="fade-down">Behind BottleSort: Our Team</h1>
  <p class="section-desc" data-aos="fade">
    Meet the team dedicated to transforming plastic waste management through
technology.
  </p>
```

```
  <div class="team-grid" data-aos="fade-up">
    <div class="team-card">
      <img src="img/cherny.png" alt="Cherny Gain Simbajon" class="profile-img">
      <h3>Cherny Gain Simbajon</h3>
      <p>BS Computer Engineering - 4th Year<br><br>La Salle University -
Ozamiz</p>
    </div>

    <div class="team-card">
      <img src="img/hannah.png" alt="Hannah Joy Almendras" class="profile-img">
      <h3>Hannah Joy Almendras</h3>
      <p>BS Computer Engineering - 4th Year<br><br>La Salle University -
Ozamiz</p>
    </div>

    <div class="team-card">
      <img src="img/jasper.png" alt="Jasper John Paitan" class="profile-img">
      <h3>Jasper John Paitan</h3>
      <p>BS Computer Engineering - 4th Year<br><br>La Salle University -
Ozamiz</p>
    </div>
  </div>
</section>

<?php include 'footer.php'; ?>

<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });
</script>

</body>
```

```
</html>
```

prototype.php

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Prototype | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">

  <style>
    #prototype {
      position: relative;
      min-height: 40vh;
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      text-align: center;
      padding: 7rem 2rem 5rem;
    }

    #prototype h1 {
      font-size: 2.6rem;
      margin-bottom: 1rem;
      color: #ffffff;
```

```
  text-shadow: 0 2px 10px rgba(0,0,0,0.3);
}

#prototype .section-desc {
  max-width: 850px;
  line-height: 1.7;
  font-size: 1.1rem;
  margin-bottom: 3rem;
  color: #eafaf4;
}

.cards-grid {
  display: flex;
  gap: 25px;
  justify-content: center;
  flex-wrap: wrap;
  width: 100%;
  max-width: 1200px;
}

.card {
  background: rgba(255,255,255,0.12);
  color: #ffffff;
  padding: 30px;
  border-radius: 12px;
  backdrop-filter: blur(10px);
  transition: all 0.3s ease;
  width: 380px;
  cursor: pointer;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  border: 1px solid rgba(255,255,255,0.15);
  display: flex;
```

```
  flex-direction: column;
  min-height: 480px;
}

.card:hover {
  transform: translateY(-5px);
  background: rgba(255,255,255,0.22);
  box-shadow: 0 6px 20px rgba(0,0,0,0.3);
}

.card img {
  width: 100%;
  height: 350px;
  object-fit: contain;
  margin-bottom: 10px;
}

.card-content {
  margin-top: auto;
}

.card h3 {
  color: #ffffff;
  margin-bottom: 12px;
  font-size: 1.5rem;
}

.card p {
  color: #d9fff2;
  font-size: 1.05rem;
  line-height: 1.5;
}
```

```
.modal {
  display: none;
  position: fixed;
  z-index: 2000;
  inset: 0;
  background: rgba(0,0,0,0.75);
  justify-content: center;
  align-items: center;
  padding: 20px;
}

.modal-content {
  background: #ffffff;
  color: #004e44;
  padding: 2rem;
  border-radius: 14px;
  box-shadow: 0 0 30px rgba(0,0,0,0.5);
  width: 90%;
  max-width: 900px;
  max-height: 90vh;
  overflow-y: auto;
  position: relative;
}

.modal-content h2 {
  text-align: center;
  margin-bottom: 1rem;
  padding: 0.8rem 0;
  font-size: 1.8rem;
  font-weight: 700;
  color: #006b5f;
  border-bottom: 2px solid #c5e8d7;
```

```
}

.modal-content > p {
  font-size: 1rem;
  line-height: 1.7;
  margin-bottom: 1.5rem;
}

.modal-close {
  position: absolute;
  top: 15px;
  right: 20px;
  font-size: 2rem;
  color: #006b5f;
  cursor: pointer;
  transition: color 0.2s ease;
  background: none;
  border: none;
}

.modal-close:hover {
  color: #00bfa6;
}

.component-row {
  display: flex;
  align-items: center;
  gap: 2rem;
  margin: 2rem 0;
  padding: 1.5rem;
  background: #f8fffe;
  border-radius: 10px;
```

```
  box-shadow: 0 2px 8px rgba(0,0,0,0.05);
}

.component-row.reverse {
  flex-direction: row-reverse;
}

.component-row img {
  width: 240px;
  height: 240px;
  object-fit: contain;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.1);
  flex-shrink: 0;
  background: #ffffff;
  padding: 10px;
}

.component-row .text-content {
  flex: 1;
}

.component-row h4 {
  color: #006b5f;
  margin-bottom: 0.8rem;
  font-size: 1.3rem;
  font-weight: 600;
}

.component-row p {
  font-size: 1rem;
  line-height: 1.7;
```

```
      color: #004e44;
    }

    @media (max-width: 900px) {
      #prototype h1 { font-size: 2.2rem; }
      #prototype .section-desc { font-size: 1rem; }
      .card { width: 320px; min-height: 440px; }
      .card img { height: 240px; }

      .component-row,
      .component-row.reverse {
        flex-direction: column;
      }

      .component-row .text-content,
      .component-row img {
        width: 200px;
        height: 200px;
      }
      .modal-content h2 { font-size: 1.5rem; }
    }
  </style>
</head>
<body>

<?php include 'navbar.php'; ?>

<section id="prototype">
  <h1 data-aos="fade-down">BottleSort System Design</h1>
  <p class="section-desc" data-aos="fade" data-aos-delay="300">
```

```
    This system integrates hardware, software, and intelligent control systems to
perform accurate plastic classification and waste segregation. The following sections
highlight the core elements of its design and functionality.
  </p>

  <div class="cards-grid" data-aos="fade-up" data-aos-delay="600">
    <div class="card" data-modal="hardware">
      <img src="img/hardware.png" alt="Hardware Design">
      <div class="card-content">
        <h3>Hardware Design</h3>
        <p>Physical assembly and integration of sensors, actuators, and the processing
unit that enable automation.</p>
      </div>
    </div>

    <div class="card" data-modal="classification">
      <img src="img/classification.png" alt="Classification Area">
      <div class="card-content">
        <h3>Classification Area</h3>
        <p>Core area where intelligent analysis identifies the plastic type using sensor
data and machine learning models.</p>
      </div>
    </div>

    <div class="card" data-modal="bins">
      <img src="img/bins.png" alt="Bin Receptacles">
      <div class="card-content">
        <h3>Bin Receptacles</h3>
        <p>Organized collection system with color-coded bins ensuring systematic
segregation of plastic types.</p>
      </div>
    </div>
```

```
  </div>
</section>

<!-- Hardware Modal -->
<div id="modal-hardware" class="modal">
  <div class="modal-content">
    <button class="modal-close">&times;</button>
    <h2>Hardware Design</h2>

    <p data-aos="fade-up">
      The hardware design integrates detection sensors, actuation modules, power
regulation units and user interface components controlled by a central processor.
Below are the physical components used in the prototype and their roles within the
BottleSort system.
    </p>

    <div class="component-row" data-aos="fade-right">
      <img src="img/rpi.png" alt="Raspberry Pi 5">
      <div class="text-content">
        <h4>Raspberry Pi 5</h4>
        <p>Serves as the central processing unit managing sensor input, running the
classification inference and controlling motors and displays.</p>
      </div>
    </div>

    <div class="component-row reverse" data-aos="fade-left">
      <img src="img/ultrasonic.png" alt="Ultrasonic HC-SR04 sensors">
      <div class="text-content">
        <h4>Ultrasonic Sensors (HC-SR04)</h4>
        <p>One ultrasonic sensor per bin measures distance to the bin surface and
estimates fill level. The Raspberry Pi polls these sensors; when a bin reaches the
```

```
configured threshold the system marks it as full and prevents further routing to that
receptacle.</p>
      </div>
    </div>

    <div class="component-row" data-aos="fade-right">
      <img src="img/tb6600.png" alt="TB6600 stepper driver">
      <div class="text-content">
        <h4>TB6600 Stepper Motor Driver</h4>
        <p>Drives the stepper motor with microstepping capability to achieve smooth
rotation and precise positioning of the bin carousel. The TB6600 accepts step/dir
signals from the controller.</p>
      </div>
    </div>

    <div class="component-row reverse" data-aos="fade-left">
      <img src="img/steppermotor.png" alt="Wantai NEMA 17 stepper motor">
      <div class="text-content">
        <h4>Wantai Stepper Motor — NEMA 17</h4>
        <p>Rotates the bin platform. The control logic divides the full rotation into five
equal positions (one per bin). Using the TB6600 driver, the system performs precise
indexing so the correct bin aligns under the chute for each classified item.</p>
      </div>
    </div>

    <div class="component-row" data-aos="fade-right">
      <img src="img/servo.png" alt="MG996R Servo Motor">
      <div class="text-content">
        <h4>Tower Pro MG996R Servo Motor</h4>
        <p>Operates the chute door (open/close) to release a single bottle into the
selected bin. The servo's timing is coordinated with the stepper's position to ensure
reliable drops without collisions.</p>
```

```
    </div>
  </div>

  <div class="component-row reverse" data-aos="fade-left">
    <img src="img/lcd.png" alt="16x2 I2C LCD">
    <div class="text-content">
      <h4>16×2 I2C LCD Display</h4>
      <p>Displays real-time information such as detected plastic type, bin fill
percentages, and system messages — useful for on-site monitoring and debugging
during testing phases.</p>
    </div>
  </div>

  <div class="component-row" data-aos="fade-right">
    <img src="img/microwave.png" alt="HFS-DC06 Microwave Radar Sensor">
    <div class="text-content">
      <h4>HFS-DC06 Microwave Radar Sensor</h4>
      <p>Detects motion/presence when a bottle is thrown into the chute, serving as
the trigger to start image/sensor acquisition and the classification workflow. Works
reliably in varying light conditions.</p>
    </div>
  </div>

  <div class="component-row reverse" data-aos="fade-left">
    <img src="img/hall.png" alt="KY-003 Hall magnetic sensor">
    <div class="text-content">
      <h4>KY-003 Hall Magnetic Sensor Module</h4>
      <p>Detects a small magnet attached to the PET bin to establish a "home" or
reference position. This allows the controller to recalibrate the carousel and recover
from step drift or missed steps.</p>
    </div>
  </div>
```

```
    <div class="component-row" data-aos="fade-right">
      <img src="img/buck.png" alt="XY-3606 buck converter">
      <div class="text-content">
        <h4>XY-3606 DC-DC Buck Converters</h4>
        <p>Two buck converters step 12V down to stable 5V rails. One converter powers
the 5 ultrasonic sensors; the other powers the microwave sensor, Hall sensor, and
servo motor — separating loads improves reliability and reduces electrical noise.</p>
      </div>
    </div>

    <div class="component-row reverse" data-aos="fade-left">
      <img src="img/powersupply.png" alt="12V 5A Power Adapter">
      <div class="text-content">
        <h4>12V 5A Adapter</h4>
        <p>Provides the primary power source for the stepper driver and buck
converters. A true-rated supply ensures sufficient overhead for motor surges and stable
operation of the entire system.</p>
      </div>
    </div>
  </div>
</div>

<!-- Classification Modal -->
<div id="modal-classification" class="modal">
  <div class="modal-content">
    <button class="modal-close">&times;</button>
    <h2>Classification Area</h2>

    <p data-aos="fade-up">
      The classification area combines sensor inputs and a trained machine learning
model to identify the plastic type (PET, HDPE, LDPE, PP, Unclassified). The system
```

uses an AI model trained with a comprehensive dataset at Edge Impulse, deployed on the Raspberry Pi for real-time inference.

```
  </p>

  <div class="component-row" data-aos="fade-right">
    <img src="img/sensor.jpg" alt="Sensor array">
    <div class="text-content">
      <h4>Sensor Array & Object Detection</h4>
      <p>The combined data sources including camera imagery and object detection
from the microwave radar sensor provide robust features that help the model
distinguish plastic bottles from other plastics even when visual cues are ambiguous.
The microwave sensor ensures reliable detection regardless of lighting
conditions.</p>
    </div>
  </div>

  <div class="component-row reverse" data-aos="fade-left">
    <img src="img/ml.jpg" alt="Machine learning">
    <div class="text-content">
      <h4>Machine Learning Algorithm & Training</h4>
      <p>The AI model was trained using Edge Impulse's machine learning platform
with a comprehensive dataset of plastic bottle images representing different plastic
types. The model uses a convolutional neural network (CNN) architecture optimized
for edge devices. The dataset was split following an 80/20 ratio, where 80% of the
images were used for training the model and 20% were reserved for testing. This ratio
ensures that the model learns effectively from the majority of the data while
maintaining a separate portion to evaluate its accuracy and generalization. Once
trained, it was converted to TensorFlow Lite format and deployed on the Raspberry Pi
for real-time classification. When the model produces a classification result, the
control logic triggers the stepper/servo routine to route the bottle to the correct
bin.</p>
    </div>
```

```
    </div>
  </div>
</div>

<!-- Bins Modal -->
<div id="modal-bins" class="modal">
  <div class="modal-content">
    <button class="modal-close">&times;</button>
    <h2>Bin Receptacles</h2>

    <p data-aos="fade-up">
      Bin receptacles are color-coded following the Resin Identification Code (RIC)
standard to provide visual consistency with international plastic sorting conventions.
Each bin is instrumented with an ultrasonic sensor for bin-level monitoring. The
system routes bottles to: PET, HDPE, LDPE, PP, and Unclassified bins; bins reaching
capacity are flagged on the LCD and analytics dashboard.
    </p>

    <div class="component-row" data-aos="fade-up" style="justify-content: center;">
      <img src="img/colorcodeRIC.png" alt="Resin Identification Code color coding
reference" style="width: 100%; max-width: 600px; height: auto;">
    </div>

    <p data-aos="fade-up" style="text-align: center; font-style: italic; color: #006b5f;
margin-bottom: 2rem;">
      Color coding follows the standardized Resin Identification Code (RIC) system for
plastic classification.
    </p>
  </div>
</div>

<?php include 'footer.php'; ?>
```

```
<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });

  // Modal handling
  document.querySelectorAll('.card').forEach(card => {
    const modalId = 'modal-' + card.dataset.modal;
    const modal = document.getElementById(modalId);

    card.addEventListener('click', () => modal.style.display = 'flex');

    modal.querySelector('.modal-close').addEventListener('click', () => {
      modal.style.display = 'none';
    });

    modal.addEventListener('click', e => {
      if (e.target === modal) modal.style.display = 'none';
    });
  });
</script>

</body>
</html>
```

logout.php

```
<?php
session_start();
session_destroy();
header("Location: admin-login.php");
exit();
```

navbar.php

```
<?php
// navbar.php - Main navigation bar
$current_page = basename($_SERVER['PHP_SELF']);
?>
<nav>
  <div class="nav-left">
    <a href="index.php">
      <img src="img/logo1.png" alt="BottleSort" class="nav-logo">
    </a>
  </div>

  <!-- Hamburger menu button (hidden on desktop) -->
  <button class="menu-toggle" aria-label="Toggle navigation menu"
aria-expanded="false">
    <span></span>
    <span></span>
    <span></span>
  </button>
  <div class="nav-center">
    <a href="index.php" class="<?= $current_page == 'index.php' ? 'active' : '' ?>">
      <img src="img/home.png" class="nav-icon" alt="Home"> Home
    </a>
    <a href="about.php" class="<?= $current_page == 'about.php' ? 'active' : '' ?>">
      <img src="img/about.png" class="nav-icon" alt="About"> About
    </a>
    <a href="analytics.php" class="<?= $current_page == 'analytics.php' ? 'active' : ''
?>">
      <img src="img/analytics.png" class="nav-icon" alt="Analytics"> Analytics
    </a>
    <a href="prototype.php" class="<?= $current_page == 'prototype.php' ? 'active' : ''
?>">
```

```
      <img src="img/prototype.png" class="nav-icon" alt="System"> System
    </a>
    <a href="team.php" class="<?= $current_page == 'team.php' ? 'active' : '' ?>">
      <img src="img/team.png" class="nav-icon" alt="Team"> Team
    </a>
    <a href="contact.php" class="<?= $current_page == 'contact.php' ? 'active' : '' ?>">
      <img src="img/contact.png" class="nav-icon" alt="Contact"> Contact Us
    </a>

    <!-- Admin link inside menu for mobile -->
    <a href="admin-login.php" class="nav-mobile-admin <?= $current_page ==
'admin-login.php' ? 'active' : '' ?>">
      <img src="img/admin.png" class="nav-icon" alt="Admin"> Admin
    </a>
  </div>

  <div class="nav-right">
    <a href="admin-login.php" class="nav-right-link">
      <img src="img/admin.png" class="nav-icon" alt="Admin"> Admin
    </a>
  </div>
</nav>

<style>
/* Hamburger menu button - completely removed from layout on desktop */
.menu-toggle {
  display: none;
  flex-direction: column;
  gap: 5px;
  background: none;
  border: none;
  cursor: pointer;
```

```
  padding: 10px;
  z-index: 1001;
  position: absolute;
  right: 20px;
  top: 50%;
  transform: translateY(-50%);
}

.menu-toggle span {
  display: block;
  width: 25px;
  height: 3px;
  background-color: #333;
  transition: all 0.3s ease;
  border-radius: 2px;
}

/* Animate hamburger to X when active */
.menu-toggle.active span:nth-child(1) {
  transform: rotate(45deg) translate(7px, 7px);
}

.menu-toggle.active span:nth-child(2) {
  opacity: 0;
}

.menu-toggle.active span:nth-child(3) {
  transform: rotate(-45deg) translate(7px, -7px);
}

/* Hide mobile admin link on desktop */
.nav-mobile-admin {
```

```
  display: none !important;
}

/* Mobile styles - apply when screen is 768px or smaller */
@media screen and (max-width: 768px) {
  .menu-toggle {
    display: flex;
  }

  .nav-center {
    position: fixed;
    top: 0;
    right: -100%;
    width: 70%;
    max-width: 300px;
    height: 100vh;
    background-color: #fff;
    flex-direction: column;
    align-items: flex-start;
    padding: 80px 20px 20px;
    box-shadow: -2px 0 10px rgba(0,0,0,0.1);
    transition: right 0.3s ease;
    z-index: 1000;
    overflow-y: auto;
  }

  .nav-center.active {
    right: 0;
  }

  .nav-center a {
    width: 100%;
```

```
    padding: 15px 10px;
    border-bottom: 1px solid #eee;
  }

  /* Hide desktop admin button on mobile */
  .nav-right {
    display: none;
  }

  /* Show mobile admin link inside menu */
  .nav-mobile-admin {
    display: flex !important;
  }

  /* Overlay when menu is open */
  .nav-center.active::before {
    content: '';
    position: fixed;
    top: 0;
    left: 0;
    width: 100vw;
    height: 100vh;
    background-color: rgba(0, 0, 0, 0.5);
    z-index: -1;
  }
}
</style>

<script>
(function() {
  'use strict';
```

```
// Mobile menu toggle
document.addEventListener('DOMContentLoaded', function() {
  const menuToggle = document.querySelector('.menu-toggle');
  const navCenter = document.querySelector('.nav-center');

  if (!menuToggle || !navCenter) return;

  // Toggle menu
  menuToggle.addEventListener('click', function() {
    const isActive = navCenter.classList.toggle('active');
    menuToggle.classList.toggle('active');
    menuToggle.setAttribute('aria-expanded', isActive);

    // Prevent body scroll when menu is open
    if (isActive) {
      document.body.style.overflow = 'hidden';
    } else {
      document.body.style.overflow = '';
    }
  });

  // Close menu when clicking on overlay
  navCenter.addEventListener('click', function(e) {
    if (e.target === navCenter && navCenter.classList.contains('active')) {
      menuToggle.click();
    }
  });

  // Close menu when clicking a link
  const navLinks = navCenter.querySelectorAll('a');
  navLinks.forEach(link => {
    link.addEventListener('click', function() {
```

```
      if (window.innerWidth <= 768) {
        menuToggle.click();
      }
    });
  });
});

// Hide/show navbar on scroll
let lastScroll = 0;
const nav = document.querySelector('nav');

if (nav) {
  window.addEventListener('scroll', function() {
    const currentScroll = window.pageYOffset;

    if (currentScroll <= 0) {
      nav.classList.remove('nav-hidden');
      return;
    }

    if (currentScroll > lastScroll && currentScroll > 100) {
      nav.classList.add('nav-hidden');
    } else {
      nav.classList.remove('nav-hidden');
    }

    lastScroll = currentScroll;
  });
}
})();
</script>
```

navbar-admin.php

```
<?php
// navbar-admin.php
$current_page = basename($_SERVER['PHP_SELF']);
?>
<nav>
  <div class="nav-left">
    <a href="index.php">
      <img src="img/logo1.png" alt="BottleSort" class="nav-logo">
    </a>
  </div>

  <button class="menu-toggle" aria-label="Toggle navigation menu"
aria-expanded="false">
    <span></span>
    <span></span>
    <span></span>
  </button>

  <div class="nav-center">
    <a href="index.php" class="<?= $current_page == 'index.php' ? 'active' : '' ?>">
      <img src="img/home.png" class="nav-icon" alt="Home"> Home
    </a>
    <a href="about.php" class="<?= $current_page == 'about.php' ? 'active' : '' ?>">
      <img src="img/about.png" class="nav-icon" alt="About"> About
    </a>
    <a href="analytics.php" class="<?= $current_page == 'analytics.php' ? 'active' : ''
?>">
      <img src="img/analytics.png" class="nav-icon" alt="Analytics"> Analytics
    </a>
    <a href="prototype.php" class="<?= $current_page == 'prototype.php' ? 'active' : ''
?>">
```

```
      <img src="img/prototype.png" class="nav-icon" alt="System"> System
    </a>
    <a href="team.php" class="<?= $current_page == 'team.php' ? 'active' : '' ?>">
      <img src="img/team.png" class="nav-icon" alt="Team"> Team
    </a>
    <a href="contact.php" class="<?= $current_page == 'contact.php' ? 'active' : '' ?>">
      <img src="img/contact.png" class="nav-icon" alt="Contact"> Contact Us
    </a>

    <a href="#" onclick="event.preventDefault(); showSignOutWarning();"
class="nav-mobile-admin">
      <img src="img/signout.png" class="nav-icon" alt="Sign Out"> Sign Out
    </a>
  </div>

  <div class="nav-right">
    <a href="#" onclick="event.preventDefault(); showSignOutWarning();"
class="nav-right-link">
      <img src="img/signout.png" class="nav-icon" alt="Sign Out"> Sign Out
    </a>
  </div>
</nav>

<style>
.menu-toggle {
  display: none;
  flex-direction: column;
  gap: 5px;
  background: none;
  border: none;
  cursor: pointer;
  padding: 10px;
```

```
  z-index: 1001;
  position: absolute;
  right: 20px;
  top: 50%;
  transform: translateY(-50%);
}

.menu-toggle span {
  display: block;
  width: 25px;
  height: 3px;
  background-color: white;
  transition: all 0.3s ease;
  border-radius: 2px;
}

.menu-toggle.active span:nth-child(1) {
  transform: rotate(45deg) translate(7px, 7px);
}

.menu-toggle.active span:nth-child(2) {
  opacity: 0;
}

.menu-toggle.active span:nth-child(3) {
  transform: rotate(-45deg) translate(7px, -7px);
}

.nav-mobile-admin {
  display: none !important;
}
```

```
@media screen and (max-width: 768px) {
  .menu-toggle {
    display: flex;
  }

  .nav-center {
    position: fixed;
    top: 0;
    right: -100%;
    width: 70%;
    max-width: 300px;
    height: 100vh;
    background-color: #fff;
    flex-direction: column;
    align-items: flex-start;
    padding: 80px 20px 20px;
    box-shadow: -2px 0 10px rgba(0,0,0,0.1);
    transition: right 0.3s ease;
    z-index: 1000;
    overflow-y: auto;
  }

  .nav-center.active {
    right: 0;
  }

  .nav-center a {
    width: 100%;
    padding: 15px 10px;
    border-bottom: 1px solid #eee;
  }
```

```
  .nav-right {
    display: none;
  }

  .nav-mobile-admin {
    display: flex !important;
  }

  .nav-center.active::before {
    content: '';
    position: fixed;
    top: 0;
    left: 0;
    width: 100vw;
    height: 100vh;
    background-color: rgba(0, 0, 0, 0.5);
    z-index: -1;
  }
}
</style>

<script>
(function() {
  document.addEventListener('DOMContentLoaded', function() {
    const menuToggle = document.querySelector('.menu-toggle');
    const navCenter = document.querySelector('.nav-center');
    if (!menuToggle || !navCenter) return;

    menuToggle.addEventListener('click', function() {
      const isActive = navCenter.classList.toggle('active');
      menuToggle.classList.toggle('active');
      menuToggle.setAttribute('aria-expanded', isActive);
```

```
    document.body.style.overflow = isActive ? 'hidden' : '';
  });

  navCenter.addEventListener('click', function(e) {
    if (e.target === navCenter && navCenter.classList.contains('active')) {
      menuToggle.click();
    }
  });

  navCenter.querySelectorAll('a').forEach(link => {
    link.addEventListener('click', function() {
      if (window.innerWidth <= 768) {
        menuToggle.click();
      }
    });
  });
});

let lastScroll = 0;
const nav = document.querySelector('nav');

if (nav) {
  window.addEventListener('scroll', function() {
    const currentScroll = window.pageYOffset;

    if (currentScroll <= 0) {
      nav.classList.remove('nav-hidden');
      return;
    }

    if (currentScroll > lastScroll && currentScroll > 100) {
      nav.classList.add('nav-hidden');
```

```
      } else {
        nav.classList.remove('nav-hidden');
      }

      lastScroll = currentScroll;
    });
  }
})();
</script>
```

index.php

```
<?php
// index.php
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Home | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <!-- Fonts -->
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">

  <!-- AOS -->
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">

  <!-- Main CSS -->
  <link rel="stylesheet" href="css/style.css">
```

```
</head>
<body>
<!-- NAVBAR -->
<?php include 'navbar.php'; ?>

<!-- MAIN CONTENT WRAPPER -->
<main>
  <!-- HERO SECTION -->
  <section id="intro">
    <div class="intro-content" data-aos="fade-right">
      <img src="img/logo2.png" alt="BottleSort Logo">
      <h1>Smart Sorting for a Sustainable Future</h1>
      <p><em>BottleSort is an automated plastic bottle sorting system that uses sensor
technology and intelligent detection to identify and separate plastic bottles by material
type — including PET, HDPE, LDPE, and PP. By improving the efficiency of plastic
segregation, BottleSort aims to reduce manual sorting effort, enhance recycling
accuracy, and contribute to better waste management practices.</em></p>
      <a href="about.php" class="cta-btn">Learn More</a>
    </div>
  </section>
</main>

<!-- FOOTER -->
<?php include 'footer.php'; ?>

<!-- AOS -->
<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });
</script>

</body>
```

```
</html>
```

get-dashboard-data.php

```
<?php
header('Content-Type: application/json');

// Database connection
$servername = "127.0.0.1";
$username = "root";
$password = "root";
$dbname = "bottlesort";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die(json_encode(['error' => 'Connection failed: ' . $conn->connect_error]));
}

$action = isset($_GET['action']) ? $_GET['action'] : 'getBinData';

// ============ GET BIN DATA ============
if ($action === 'getBinData') {
    $sql = "SELECT
                BinLevel_PET,
                BinLevel_HDPE,
                BinLevel_LDPE,
                BinLevel_PP,
                BinLevel_Unclassified,
                record_time
            FROM plastic_bottle_bin_level
            ORDER BY record_time DESC
            LIMIT 1";
```

```
    $result = $conn->query($sql);

    $bins = [];
    $fullBinsCount = 0;

    if ($result->num_rows > 0) {
        $row = $result->fetch_assoc();
        $timestamp = $row['record_time'];

        $binTypes = [
            1 => ['type' => 'PET', 'level' => (int)$row['BinLevel_PET']],
            2 => ['type' => 'HDPE', 'level' => (int)$row['BinLevel_HDPE']],
            3 => ['type' => 'LDPE', 'level' => (int)$row['BinLevel_LDPE']],
            4 => ['type' => 'PP', 'level' => (int)$row['BinLevel_PP']],
            5 => ['type' => 'Unclassified', 'level' => (int)$row['BinLevel_Unclassified']]
        ];

        foreach ($binTypes as $id => $binData) {
            $bins[] = [
                'id' => $id,
                'type' => $binData['type'],
                'level' => $binData['level'],
                'status' => $binData['level'] >= 95 ? 'FULL' : ($binData['level'] >= 70 ?
'HIGH' : 'OK'),
                'statusClass' => $binData['level'] >= 95 ? 'status-full' : ($binData['level'] >=
70 ? 'status-warning' : 'status-ok'),
                'progressColor' => $binData['level'] >= 95 ? '#e74c3c' : ($binData['level'] >=
70 ? '#f39c12' : '#2ecc71'),
                'lastUpdated' => date('Y-m-d H:i:s', strtotime($timestamp))
            ];

            if ($binData['level'] >= 95) {
```

```
            $fullBinsCount++;
        }
    }
}

echo json_encode([
    'bins' => $bins,
    'fullBinsCount' => $fullBinsCount
]);
}

// ============ GET BOTTLE STATISTICS ============
else if ($action === 'getBottleStats') {
    $period = isset($_GET['period']) ? $_GET['period'] : 'today';

    $dateCondition = "";
    switch($period) {
        case 'today':
            $dateCondition = "DATE(record_time) = CURDATE()";
            break;
        case 'weekly':
            $dateCondition = "record_time >= DATE_SUB(CURDATE(), INTERVAL 6
DAY)";
            break;
        case 'monthly':
            $dateCondition = "record_time >= DATE_SUB(CURDATE(), INTERVAL 29
DAY)";
            break;
    }

    // Get total count
    $totalSql = "SELECT COUNT(*) as total
```

```
            FROM plastic_bottle_logs
            WHERE $dateCondition";
$totalResult = $conn->query($totalSql);
$totalCount = 0;
if ($totalResult->num_rows > 0) {
    $totalCount = (int)$totalResult->fetch_assoc()['total'];
}

// Get breakdown by type
$breakdownSql = "SELECT
                  bottle_type,
                  COUNT(*) as count
                FROM plastic_bottle_logs
                WHERE $dateCondition
                GROUP BY bottle_type";

$breakdownResult = $conn->query($breakdownSql);
$bottleData = [
    'PET' => 0,
    'HDPE' => 0,
    'LDPE' => 0,
    'PP' => 0,
    'Unclassified' => 0
];

if ($breakdownResult->num_rows > 0) {
    while($row = $breakdownResult->fetch_assoc()) {
        $type = $row['bottle_type'];
        if (isset($bottleData[$type])) {
            $bottleData[$type] = (int)$row['count'];
        }
    }
```

```
    }

    echo json_encode([
        'totalCount' => $totalCount,
        'bottleData' => $bottleData,
        'period' => $period
    ]);
}

// ============ GET FILTERED DATA ============
else if ($action === 'getFilteredData') {
    $startDate = isset($_GET['startDate']) ? $_GET['startDate'] : date('Y-m-d',
strtotime('-30 days'));
    $endDate = isset($_GET['endDate']) ? $_GET['endDate'] : date('Y-m-d');
    $binType = isset($_GET['binType']) ? $_GET['binType'] : 'all';

    $whereClause = "DATE(record_time) BETWEEN '$startDate' AND '$endDate'";
    if ($binType !== 'all') {
        $whereClause .= " AND bottle_type = '$binType'";
    }

    // Get daily counts for chart
    $chartSql = "SELECT
                    DATE(record_time) as date,
                    bottle_type,
                    COUNT(*) as count
                FROM plastic_bottle_logs
                WHERE $whereClause
                GROUP BY DATE(record_time), bottle_type
                ORDER BY date ASC";

    $chartResult = $conn->query($chartSql);
```

```
$chartData = [];
$dates = [];

if ($chartResult->num_rows > 0) {
    while($row = $chartResult->fetch_assoc()) {
        $date = $row['date'];
        $type = $row['bottle_type'];
        $count = (int)$row['count'];

        if (!in_array($date, $dates)) {
            $dates[] = $date;
        }

        if (!isset($chartData[$type])) {
            $chartData[$type] = [];
        }

        $chartData[$type][$date] = $count;
    }
}

// Fill in missing dates with 0
foreach ($chartData as $type => $data) {
    foreach ($dates as $date) {
        if (!isset($chartData[$type][$date])) {
            $chartData[$type][$date] = 0;
        }
    }
    ksort($chartData[$type]);
}

// Get summary for export
```

```
    $summarySql = "SELECT
                    bottle_type,
                    COUNT(*) as count,
                    MIN(record_time) as first_record,
                    MAX(record_time) as last_record
                  FROM plastic_bottle_logs
                  WHERE $whereClause
                  GROUP BY bottle_type";

    $summaryResult = $conn->query($summarySql);

    $summary = [];
    if ($summaryResult->num_rows > 0) {
        while($row = $summaryResult->fetch_assoc()) {
            $summary[] = [
                'type' => $row['bottle_type'],
                'count' => (int)$row['count'],
                'firstRecord' => $row['first_record'],
                'lastRecord' => $row['last_record']
            ];
        }
    }

    echo json_encode([
        'dates' => $dates,
        'chartData' => $chartData,
        'summary' => $summary,
        'startDate' => $startDate,
        'endDate' => $endDate,
        'binType' => $binType
    ]);
}
```

```
// ============ EMPTY BIN ============
else if ($action === 'emptyBin') {
    $binId = isset($_POST['binId']) ? (int)$_POST['binId'] : 0;

    if ($binId < 1 || $binId > 5) {
        echo json_encode(['success' => false, 'message' => 'Invalid bin ID']);
        exit;
    }

    $binColumns = [
        1 => 'BinLevel_PET',
        2 => 'BinLevel_HDPE',
        3 => 'BinLevel_LDPE',
        4 => 'BinLevel_PP',
        5 => 'BinLevel_Unclassified'
    ];

    $column = $binColumns[$binId];

    // Get current level to check if it's full
    $checkSql = "SELECT $column as level FROM plastic_bottle_bin_level ORDER
BY record_time DESC LIMIT 1";
    $checkResult = $conn->query($checkSql);

    if ($checkResult->num_rows > 0) {
        $currentLevel = (int)$checkResult->fetch_assoc()['level'];

        if ($currentLevel < 95) {
            echo json_encode(['success' => false, 'message' => 'Bin must be at least 95%
full to empty']);
            exit;
        }
```

```
    }

    // Insert new row with emptied bin
    $getSql = "SELECT * FROM plastic_bottle_bin_level ORDER BY record_time
DESC LIMIT 1";
    $getResult = $conn->query($getSql);

    if ($getResult->num_rows > 0) {
        $currentData = $getResult->fetch_assoc();

        $insertSql = "INSERT INTO plastic_bottle_bin_level
                    (BinLevel_PET, BinLevel_HDPE, BinLevel_LDPE, BinLevel_PP,
BinLevel_Unclassified, record_time)
                    VALUES (?, ?, ?, ?, ?, NOW())";

        $stmt = $conn->prepare($insertSql);

        $levels = [
            (int)$currentData['BinLevel_PET'],
            (int)$currentData['BinLevel_HDPE'],
            (int)$currentData['BinLevel_LDPE'],
            (int)$currentData['BinLevel_PP'],
            (int)$currentData['BinLevel_Unclassified']
        ];

        // Set the emptied bin to 0
        $levels[$binId - 1] = 0;

        $stmt->bind_param("iiiii", $levels[0], $levels[1], $levels[2], $levels[3],
$levels[4]);

        if ($stmt->execute()) {
```

```
            // Log the activity
            $activitySql = "INSERT INTO admin_activity (activity_type, description,
timestamp)
                        VALUES ('bin_emptied', 'Emptied bin " . $binId . "', NOW())";
            $conn->query($activitySql);

            echo json_encode(['success' => true, 'message' => 'Bin emptied successfully']);
        } else {
            echo json_encode(['success' => false, 'message' => 'Failed to empty bin']);
        }

        $stmt->close();
    } else {
        echo json_encode(['success' => false, 'message' => 'No bin data found']);
    }
}

$conn->close();
?>
```

get-admin-data.php

```
<?php
$servername = "127.0.0.1";
$port = 3307;
$username = "root";
$password = "root";
$dbname = "bottlesort";

$conn = new mysqli($servername, $username, $password, $dbname, $port);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
```

```
}
?>
```

footer.php

```
<?php
// footer.php
?>
<footer class="footer">
  © <?php echo date("Y"); ?> BottleSort — Innovating for a Sustainable Future
</footer>
```

get-analytics-data.php

```
<?php
header('Content-Type: application/json');

// Database connection
$servername = "127.0.0.1";
$port = 3307;
$username = "root";
$password = "root";
$dbname = "bottlesort";

$conn = new mysqli($servername, $username, $password, $dbname, $port);

if ($conn->connect_error) {
    die(json_encode(['error' => 'Connection failed: ' . $conn->connect_error]));
}

$period = isset($_GET['period']) ? $_GET['period'] : 'today';

// Calculate date range based on period
$dateCondition = "";
```

```
switch($period) {
    case 'today':
        $dateCondition = "DATE(record_time) = CURDATE()";
        break;
    case 'week':
        $dateCondition = "record_time >= DATE_SUB(CURDATE(), INTERVAL 6
DAY)";
        break;
    case 'month':
        $dateCondition = "record_time >= DATE_SUB(CURDATE(), INTERVAL 29
DAY)";
        break;
}

// Get plastic bottle counts by type
$sql = "SELECT
            bottle_type,
            COUNT(*) as count
        FROM plastic_bottle_logs
        WHERE $dateCondition
        GROUP BY bottle_type";

$result = $conn->query($sql);
$bottleData = [
    'PET' => 0,
    'HDPE' => 0,
    'LDPE' => 0,
    'PP' => 0,
    'Unclassified' => 0
];

$totalCount = 0;
```

```
if ($result->num_rows > 0) {
    while($row = $result->fetch_assoc()) {
        $type = $row['bottle_type'];
        $count = (int)$row['count'];
        if (isset($bottleData[$type])) {
            $bottleData[$type] = $count;
            $totalCount += $count;
        }
    }
}

// Get bin levels from plastic_bottle_bin_level
$binSql = "SELECT
            BinLevel_PET,
            BinLevel_HDPE,
            BinLevel_LDPE,
            BinLevel_PP,
            BinLevel_Unclassified,
            record_time
        FROM plastic_bottle_bin_level
        ORDER BY record_time DESC
        LIMIT 1";

$binResult = $conn->query($binSql);

$binLevels = [
    'PET' => 0,
    'HDPE' => 0,
    'LDPE' => 0,
    'PP' => 0,
    'Unclassified' => 0
];
```

```
$lastUpdated = date('Y-m-d H:i');

if ($binResult->num_rows > 0) {
    $binRow = $binResult->fetch_assoc();
    $binLevels['PET'] = (int)$binRow['BinLevel_PET'];
    $binLevels['HDPE'] = (int)$binRow['BinLevel_HDPE'];
    $binLevels['LDPE'] = (int)$binRow['BinLevel_LDPE'];
    $binLevels['PP'] = (int)$binRow['BinLevel_PP'];
    $binLevels['Unclassified'] = (int)$binRow['BinLevel_Unclassified'];
    $lastUpdated = date('Y-m-d H:i', strtotime($binRow['record_time']));
}

$response = [
    'bottleData' => array_values($bottleData),
    'totalCount' => $totalCount,
    'binLevels' => $binLevels,
    'lastUpdated' => $lastUpdated
];

echo json_encode($response);

$conn->close();
?>
```

auth-check.php

```
<?php
session_start();

// Redirect if not logged in
if (!isset($_SESSION['admin_logged_in'])) {
  header("Location: admin-login.php");
  exit();
```

```
}

// Session timeout: 1 hour (3600 seconds)
$timeout_duration = 3600;

if (isset($_SESSION['last_activity'])) {
  $elapsed_time = time() - $_SESSION['last_activity'];

  if ($elapsed_time > $timeout_duration) {
    session_unset();
    session_destroy();
    header("Location: admin-login.php?msg=session_expired");
    exit();
  }
}

// Update last activity time
$_SESSION['last_activity'] = time();
?>
```

contact.php

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Contact | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">
```

```
<link rel="stylesheet" href="css/style.css">

<style>
  #contact {
    min-height: 40vh;
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
    padding: 7rem 2rem 5rem;
  }

  .contact-header {
    display: flex;
    gap: 3rem;
    align-items: center;
    justify-content: center;
    max-width: 1200px;
    width: 100%;
    margin-bottom: 3rem;
    text-align: center;
  }

  .contact-intro {
    flex: 1;
    color: #ffffff;
  }

  .contact-intro h1 {
    font-size: 2.6rem;
    margin-bottom: 1rem;
    color: #ffffff;
```

```
  text-shadow: 0 2px 10px rgba(0,0,0,0.3);
}

.contact-intro p {
  line-height: 1.7;
  font-size: 1.1rem;
  color: #eafaf4;
}

.contact-container {
  display: flex;
  gap: 2rem;
  justify-content: center;
  align-items: flex-start;
  max-width: 1200px;
  width: 100%;
}

form {
  background: rgba(255, 255, 255, 0.12);
  color: #ffffff;
  border-radius: 12px;
  padding: 2.5rem;
  flex: 1;
  max-width: 650px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  backdrop-filter: blur(10px);
  border: 1px solid rgba(255,255,255,0.15);
}

form label {
  display: block;
```

```
  font-weight: 600;
  margin-top: 1rem;
  color: #ffffff;
}

form input,
form textarea,
form select {
  width: 100%;
  padding: 0.9rem;
  border: 1px solid rgba(255,255,255,0.3);
  border-radius: 6px;
  margin-top: 0.5rem;
  font-family: 'Open Sans', sans-serif;
  font-size: 1rem;
  transition: border-color 0.3s ease;
  background: rgba(255,255,255,0.1);
  color: #ffffff;
}

form input::placeholder,
form textarea::placeholder {
  color: rgba(255,255,255,0.6);
}

form select option {
  background: #004e44;
  color: #ffffff;
}

form input:focus,
form textarea:focus,
```

```
form select:focus {
  outline: none;
  border-color: rgba(255,255,255,0.6);
  background: rgba(255,255,255,0.15);
}

#other_inquiry_container {
  display: none;
}

form textarea {
  resize: vertical;
  min-height: 180px;
}

form button {
  background: rgba(255,255,255,0.2);
  color: white;
  border: 1px solid rgba(255,255,255,0.3);
  padding: 1rem 2rem;
  border-radius: 6px;
  margin-top: 1.5rem;
  cursor: pointer;
  font-weight: 600;
  font-size: 1rem;
  transition: all 0.3s ease;
  width: 100%;
}

form button:hover {
  background: rgba(255,255,255,0.3);
  border-color: rgba(255,255,255,0.5);
```

```
}

.message {
  margin-top: 1rem;
  padding: 1rem;
  border-radius: 6px;
  font-weight: 600;
  text-align: center;
}

.success {
  background: #c5e8d7;
  color: #006b5f;
}

.error {
  background: #f8d7da;
  color: #721c24;
}

.privacy {
  max-width: 1200px;
  width: 100%;
  text-align: center;
  font-size: 0.9rem;
  line-height: 1.7;
  margin-top: 2rem;
  color: #ffffff;
  padding: 0 2rem;
}

.privacy h3 {
```

```
      margin-bottom: 0.8rem;
      font-size: 1.2rem;
      font-weight: 600;
    }

    @media (max-width: 900px) {
      .contact-header {
        flex-direction: column;
        gap: 2rem;
      }

      .contact-intro h1 { font-size: 2.2rem; }
      .contact-intro p { font-size: 1rem; }

      .contact-container {
        flex-direction: column;
        align-items: center;
      }

      form { max-width: 100%; }
    }

    @media (max-width: 480px) {
      .contact-intro h1 { font-size: 1.9rem; }
      form { padding: 1.5rem; }
      .privacy { font-size: 0.85rem; }
      .privacy h3 { font-size: 1.1rem; }
    }
  </style>
</head>
<body>
```

```
<?php include 'navbar.php'; ?>

<section id="contact">
  <div class="contact-header" data-aos="fade-down">
    <div class="contact-intro">
      <h1>Contact BottleSort</h1>
      <p>We'd love to hear from you! Please select your inquiry type and send us a
message. Your feedback helps us improve and innovate our plastic waste management
solutions.</p>
    </div>
  </div>

  <div class="contact-container">
    <!-- REPLACE 'YOUR_FORM_ID' with your actual Formspree form ID -->
    <form method="POST" action="https://formspree.io/f/xblpgzdz"
data-aos="fade-up">
      <label for="name">Full Name</label>
      <input type="text" id="name" name="name" placeholder="Your name" required>

      <label for="email">Email Address</label>
      <input type="email" id="email" name="email" placeholder="your@email.com"
required>

      <label for="inquiry_type">Inquiry Type</label>
      <select id="inquiry_type" name="inquiry_type" required>
        <option value="">-- Select an Option --</option>
        <option value="Partnership or Collaboration">Partnership or
Collaboration</option>
        <option value="Technical Support">Technical Support</option>
        <option value="Feedback or Suggestions">Feedback or Suggestions</option>
        <option value="General Question">General Question</option>
        <option value="Other">Other (Please specify below)</option>
```

```
      </select>

      <div id="other_inquiry_container">
        <label for="other_inquiry">Please specify</label>
        <input type="text" id="other_inquiry" name="other_inquiry" placeholder="Your
inquiry type">
      </div>

      <label for="message">Message</label>
      <textarea id="message" name="message" placeholder="Write your message
here..." required></textarea>

      <button type="submit">Send Message</button>
    </form>
  </div>

  <div class="privacy" data-aos="fade-up">
    <h3>Privacy Notice</h3>
    <p>
      By submitting this form, you consent to BottleSort collecting and processing your
personal data
      (name, email, and message) solely for the purpose of addressing your inquiry.
      Your data will not be shared with third parties and will be deleted after your
request is resolved.
    </p>
  </div>
</section>

<?php include 'footer.php'; ?>

<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
```

```
  AOS.init({ duration: 1000, once: true });

  const inquirySelect = document.getElementById('inquiry_type');
  const otherContainer = document.getElementById('other_inquiry_container');
  const otherInput = document.getElementById('other_inquiry');

  inquirySelect.addEventListener('change', function() {
    if (this.value === 'Other') {
      otherContainer.style.display = 'block';
    } else {
      otherContainer.style.display = 'none';
      otherInput.value = '';
    }
  });
</script>

</body>
</html>
```

charts.php

```
<?php
// charts.php - Chart initialization and management
// This file contains all chart-related functionality
?>

<script>
  // charts.php - Chart initialization and management

  let bottleChart;
  let filteredLineChart = null;

  // Initialize Charts
```

```
function initCharts() {
  try {
    initBottleChart();
  } catch (error) {
    console.error('Error creating charts:', error);
  }
}

// Initialize bottle breakdown chart (Plastic Type Breakdown)
function initBottleChart() {
  try {
    const bottleCtx = document.getElementById('bottleChart');
    if (bottleCtx) {
      bottleChart = new Chart(bottleCtx.getContext('2d'), {
        type: 'doughnut',
        data: {
          labels: ['PET', 'HDPE', 'LDPE', 'PP', 'Unclassified'],
          datasets: [{
            data: [450, 320, 280, 150, 47],
            backgroundColor: [
              '#ffff00',  // Yellow for PET
              '#FFA500',  // Orange for HDPE
              '#008000',  // Green for LDPE
              '#808080',  // Gray for PP
              '#0000FF'   // Blue for Unclassified
            ],
            borderColor: '#fff',
            borderWidth: 3,
            hoverOffset: 15,
            hoverBackgroundColor: [
              '#ffff33',
              '#ffb84d',
```

```
      '#00b300',
      '#999999',
      '#3333ff'
    ]
  }]
},
options: {
  responsive: true,
  maintainAspectRatio: true,
  plugins: {
    legend: {
      position: 'bottom',
      labels: {
        padding: 15,
        font: {
          size: 13,
          weight: 600
        },
        color: '#004e44',
        usePointStyle: true,
        pointStyle: 'circle'
      }
    },
    tooltip: {
      backgroundColor: 'rgba(0, 0, 0, 0.8)',
      titleFont: {
        size: 14,
        weight: 700
      },
      bodyFont: {
        size: 13,
        weight: 600
```

```
            },
            padding: 12,
            cornerRadius: 8,
            displayColors: true,
            borderColor: 'rgba(0, 0, 0, 0.2)',
            borderWidth: 1,
            callbacks: {
              label: function(context) {
                const label = context.label || '';
                const value = context.parsed || 0;
                const total = context.dataset.data.reduce((a, b) => a + b, 0);
                const percentage = ((value / total) * 100).toFixed(1);
                return label + ': ' + value.toLocaleString() + ' bottles (' + percentage +
'%)';
              }
            }
          }
        }
      }
    });
  }
} catch (error) {
  console.error('Error creating bottle chart:', error);
}
}

// Update bottle breakdown chart with new time range
function updateBottleChart(range) {
  if (!bottleChart) return;

  // Generate random data based on time range
  const multiplier = range === 'today' ? 1 : (range === 'weekly' ? 7 : 30);
```

```
  // Base distribution percentages (approximate realistic ratios)
  const petBase = Math.floor((Math.random() * 100 + 350) * multiplier);
  const hdpeBase = Math.floor((Math.random() * 80 + 250) * multiplier);
  const ldpeBase = Math.floor((Math.random() * 70 + 220) * multiplier);
  const ppBase = Math.floor((Math.random() * 50 + 120) * multiplier);
  const unclassifiedBase = Math.floor((Math.random() * 20 + 30) * multiplier);

  const newData = [petBase, hdpeBase, ldpeBase, ppBase, unclassifiedBase];

  bottleChart.data.datasets[0].data = newData;
  bottleChart.update('active');
}

// ==================== FILTERED LINE CHART FUNCTIONS
====================

// Generate filtered data based on date range and bin type
function generateFilteredData(startDate, endDate, binType) {
  const start = new Date(startDate);
  const end = new Date(endDate);
  const daysDiff = Math.ceil((end - start) / (1000 * 60 * 60 * 24)) + 1;

  const labels = [];
  const datasets = [];

  // Generate date labels
  for (let i = 0; i < daysDiff; i++) {
    const date = new Date(start);
    date.setDate(start.getDate() + i);
    labels.push(date.toLocaleDateString('en-US', { month: 'short', day: 'numeric' }));
  }
```

```
// Get bins to display
const rows = document.querySelectorAll('tr[data-bin-id]');
const binsToShow = [];

rows.forEach(row => {
  const type = row.dataset.type;
  if (binType === 'all' || type === binType) {
    binsToShow.push({
      id: row.dataset.binId,
      type: type
    });
  }
});

// Color palette matching theme - shades of teal/green
const colors = [
  { border: 'rgba(0, 78, 68, 1)', bg: 'rgba(0, 78, 68, 0.1)' },        // Dark Teal
  { border: 'rgba(0, 122, 110, 1)', bg: 'rgba(0, 122, 110, 0.1)' },    // Medium Teal
  { border: 'rgba(0, 150, 136, 1)', bg: 'rgba(0, 150, 136, 0.1)' },    // Light Teal
  { border: 'rgba(72, 177, 191, 1)', bg: 'rgba(72, 177, 191, 0.1)' },  // Blue-Teal
  { border: 'rgba(46, 125, 50, 1)', bg: 'rgba(46, 125, 50, 0.1)' }     // Forest Green
];

// Generate data for each bin
binsToShow.forEach((bin, index) => {
  const data = [];
  let currentLevel = Math.floor(Math.random() * 30) + 20; // Start between 20-50%

  for (let i = 0; i < daysDiff; i++) {
    // Simulate gradual filling with occasional emptying
    if (currentLevel >= 95 && Math.random() > 0.7) {
      currentLevel = Math.floor(Math.random() * 20) + 5; // Empty to 5-25%
```

```
      } else {
        currentLevel = Math.min(100, currentLevel + Math.floor(Math.random() * 15)
+ 5);
      }
      data.push(currentLevel);
    }

    const color = colors[index % colors.length];
    datasets.push({
      label: `${bin.type} Bin (ID: ${bin.id})`,
      data: data,
      borderColor: color.border,
      backgroundColor: color.bg,
      borderWidth: 3,
      tension: 0.4,
      fill: true,
      pointRadius: 5,
      pointHoverRadius: 7,
      pointBackgroundColor: color.border,
      pointBorderColor: '#fff',
      pointBorderWidth: 2,
      pointHoverBackgroundColor: '#fff',
      pointHoverBorderColor: color.border,
      pointHoverBorderWidth: 3
    });
  });

  return { labels, datasets };
}

// Apply filter and display line chart
function applyFilter() {
```

```
    const startDate = document.getElementById('startDate').value;
    const endDate = document.getElementById('endDate').value;
    const binType = document.getElementById('binType').value;

    if (!startDate || !endDate) {
      alert('Please select both start and end dates.');
      return;
    }

    const start = new Date(startDate);
    const end = new Date(endDate);

    if (start > end) {
      alert('Start date must be before end date.');
      return;
    }

    // Generate and display chart
    const chartData = generateFilteredData(startDate, endDate, binType);

    // Update chart title
    const binTypeText = binType === 'all' ? 'All Bins' : `${binType} Bin`;
    const dateRange = `${start.toLocaleDateString('en-US', { month: 'short', day:
'numeric', year: 'numeric' })} - ${end.toLocaleDateString('en-US', { month: 'short',
day: 'numeric', year: 'numeric' })}`;
    document.getElementById('filteredChartTitle').textContent = `${binTypeText}
Level Trends (${dateRange})`;

    // Show chart container
    const chartContainer = document.getElementById('filteredChartContainer');
    chartContainer.classList.add('active');
```

```
// Destroy previous chart if exists
if (filteredLineChart) {
  filteredLineChart.destroy();
}

// Create new chart
const ctx = document.getElementById('filteredLineChart').getContext('2d');
filteredLineChart = new Chart(ctx, {
  type: 'line',
  data: chartData,
  options: {
    responsive: true,
    maintainAspectRatio: true,
    plugins: {
      legend: {
        display: true,
        position: 'top',
        labels: {
          usePointStyle: true,
          padding: 20,
          font: {
            size: 13,
            weight: '600',
            family: "'Open Sans', sans-serif"
          },
          color: '#004e44'
        }
      },
      tooltip: {
        mode: 'index',
        intersect: false,
        backgroundColor: 'rgba(0, 78, 68, 0.95)',
```

```
      titleFont: {
        size: 14,
        weight: 'bold',
        family: "'Open Sans', sans-serif"
      },
      bodyFont: {
        size: 13,
        family: "'Open Sans', sans-serif"
      },
      padding: 12,
      borderColor: 'rgba(0, 122, 110, 0.5)',
      borderWidth: 1,
      callbacks: {
        label: function(context) {
          return context.dataset.label + ': ' + context.parsed.y + '%';
        }
      }
    }
  },
  scales: {
    y: {
      beginAtZero: true,
      max: 100,
      ticks: {
        callback: function(value) {
          return value + '%';
        },
        font: {
          size: 12,
          family: "'Open Sans', sans-serif"
        },
        color: '#004e44'
```

```
    },
    grid: {
      color: 'rgba(0, 78, 68, 0.08)',
      drawBorder: false
    },
    title: {
      display: true,
      text: 'Fill Level (%)',
      font: {
        size: 14,
        weight: 'bold',
        family: "'Open Sans', sans-serif"
      },
      color: '#004e44',
      padding: { top: 10, bottom: 10 }
    }
  },
  x: {
    ticks: {
      font: {
        size: 11,
        family: "'Open Sans', sans-serif"
      },
      color: '#004e44',
      maxRotation: 45,
      minRotation: 45
    },
    grid: {
      color: 'rgba(0, 78, 68, 0.05)',
      drawBorder: false
    },
    title: {
```

```
            display: true,
            text: 'Date',
            font: {
              size: 14,
              weight: 'bold',
              family: "'Open Sans', sans-serif"
            },
            color: '#004e44',
            padding: { top: 10, bottom: 0 }
          }
        }
      },
      interaction: {
        mode: 'nearest',
        axis: 'x',
        intersect: false
      }
    }
  });

  // Scroll to chart
  setTimeout(() => {
    chartContainer.scrollIntoView({
      behavior: 'smooth',
      block: 'nearest'
    });
  }, 100);
}

// ==================== CHART CLEANUP ====================

// Destroy charts (if needed for cleanup)
```

```
function destroyCharts() {
  if (bottleChart) {
    bottleChart.destroy();
    bottleChart = null;
  }
  if (filteredLineChart) {
    filteredLineChart.destroy();
    filteredLineChart = null;
  }
}

// ==================== AUTO-UPDATE ====================

// Auto-update chart data every minute to reflect real-time changes
setInterval(() => {
  if (bottleChart && currentTimeRange) {
    // Slightly adjust the data to simulate real-time updates
    const currentData = bottleChart.data.datasets[0].data;
    const newData = currentData.map(value => {
      const change = Math.floor(Math.random() * 21) - 10; // Random change between
-10 and +10
      return Math.max(0, value + change); // Ensure non-negative
    });

    bottleChart.data.datasets[0].data = newData;
    bottleChart.update('none'); // Update without animation for smoother real-time
feel
  }
}, 60000); // Update every minute
</script>
```

analytics.php

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Analytics | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    .hero-header {
      min-height: 40vh;
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      text-align: center;
      padding: 7rem 2rem 5rem;
      color: #fff;
    }

    .hero-header h1 {
      font-size: 2.6rem;
      margin-bottom: 1rem;
      text-shadow: 0 2px 10px rgba(0,0,0,0.3);
    }
```

```
.hero-header p {
  max-width: 850px;
  line-height: 1.7;
  font-size: 1.1rem;
  color: #eafaf4;
}

.stats-card {
  background: rgba(255,255,255,0.12);
  border-radius: 12px;
  padding: 2rem;
  color: #fff;
  max-width: 900px;
  margin: 0.5rem auto;
  box-shadow: 0 8px 20px rgba(0,0,0,0.2);
  backdrop-filter: blur(10px);
  border: 1px solid rgba(255,255,255,0.15);
  display: grid;
  grid-template-columns: 1fr 2fr 1fr;
  gap: 1rem;
  align-items: center;
}

.chart-buttons {
  display: flex;
  flex-direction: column;
  gap: 0.75rem;
}

.chart-buttons button {
  padding: 0.8rem 1.5rem;
  border: none;
```

```
  border-radius: 30px;
  cursor: pointer;
  font-weight: 600;
  background: rgba(255,255,255,0.12);
  color: #fff;
  transition: all 0.2s ease;
  width: 100%;
}

.chart-buttons button:hover {
  background: rgba(255,255,255,0.4);
  transform: scale(1.05);
}

.chart-buttons button.active {
  background: #fff;
  color: #004e44;
}

.stat-box {
  text-align: center;
}

.stat-box h3 {
  font-weight: 600;
  margin-bottom: 0.5rem;
  font-size: 1.2rem;
}

.stat-box p {
  font-size: 3rem;
  font-weight: bold;
```

```
  margin: 0.25rem 0;
}

.date-range {
  text-align: center;
  min-width: 220px;
}

.date-range h4 {
  font-size: 1rem;
  font-weight: 600;
  margin-bottom: 0.5rem;
  opacity: 0.8;
}

.date-range p {
  font-size: 1.1rem;
  line-height: 1.6;
  white-space: nowrap;
}

.chart-card, .bin-status-card {
  background: #fff !important;
  border-radius: 12px;
  padding: 2.5rem;
  color: #004e44;
  max-width: 900px;
  margin: 2rem auto;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
}

.bin-status-card {
```

```
  margin-bottom: 4rem;
}

.bin-status-card h3, .chart-card h3 {
  text-align: center;
  margin-bottom: 1.5rem;
}

.table-wrapper {
  overflow-x: auto;
  -webkit-overflow-scrolling: touch;
}

#plasticChart {
  background: #fff;
  border-radius: 8px;
}

.bin-status-card table {
  width: 100%;
  border-collapse: collapse;
  text-align: center;
  min-width: 600px;
}

.bin-status-card th, .bin-status-card td {
  padding: 0.75rem;
  border-bottom: 1px solid rgba(0,78,68,0.2);
}

.bin-status-card th {
  background: rgba(0,78,68,0.1);
```

```
  font-weight: 600;
}

.bin-status-card tr:hover {
  background: rgba(0,78,68,0.05);
}

.status-empty { color: #2E8B57; font-weight: bold; }
.status-half { color: #DAA520; font-weight: bold; }
.status-full { color: #C0392B; font-weight: bold; }

.loading {
  text-align: center;
  padding: 2rem;
  font-style: italic;
  color: #666;
}

@media (max-width: 900px) {
  .hero-header h1 { font-size: 2.2rem; }
  .hero-header p { font-size: 1rem; }
  .stats-card, .chart-card, .bin-status-card {
    margin: 2rem 1rem;
    padding: 2rem 1rem;
  }
  .stats-card {
    grid-template-columns: 1fr;
    gap: 1.5rem;
  }
  .chart-buttons {
    width: 100%;
    max-width: 300px;
```

```
        margin: 0 auto;
      }
    }
  </style>
</head>
<body>
  <?php include 'navbar.php'; ?>

  <section class="hero-header" data-aos="fade-down">
    <h1>Plastic Bottle Classification Analytics</h1>
    <p>Track the number of plastic bottles thrown and see their type breakdown
below.</p>
  </section>

  <section class="stats-card" data-aos="fade-up">
    <div class="chart-buttons">
      <button class="active" onclick="updateChart('today', this)">Today</button>
      <button onclick="updateChart('week', this)">This Week</button>
      <button onclick="updateChart('month', this)">This Month</button>
    </div>

    <div class="stat-box">
      <h3>Recycled</h3>
      <p id="recycledCount">-</p>
      <small id="recycledPeriod">Today</small>
    </div>

    <div class="date-range">
      <h4>Date Range</h4>
      <p id="dateRangeText">Loading...</p>
    </div>
  </section>
```

```
<section class="chart-card" data-aos="fade-up">
  <h3>Plastic Type Breakdown</h3>
  <canvas id="plasticChart" width="600" height="350"></canvas>
</section>

<section class="bin-status-card" data-aos="fade-up">
  <h3>Bin Status</h3>
  <div class="table-wrapper">
    <table>
      <thead>
        <tr>
          <th>Plastic Type</th>
          <th>Fill Level (%)</th>
          <th>Status</th>
          <th>Last Updated</th>
        </tr>
      </thead>
      <tbody id="binStatusTable">
        <tr><td colspan="4" class="loading">Loading bin status...</td></tr>
      </tbody>
    </table>
  </div>
</section>

<?php include 'footer.php'; ?>

<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });

  let currentPeriod = 'today';
  let binLevelsData = {};
```

```
let lastUpdatedTime = '';

const ctx = document.getElementById('plasticChart').getContext('2d');

// Plugin to set white background
const whiteBackgroundPlugin = {
  id: 'whiteBackground',
  beforeDraw: (chart) => {
    const ctx = chart.ctx;
    ctx.save();
    ctx.fillStyle = 'white';
    ctx.fillRect(0, 0, chart.width, chart.height);
    ctx.restore();
  }
};

const plasticChart = new Chart(ctx, {
  type: 'bar',
  data: {
    labels: ['PET', 'HDPE', 'LDPE', 'PP', 'Unclassified'],
    datasets: [{
      data: [0, 0, 0, 0, 0],
      backgroundColor: ['#ffff00', '#FFA500', '#008000', '#808080', '#0000FF']
    }]
  },
  options: {
    responsive: true,
    plugins: {
      legend: { display: false }
    },
    scales: {
      y: {
```

```
        beginAtZero: true,
        grid: { color: 'rgba(0,0,0,0.1)' },
        ticks: {
          color: '#333',
          stepSize: 1
        }
      },
      x: {
        grid: { color: 'rgba(0,0,0,0.05)' },
        ticks: { color: '#333' }
      }
    },
    layout: {
      padding: 10
    }
  },
  plugins: [whiteBackgroundPlugin]
});

async function fetchAnalyticsData(period) {
  try {
    const response = await fetch(`get-analytics-data.php?period=${period}`);
    const data = await response.json();

    if (data.error) {
      console.error('Error:', data.error);
      return;
    }

    // Update chart
    plasticChart.data.datasets[0].data = data.bottleData;
    plasticChart.update();
```

```
    // Update total count
    document.getElementById('recycledCount').textContent = data.totalCount;

    // Store bin levels data
    binLevelsData = data.binLevels;
    lastUpdatedTime = data.lastUpdated;

    // Update bin table
    renderBinTable();

  } catch (error) {
    console.error('Fetch error:', error);
    document.getElementById('recycledCount').textContent = 'Error';
  }
}

function updateChart(period, button) {
  currentPeriod = period;

  document.getElementById('recycledPeriod').textContent =
    period === 'today' ? 'Today' : period === 'week' ? 'This Week' : 'This Month';

  updateDateRange(period);

  document.querySelectorAll('.chart-buttons button').forEach(btn =>
    btn.classList.remove('active')
  );
  button.classList.add('active');

  fetchAnalyticsData(period);
}
```

```
function updateDateRange(period) {
  const today = new Date();
  let dateText = '';

  if (period === 'today') {
    dateText = today.toLocaleDateString('en-US', { month: 'long', day: 'numeric',
year: 'numeric' });
  } else if (period === 'week') {
    const weekStart = new Date(today);
    weekStart.setDate(today.getDate() - 6);
    dateText = `${weekStart.toLocaleDateString('en-US', { month: 'short', day:
'numeric' })} - ${today.toLocaleDateString('en-US', { month: 'short', day: 'numeric',
year: 'numeric' })}`;
  } else {
    const monthStart = new Date(today);
    monthStart.setDate(today.getDate() - 29);
    dateText = `${monthStart.toLocaleDateString('en-US', { month: 'short', day:
'numeric' })} - ${today.toLocaleDateString('en-US', { month: 'short', day: 'numeric',
year: 'numeric' })}`;
  }

  document.getElementById('dateRangeText').textContent = dateText;
}

function getStatus(level) {
  if (level < 30) return '<span class="status-empty">Empty</span>';
  if (level < 90) return '<span class="status-half">Half-Full</span>';
  return '<span class="status-full">Full</span>';
}

function renderBinTable() {
  const binTypes = ['PET', 'HDPE', 'LDPE', 'PP', 'Unclassified'];
```

```
    document.getElementById('binStatusTable').innerHTML = binTypes.map(type =>
{
      const level = binLevelsData[type] || 0;
      return `
        <tr>
          <td>${type}</td>
          <td>${level}%</td>
          <td>${getStatus(level)}</td>
          <td>${lastUpdatedTime}</td>
        </tr>
      `;
    }).join('');
  }

  // Initial load
  updateDateRange('today');
  fetchAnalyticsData('today');

  // Auto-refresh every 30 seconds
  setInterval(() => {
    fetchAnalyticsData(currentPeriod);
  }, 30000);
 </script>
</body>
</html>
```

admin-dashboard.php

```
<?php // admin-dashboard.php
require_once 'auth-check.php';

$admin_username = isset($_SESSION['admin_username']) ?
$_SESSION['admin_username'] : 'Admin';
```

```
// Bin Simulation
$bins = [
  ['id'=>1,'type'=>'PET','level'=>95],
  ['id'=>2,'type'=>'HDPE','level'=>50],
  ['id'=>3,'type'=>'LDPE','level'=>90],
  ['id'=>4,'type'=>'PP','level'=>30],
  ['id'=>5,'type'=>'Unclassified','level'=>100]
];

// Calculate stats
$total_bins = count($bins);
$full_bins = array_filter($bins, function($bin) {
  return $bin['level'] >= 95;
});
$full_bins_count = count($full_bins);
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Admin Dashboard | BottleSort</title>
  <link rel="icon" type="image/svg+xml" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">

  <link rel="stylesheet" href="css/style.css">

  <style>
    /* ---------------------- DASHBOARD SPECIFIC STYLES ---------------------- */
```

```
/* These styles extend style.css for dashboard-specific components */

.dashboard-container {
  flex: 1;
  padding: 2rem 6%;
  max-width: 1400px;
  margin: 0 auto;
  width: 100%;
  position: relative;
  z-index: 1;
}

/* Dashboard Header */
.dashboard-header {
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  margin-bottom: 2.5rem;
  gap: 0.5rem;
  text-align: center;
}
.dashboard-title {
  color: white;
  font-size: 1.8rem;
  font-weight: 700;
  text-shadow: 0 2px 4px rgba(0,0,0,0.2);
}
.dashboard-welcome {
  color: white;
  font-size: 3.5rem;
  font-weight: 800;
```

```
  text-shadow: 0 3px 6px rgba(0,0,0,0.3);
  display: flex;
  align-items: center;
  gap: 0.5rem;
  margin-top: 1rem;
}

/* Statistics Cards - Side by Side */
.stats-grid {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 1.5rem;
  margin-bottom: 2.5rem;
}
.stat-card {
  background: rgba(255, 255, 255, 0.12);
  color: #ffffff;
  padding: 2.2rem;
  border-radius: 12px;
  backdrop-filter: blur(10px);
  transition: all 0.3s ease;
  box-shadow: 0 4px 15px rgba(0,0,0,0.2);
  border: 1px solid rgba(255,255,255,0.15);
  text-align: center;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  cursor: pointer;
}
.stat-card:hover {
  transform: translateY(-5px);
```

```
  background: rgba(255,255,255,0.22);
  box-shadow: 0 6px 20px rgba(0,0,0,0.3);
}
.stat-card:active {
  transform: scale(0.98);
}
.stat-card h3 {
  color: #ffffff;
  font-size: 0.95rem;
  margin-bottom: 0.8rem;
  text-transform: uppercase;
  font-weight: 600;
  letter-spacing: 0.5px;
}
.stat-card .stat-value {
  font-size: 4rem;
  font-weight: 800;
  color: #ffffff;
  margin: 0.5rem 0;
  line-height: 1;
  text-shadow: 0 2px 4px rgba(0,0,0,0.3);
  display: flex;
  justify-content: center;
  align-items: center;
}
.stat-card .stat-label {
  font-size: 1rem;
  color: #d9fff2;
  margin-top: 0.8rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.5px;
```

```
}
.stat-card .stat-date {
  font-size: 0.85rem;
  color: #d9fff2;
  margin-top: 0.5rem;
  font-style: italic;
}

/* Time Range Selector */
.time-range-selector {
  display: flex;
  gap: 0.5rem;
  justify-content: center;
  margin-top: 1rem;
}
.time-range-btn {
  padding: 0.4rem 1rem;
  background: rgba(255, 255, 255, 0.2);
  border: 1px solid rgba(255, 255, 255, 0.3);
  border-radius: 6px;
  cursor: pointer;
  font-size: 0.85rem;
  font-weight: 600;
  transition: all 0.3s ease;
  color: #ffffff;
}
.time-range-btn.active {
  background: rgba(255, 255, 255, 0.4);
  color: #004e44;
  font-weight: 700;
}
.time-range-btn:hover {
```

```
  background: rgba(255, 255, 255, 0.35);
}

/* Charts Section */
.charts-section {
  display: grid;
  grid-template-columns: 1fr;
  gap: 2rem;
  margin-bottom: 2.5rem;
}
.chart-container {
  background: rgba(255, 255, 255, 0.95);
  padding: 2rem;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.08);
  border: 1px solid rgba(0, 78, 68, 0.1);
}
.chart-container h3 {
  color: #004e44;
  margin-bottom: 1.5rem;
  text-align: center;
  font-size: 1.2rem;
  font-weight: 700;
}
.chart-container canvas {
  max-height: 350px;
}

/* Filter Section */
.filter-section {
  background: rgba(255, 255, 255, 0.95);
  padding: 2rem;
```

```
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.08);
  margin-bottom: 2.5rem;
  border: 1px solid rgba(0, 78, 68, 0.1);
}
.filter-section h2 {
  color: #004e44;
  text-align: center;
  margin-bottom: 1.5rem;
  font-size: 1.4rem;
  font-weight: 700;
}
.filter-controls {
  display: flex;
  flex-wrap: wrap;
  gap: 1rem;
  align-items: flex-end;
}
.filter-group {
  flex: 1;
  min-width: 200px;
}
.filter-group label {
  display: block;
  color: #004e44;
  font-weight: 600;
  margin-bottom: 0.5rem;
  font-size: 0.9rem;
}
.filter-group input, .filter-group select {
  width: 100%;
  padding: 0.75rem;
```

```
  border: 2px solid #e0e0e0;
  border-radius: 8px;
  font-family: 'Open Sans', sans-serif;
  font-size: 1rem;
  transition: all 0.3s ease;
}
.filter-group input:focus, .filter-group select:focus {
  outline: none;
  border-color: #004e44;
  box-shadow: 0 0 0 3px rgba(0, 78, 68, 0.1);
}

/* Filtered Chart Container */
.filtered-chart-container {
  margin-top: 2rem;
  padding-top: 2rem;
  border-top: 2px solid rgba(0, 78, 68, 0.1);
  display: none;
  opacity: 0;
  transform: translateY(20px);
  transition: all 0.5s ease;
}
.filtered-chart-container.active {
  display: block;
  opacity: 1;
  transform: translateY(0);
}
.filtered-chart-container h3 {
  color: #004e44;
  margin-bottom: 1.5rem;
  text-align: center;
  font-size: 1.2rem;
```

```
  font-weight: 700;
}
.filtered-chart-container canvas {
  max-height: 500px;
  min-height: 400px;
}

/* Buttons */
.btn {
  padding: 0.75rem 1.8rem;
  background: linear-gradient(135deg, #004e44 0%, #007a6e 100%);
  color: white;
  border: none;
  border-radius: 8px;
  cursor: pointer;
  font-weight: 600;
  font-family: 'Open Sans', sans-serif;
  transition: all 0.3s ease;
  font-size: 1rem;
  box-shadow: 0 2px 8px rgba(0, 78, 68, 0.2);
}
.btn:hover {
  background: linear-gradient(135deg, #007a6e 0%, #009688 100%);
  transform: translateY(-2px);
  box-shadow: 0 4px 12px rgba(0, 78, 68, 0.3);
}
.btn-secondary {
  background: linear-gradient(135deg, #48B1BF 0%, #3a8f9c 100%);
}
.btn-secondary:hover {
  background: linear-gradient(135deg, #3a8f9c 0%, #2d7380 100%);
}
```

```
.btn-small {
  padding: 0.5rem 1rem;
  font-size: 0.85rem;
}
.btn-small:disabled {
  opacity: 0.5;
  cursor: not-allowed;
}
.btn-small:disabled:hover {
  transform: none;
}
.btn-signout {
  background: linear-gradient(135deg, #e74c3c 0%, #c0392b 100%) !important;
}
.btn-signout:hover {
  background: linear-gradient(135deg, #c0392b 0%, #a93226 100%) !important;
}

/* Table Section */
.table-section {
  background: rgba(255, 255, 255, 0.95);
  padding: 2rem;
  border-radius: 12px;
  box-shadow: 0 4px 15px rgba(0,0,0,0.08);
  margin-bottom: 2.5rem;
  overflow-x: auto;
  border: 1px solid rgba(0, 78, 68, 0.1);
}
.table-section h2 {
  color: #004e44;
  text-align: center;
  margin-bottom: 1.5rem;
```

```
  font-size: 1.4rem;
  font-weight: 700;
}
table {
  width: 100%;
  border-collapse: collapse;
  background: white;
  border-radius: 8px;
  overflow: hidden;
}
th, td {
  padding: 1rem;
  text-align: center;
  border-bottom: 1px solid #e0e0e0;
}
th {
  background: linear-gradient(135deg, #004e44 0%, #007a6e 100%);
  color: white;
  font-weight: 600;
  text-transform: uppercase;
  font-size: 0.85rem;
  letter-spacing: 0.5px;
}
tbody tr {
  transition: background 0.2s ease;
}
tbody tr:hover {
  background: #f0f8f7;
}
tbody tr:last-child td {
  border-bottom: none;
}
```

```
.status-full {
  color: #e74c3c;
  font-weight: bold;
}
.status-ok {
  color: #2ecc71;
  font-weight: bold;
}
.status-warning {
  color: #f39c12;
  font-weight: bold;
}
.progress-container {
  background: #eee;
  border-radius: 12px;
  overflow: hidden;
  height: 26px;
  width: 100%;
  position: relative;
}
.progress-bar {
  height: 100%;
  display: flex;
  align-items: center;
  justify-content: center;
  color: white;
  font-weight: bold;
  font-size: 0.85rem;
  transition: width 1s ease;
  border-radius: 12px;
}
.action-buttons {
```

```
  display: flex;
  gap: 0.5rem;
  justify-content: center;
}

/* Modal Styles */
.modal {
  display: none;
  position: fixed;
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
  background: rgba(0,0,0,0.6);
  z-index: 2000;
  animation: fadeIn 0.3s ease;
}
.modal.active {
  display: flex;
  justify-content: center;
  align-items: center;
}
.modal-content {
  background: white;
  padding: 2.5rem;
  border-radius: 16px;
  max-width: 500px;
  width: 90%;
  box-shadow: 0 10px 40px rgba(0,0,0,0.3);
  animation: slideUp 0.3s ease;
}
.modal-content h2 {
```

```
  color: #004e44;
  margin-bottom: 1.5rem;
  font-size: 1.6rem;
  text-align: center;
}
.modal-content .detail-row {
  display: flex;
  justify-content: space-between;
  padding: 0.8rem 0;
  border-bottom: 1px solid #e0e0e0;
}
.modal-content .detail-row:last-child {
  border-bottom: none;
}
.modal-content .detail-label {
  font-weight: 600;
  color: #004e44;
}
.modal-content .detail-value {
  color: #666;
}

/* Notification Banner */
#popup {
  position: fixed;
  top: 80px;
  right: 20px;
  background: rgba(255, 255, 255, 0.95);
  padding: 1.2rem 1.5rem;
  border-radius: 12px;
  box-shadow: 0 4px 20px rgba(0, 78, 68, 0.15);
  text-align: left;
```

```
  z-index: 3000;
  max-width: 400px;
  width: 90%;
  display: none;
  border-left: 4px solid #007a6e;
  animation: slideInRight 0.4s ease forwards;
}
#popup h2 {
  color: #004e44;
  margin-bottom: 0.8rem;
  font-size: 1.1rem;
  font-weight: 700;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}
#popup p {
  color: #666;
  margin-bottom: 0.8rem;
  font-size: 0.9rem;
}
#popup ul {
  list-style: none;
  margin: 0.5rem 0 1rem 0;
  padding: 0;
}
#popup ul li {
  margin: 0.4rem 0;
  font-weight: 600;
  color: #007a6e;
  padding: 0.5rem;
  background: rgba(0, 122, 110, 0.05);
```

```
  border-radius: 6px;
  font-size: 0.9rem;
}
#popup button {
  width: 100%;
  padding: 0.6rem;
  background: #004e44;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
  font-weight: 600;
  font-size: 0.9rem;
  transition: all 0.3s ease;
}
#popup button:hover {
  background: #007a6e;
}

/* Animations */
@keyframes fadeIn {
  from { opacity: 0; }
  to { opacity: 1; }
}
@keyframes slideUp {
  from { transform: translateY(50px); opacity: 0; }
  to { transform: translateY(0); opacity: 1; }
}
@keyframes slideInRight {
  from { opacity: 0; transform: translateX(100px); }
  to { opacity: 1; transform: translateX(0); }
}
```

```
    /* ---------------------- RESPONSIVE ---------------------- */
    @media (max-width: 1100px) {
      .stats-grid {
        grid-template-columns: 1fr;
      }
    }

    @media (max-width: 768px) {
      .dashboard-container { padding: 2rem 4%; }
      .dashboard-title { font-size: 1.4rem; }
      .dashboard-welcome { font-size: 2.5rem; }
      .stat-card .stat-value { font-size: 3rem; }
      .filter-controls { flex-direction: column; }
      .filter-group { width: 100%; }
      .stats-grid { grid-template-columns: 1fr; }
      table { font-size: 0.85rem; }
      th, td { padding: 0.7rem 0.4rem; }
      .action-buttons { flex-direction: column; }
      .time-range-selector { flex-wrap: wrap; }
      #popup { top: 70px; right: 10px; left: 10px; width: auto; }
    }
  </style>
</head>
<body>

  <?php include 'navbar-admin.php'; ?>

  <main>
    <div class="dashboard-container">
      <div class="dashboard-header">
        <h1 class="dashboard-title">Admin Dashboard</h1>
        <div class="dashboard-welcome">
```

```
        Welcome, <strong><?= htmlspecialchars($admin_username) ?>!</strong>
      </div>
    </div>

    <div class="stats-grid">
      <div class="stat-card" onclick="scrollToBinMonitoring()" title="Click to view
bin monitoring">
        <h3>Full Bins</h3>
        <div class="stat-value" id="fullBinsCount"><?= $full_bins_count ?></div>
        <div class="stat-label">Needs Attention</div>
      </div>

      <div class="stat-card" onclick="scrollToPlasticBreakdown()" title="Click to
view plastic breakdown">
        <h3>Total Bottles</h3>
        <div class="stat-value" id="totalBottles">1,247</div>
        <div class="stat-label" id="bottleTimeRange">Collected Today</div>
        <div class="stat-date" id="bottleTimeDate"></div>
        <div class="time-range-selector">
          <button class="time-range-btn active" onclick="changeTimeRange('today');
event.stopPropagation();">Today</button>
          <button class="time-range-btn" onclick="changeTimeRange('weekly');
event.stopPropagation();">Weekly</button>
          <button class="time-range-btn" onclick="changeTimeRange('monthly');
event.stopPropagation();">Monthly</button>
        </div>
      </div>
    </div>

    <div class="charts-section">
      <div class="chart-container" id="plasticBreakdownSection">
        <h3>Plastic Type Breakdown</h3>
```

```
        <canvas id="bottleChart"></canvas>
      </div>
    </div>

    <div class="table-section" id="binMonitoringSection">
      <h2>Real-Time Bin Monitoring</h2>
      <table>
        <thead>
          <tr>
            <th>Bin ID</th>
            <th>Type</th>
            <th>Level (%)</th>
            <th>Status</th>
            <th>Last Updated</th>
            <th>Actions</th>
          </tr>
        </thead>
        <tbody id="binTableBody">
          <?php foreach($bins as $bin):
            $statusClass = ($bin['level'] >= 95) ? 'status-full' : (($bin['level'] >= 70) ?
'status-warning' : 'status-ok');
            $statusText = ($bin['level'] >= 95) ? 'FULL' : (($bin['level'] >= 70) ? 'HIGH' :
'OK');
            $progressColor = ($bin['level'] >= 95) ? '#e74c3c' : (($bin['level'] >= 70) ?
'#f39c12' : '#2ecc71');
          ?>
          <tr data-bin-id="<?= $bin['id'] ?>" data-level="<?= $bin['level'] ?>"
data-type="<?= $bin['type'] ?>" data-updated="">
            <td><?= $bin['id'] ?></td>
            <td><strong><?= $bin['type'] ?></strong></td>
            <td>
              <div class="progress-container">
```

```
            <div class="progress-bar" style="background:<?= $progressColor ?>;
width:<?= $bin['level'] ?>%;"><?= $bin['level'] ?>%</div>
          </div>
        </td>
        <td class="<?= $statusClass ?>"><?= $statusText ?></td>
        <td class="timestamp">Loading...</td>
        <td>
          <div class="action-buttons">
            <button class="btn btn-small" onclick="viewDetails(<?= $bin['id']
?>)">Details</button>
            <button class="btn btn-small btn-secondary" onclick="emptyBin(<?=
$bin['id'] ?>)" id="empty-btn-<?= $bin['id'] ?>">Empty</button>
          </div>
        </td>
      </tr>
      <?php endforeach; ?>
    </tbody>
  </table>
</div>

<div class="filter-section">
  <h2>Data Filter & Export</h2>
  <div class="filter-controls">
    <div class="filter-group">
      <label for="startDate">Start Date</label>
      <input type="date" id="startDate">
    </div>
    <div class="filter-group">
      <label for="endDate">End Date</label>
      <input type="date" id="endDate">
    </div>
    <div class="filter-group">
```

```
          <label for="binType">Bin Type</label>
          <select id="binType">
            <option value="all">All Types</option>
            <option value="PET">PET</option>
            <option value="HDPE">HDPE</option>
            <option value="LDPE">LDPE</option>
            <option value="PP">PP</option>
            <option value="Unclassified">Unclassified</option>
          </select>
        </div>
        <button class="btn" onclick="applyFilter()">Apply Filter</button>
        <button class="btn btn-secondary" onclick="exportCSV()">Export
CSV</button>
      </div>

      <div class="filtered-chart-container" id="filteredChartContainer">
        <h3 id="filteredChartTitle">Bin Level Trends</h3>
        <canvas id="filteredLineChart"></canvas>
      </div>
    </div>
  </div>
 </main>

 <div id="detailsModal" class="modal">
  <div class="modal-content">
    <h2>Bin Details</h2>
    <div id="modalDetails"></div>
    <button class="btn" style="margin-top: 1.5rem; width: 100%;"
onclick="closeModal()">Close</button>
  </div>
 </div>
```

```
  <div id="popup">
    <h2>🔔 BIN ALERT!</h2>
    <p>The following bins require attention:</p>
    <ul id="popup-list"></ul>
    <button onclick="closePopup()">Dismiss</button>
  </div>

  <div id="signOutModal" class="modal">
    <div class="modal-content">
      <h2>Sign Out Confirmation</h2>
      <p>Are you sure you want to sign out? You will need to log in again to access the
admin dashboard.</p>
      <div style="display: flex; gap: 1rem; margin-top: 2rem; justify-content: center;">
        <button class="btn btn-signout" onclick="confirmSignOut()" style="width:
200px;">Yes, Sign Out</button>
        <button class="btn" onclick="closeSignOutModal()" style="width:
200px;">Cancel</button>
      </div>
    </div>
  </div>

  <?php include 'footer.php'; ?>

  <script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<script>
// Initialize AOS
if (typeof AOS !== 'undefined') {
  AOS.init({ duration: 800, once: true });
}
```

```
// Global variables
let currentTimeRange = 'today';
let updateInterval;
let bottleChartInstance;
let filteredChartInstance;

// ============ FETCH BIN DATA ============
async function fetchBinData() {
  try {
    const response = await fetch('get-dashboard-data.php?action=getBinData');
    const data = await response.json();

    if (data.error) {
      console.error('Error fetching bin data:', data.error);
      return;
    }

    updateBinTable(data.bins);
    document.getElementById('fullBinsCount').textContent = data.fullBinsCount;

    // Show popup if there are full bins
    if (data.fullBinsCount > 0) {
      showFullBinPopup(data.bins.filter(bin => bin.level >= 95));
    } else {
      closePopup();
    }

  } catch (error) {
    console.error('Error fetching bin data:', error);
  }
}
```

```
// ============ UPDATE BIN TABLE ============
function updateBinTable(bins) {
  const tbody = document.getElementById('binTableBody');
  tbody.innerHTML = '';

  bins.forEach(bin => {
    const row = document.createElement('tr');
    row.dataset.binId = bin.id;
    row.dataset.level = bin.level;
    row.dataset.type = bin.type;
    row.dataset.updated = bin.lastUpdated;

    row.innerHTML = `
      <td>${bin.id}</td>
      <td><strong>${bin.type}</strong></td>
      <td>
        <div class="progress-container">
          <div class="progress-bar" style="background:${bin.progressColor};
width:${bin.level}%;">${bin.level}%</div>
        </div>
      </td>
      <td class="${bin.statusClass}">${bin.status}</td>
      <td class="timestamp">${bin.lastUpdated}</td>
      <td>
        <div class="action-buttons">
          <button class="btn btn-small"
onclick="viewDetails(${bin.id})">Details</button>
          <button class="btn btn-small btn-secondary"
                onclick="emptyBin(${bin.id})"
                id="empty-btn-${bin.id}"
                ${bin.level < 95 ? 'disabled title="Bin must be full to empty"' : ''}>
            Empty
```

```
        </button>
      </div>
    </td>
  `;

  tbody.appendChild(row);
 });
}

// ============ FETCH BOTTLE STATISTICS ============
async function fetchBottleStats(period = 'today') {
 try {
  const response = await
fetch(`get-dashboard-data.php?action=getBottleStats&period=${period}`);
  const data = await response.json();

  if (data.error) {
    console.error('Error fetching bottle stats:', data.error);
    return;
  }

  // Update total bottles count
  document.getElementById('totalBottles').textContent =
data.totalCount.toLocaleString();

  // Update bottle chart
  updateBottleChartWithData(data.bottleData);

 } catch (error) {
  console.error('Error fetching bottle stats:', error);
 }
}
```

```
// ============ UPDATE BOTTLE CHART ============
function updateBottleChartWithData(bottleData) {
  if (bottleChartInstance) {
    bottleChartInstance.data.datasets[0].data = Object.values(bottleData);
    bottleChartInstance.update();
  }
}

// ============ INITIALIZE BOTTLE CHART ============
function initBottleChart() {
  const ctx = document.getElementById('bottleChart');
  if (!ctx) return;

  bottleChartInstance = new Chart(ctx, {
    type: 'doughnut',
    data: {
      labels: ['PET', 'HDPE', 'LDPE', 'PP', 'Unclassified'],
      datasets: [{
        data: [0, 0, 0, 0, 0],
        backgroundColor: [
          '#3498db',
          '#e74c3c',
          '#2ecc71',
          '#f39c12',
          '#95a5a6'
        ],
        borderWidth: 2,
        borderColor: '#fff'
      }]
    },
    options: {
      responsive: true,
```

```
      maintainAspectRatio: true,
      plugins: {
        legend: {
          position: 'bottom',
          labels: {
            padding: 15,
            font: { size: 12, weight: 'bold' }
          }
        },
        tooltip: {
          callbacks: {
            label: function(context) {
              const label = context.label || '';
              const value = context.parsed || 0;
              const total = context.dataset.data.reduce((a, b) => a + b, 0);
              const percentage = total > 0 ? ((value / total) * 100).toFixed(1) : 0;
              return `${label}: ${value} bottles (${percentage}%)`;
            }
          }
        }
      }
    }
  });
}

// ============ TIME RANGE FUNCTIONS ============
function getDateDisplay(range) {
  const now = new Date();
  const options = { year: 'numeric', month: 'long', day: 'numeric' };
  if (range === 'today') {
    return now.toLocaleDateString('en-US', options);
  } else if (range === 'weekly') {
```

```
    const weekStart = new Date(now);
    weekStart.setDate(now.getDate() - 6);
    return `${weekStart.toLocaleDateString('en-US', { month: 'short', day: 'numeric' })}
- ${now.toLocaleDateString('en-US', { month: 'short', day: 'numeric', year: 'numeric'
})}`;
  } else if (range === 'monthly') {
    const monthStart = new Date(now);
    monthStart.setDate(now.getDate() - 29);
    return `${monthStart.toLocaleDateString('en-US', { month: 'short', day: 'numeric'
})} - ${now.toLocaleDateString('en-US', { month: 'short', day: 'numeric', year:
'numeric' })}`;
  }
  return '';
}

function changeTimeRange(range) {
  currentTimeRange = range;
  document.querySelectorAll('.time-range-btn').forEach(btn =>
btn.classList.remove('active'));
  event.target.classList.add('active');

  const labels = {
    today: 'Collected Today',
    weekly: 'Collected This Week',
    monthly: 'Collected This Month'
  };

  document.getElementById('bottleTimeRange').textContent = labels[range];
  document.getElementById('bottleTimeDate').textContent = getDateDisplay(range);

  fetchBottleStats(range);
}
```

```
// ============ FILTER & EXPORT ============
async function applyFilter() {
  const startDate = document.getElementById('startDate').value;
  const endDate = document.getElementById('endDate').value;
  const binType = document.getElementById('binType').value;

  if (!startDate || !endDate) {
    alert('Please select both start and end dates.');
    return;
  }

  if (new Date(startDate) > new Date(endDate)) {
    alert('Start date must be before end date.');
    return;
  }

  try {
    const response = await fetch(

`get-dashboard-data.php?action=getFilteredData&startDate=${startDate}&endDate=
${endDate}&binType=${binType}`
    );
    const data = await response.json();

    if (data.error) {
      console.error('Error:', data.error);
      alert('Error fetching filtered data.');
      return;
    }

    displayFilteredChart(data);
```

```
  } catch (error) {
    console.error('Error applying filter:', error);
    alert('Error applying filter.');
  }
}

function displayFilteredChart(data) {
  const container = document.getElementById('filteredChartContainer');
  container.classList.add('active');

  const title = document.getElementById('filteredChartTitle');
  const typeLabel = data.binType === 'all' ? 'All Types' : data.binType;
  title.textContent = `Bottle Collection Trends: ${typeLabel} (${data.startDate} to
${data.endDate})`;

  const ctx = document.getElementById('filteredLineChart');

  if (filteredChartInstance) {
    filteredChartInstance.destroy();
  }

  const datasets = [];
  const colors = {
    'PET': '#3498db',
    'HDPE': '#e74c3c',
    'LDPE': '#2ecc71',
    'PP': '#f39c12',
    'Unclassified': '#95a5a6'
  };

  for (const [type, values] of Object.entries(data.chartData)) {
    datasets.push({
```

```
    label: type,
    data: Object.values(values),
    borderColor: colors[type] || '#333',
    backgroundColor: colors[type] || '#333',
    borderWidth: 2,
    fill: false,
    tension: 0.1
  });
}

filteredChartInstance = new Chart(ctx, {
  type: 'line',
  data: {
    labels: data.dates,
    datasets: datasets
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
      legend: {
        position: 'top'
      },
      tooltip: {
        mode: 'index',
        intersect: false
      }
    },
    scales: {
      y: {
        beginAtZero: true,
        title: {
```

```
            display: true,
            text: 'Number of Bottles'
          }
        },
        x: {
          title: {
            display: true,
            text: 'Date'
          }
        }
      }
    }
  });
}

async function exportCSV() {
  const startDate = document.getElementById('startDate').value;
  const endDate = document.getElementById('endDate').value;
  const binType = document.getElementById('binType').value;

  if (!startDate || !endDate) {
    alert('Please select date range before exporting.');
    return;
  }

  try {
    const response = await fetch(

`get-dashboard-data.php?action=getFilteredData&startDate=${startDate}&endDate=
${endDate}&binType=${binType}`
    );
    const data = await response.json();
```

```
    if (data.error) {
      alert('Error fetching data for export.');
      return;
    }

    let csvContent = "data:text/csv;charset=utf-8,";
    csvContent += "Date,Plastic Type,Bottle Count\n";

    for (const [type, values] of Object.entries(data.chartData)) {
      for (const [date, count] of Object.entries(values)) {
        csvContent += `${date},${type},${count}\n`;
      }
    }

    const filename = `bottle_collection_${binType}_${startDate}_to_${endDate}.csv`;
    const link = document.createElement("a");
    link.setAttribute("href", encodeURI(csvContent));
    link.setAttribute("download", filename);
    document.body.appendChild(link);
    link.click();
    document.body.removeChild(link);

  } catch (error) {
    console.error('Error exporting CSV:', error);
    alert('Error exporting data.');
  }
}

// ============ BIN ACTIONS ============
function viewDetails(binId) {
  const row = document.querySelector(`tr[data-bin-id="${binId}"]`);
  if (row) {
```

```
    const detailsHTML = `
      <div class="detail-row">
        <span class="detail-label">Bin ID:</span>
        <span class="detail-value">${binId}</span>
      </div>
      <div class="detail-row">
        <span class="detail-label">Type:</span>
        <span class="detail-value">${row.dataset.type}</span>
      </div>
      <div class="detail-row">
        <span class="detail-label">Fill Level:</span>
        <span class="detail-value">${row.dataset.level}%</span>
      </div>
      <div class="detail-row">
        <span class="detail-label">Status:</span>
        <span class="detail-value">${row.cells[3].textContent}</span>
      </div>
      <div class="detail-row">
        <span class="detail-label">Last Updated:</span>
        <span class="detail-value">${row.dataset.updated}</span>
      </div>
    `;
    document.getElementById('modalDetails').innerHTML = detailsHTML;
    document.getElementById('detailsModal').classList.add('active');
  }
}

async function emptyBin(binId) {
  const row = document.querySelector(`tr[data-bin-id="${binId}"]`);
  if (!row) return;

  const level = parseInt(row.dataset.level);
```

```
  if (level < 95) {
    alert('This bin is not full yet. Bins can only be emptied when they reach 95% or
more.');
    return;
  }

  if (!confirm(`Are you sure you want to empty ${row.dataset.type} Bin #${binId}?`))
{
    return;
  }

  try {
    const formData = new FormData();
    formData.append('binId', binId);

    const response = await fetch('get-dashboard-data.php?action=emptyBin', {
      method: 'POST',
      body: formData
    });

    const data = await response.json();

    if (data.success) {
      alert(`Bin #${binId} has been emptied successfully!`);
      fetchBinData(); // Refresh bin data
    } else {
      alert(data.message || 'Failed to empty bin.');
    }

  } catch (error) {
    console.error('Error emptying bin:', error);
    alert('Error emptying bin.');
```

```
  }
}

// ============ POPUP & MODAL ============
function showFullBinPopup(fullBins) {
  const popup = document.getElementById('popup');
  const popupList = document.getElementById('popup-list');

  if (!popup || !popupList) return;

  popupList.innerHTML = '';
  fullBins.forEach(bin => {
    const li = document.createElement('li');
    li.textContent = `${bin.type} Bin (ID: ${bin.id})`;
    popupList.appendChild(li);
  });

  if (fullBins.length > 0 && popup.style.display !== 'block') {
    popup.style.display = 'block';
    setTimeout(() => closePopup(), 15000);
  }
}

function closePopup() {
  const popup = document.getElementById('popup');
  if (popup) popup.style.display = 'none';
}

function closeModal() {
  document.getElementById('detailsModal').classList.remove('active');
}
```

```
document.getElementById('detailsModal')?.addEventListener('click', function(e) {
  if (e.target === this) closeModal();
});

// ============ SCROLL FUNCTIONS ============
function scrollToBinMonitoring() {
  document.getElementById('binMonitoringSection')?.scrollIntoView({
    behavior: 'smooth',
    block: 'start'
  });
}

function scrollToPlasticBreakdown() {
  document.getElementById('plasticBreakdownSection')?.scrollIntoView({
    behavior: 'smooth',
    block: 'start'
  });
}

// ============ SIGN OUT ============
function showSignOutWarning() {
  document.getElementById('signOutModal').classList.add('active');
}

function closeSignOutModal() {
  document.getElementById('signOutModal').classList.remove('active');
}

function confirmSignOut() {
  closeSignOutModal();
  window.location.href = 'admin-login.php?logout=1';
}
```

```
// ============ DATE INITIALIZATION ============
function initializeDates() {
  const today = new Date();
  const thirtyDaysAgo = new Date(today);
  thirtyDaysAgo.setDate(today.getDate() - 30);

  const endInput = document.getElementById('endDate');
  const startInput = document.getElementById('startDate');

  if (endInput) endInput.valueAsDate = today;
  if (startInput) startInput.valueAsDate = thirtyDaysAgo;
}

// ============ INITIALIZE ON LOAD ============
window.addEventListener('load', async () => {
  initializeDates();
  initBottleChart();

  // Initial data fetch
  await fetchBinData();
  await fetchBottleStats('today');

  document.getElementById('bottleTimeDate').textContent = getDateDisplay('today');

  // Set up auto-refresh every 30 seconds
  updateInterval = setInterval(async () => {
    await fetchBinData();
    await fetchBottleStats(currentTimeRange);
  }, 30000);
});

// Cleanup on page unload
```

```
window.addEventListener('beforeunload', () => {
  if (updateInterval) {
    clearInterval(updateInterval);
  }
});
</script>
</body>
</html>
```

admin-login.php

```
<?php
// admin-login.php
session_start();
require_once 'get-admin-data.php';

// Handle logout
if (isset($_GET['logout']) && $_GET['logout'] == '1') {
    $_SESSION = array();
    if (isset($_COOKIE[session_name()])) {
        setcookie(session_name(), '', time() - 3600, '/');
    }
    session_destroy();
    header('Location: admin-login.php?msg=logged_out');
    exit();
}

// If already logged in, redirect to dashboard
if (isset($_SESSION['admin_logged_in']) && $_SESSION['admin_logged_in'] ===
true) {
    header('Location: admin-dashboard.php');
    exit();
}
```

```
$error = '';
$success = '';

if (isset($_GET['msg'])) {
    if ($_GET['msg'] == 'logged_out') {
        $success = 'You have been successfully logged out.';
    } elseif ($_GET['msg'] == 'session_expired') {
        $error = 'Your session has expired. Please log in again.';
    }
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $username = $_POST['username'] ?? '';
    $password = $_POST['password'] ?? '';

    // Prepare query to check username or email
    $stmt = $conn->prepare("SELECT admin_id, admin_name, admin_password
FROM admins WHERE admin_name = ? OR admin_email = ?");
    $stmt->bind_param("ss", $username, $username);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows === 1) {
        $row = $result->fetch_assoc();

        // Verify password (use password_hash() when storing)
        if (password_verify($password, $row['admin_password'])) {
            $_SESSION['admin_logged_in'] = true;
            $_SESSION['admin_username'] = $row['admin_name'];
            $_SESSION['login_time'] = time();
            $_SESSION['last_activity'] = time();
            ini_set('session.gc_maxlifetime', 3600);
```

```
            header('Location: admin-dashboard.php');
            exit();
        } else {
            $error = 'Invalid password. Please try again.';
        }
    } else {
        $error = 'No account found with that username or email.';
    }

    $stmt->close();
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Administrator Login | BottleSort</title>
  <link rel="icon" type="image/png" href="img/logo8.png">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link
href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&dis
play=swap" rel="stylesheet">
  <link href="https://unpkg.com/aos@2.3.4/dist/aos.css" rel="stylesheet">
  <link rel="stylesheet" href="css/style.css">

  <style>
    body {
      display: flex;
      flex-direction: column;
      min-height: 100vh;
    }
```

```
.login-container {
  flex: 1;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  padding: 3rem 2rem;
  gap: 2rem;
}

.logo-outside {
  width: 300px;
  max-width: 90vw;
}

.login-box {
  background: rgba(255, 255, 255, 0.95);
  color: #004e44;
  padding: 2.5rem;
  border-radius: 10px;
  box-shadow: 0 4px 20px rgba(0,0,0,0.15);
  width: 100%;
  max-width: 420px;
}

.login-box h2 {
  margin-bottom: 1.5rem;
  color: #004e44;
  font-size: 1.8rem;
  text-align: center;
}
```

```
.login-box input {
  width: 100%;
  padding: 0.9rem;
  margin-bottom: 1rem;
  border: 1px solid #ccc;
  border-radius: 5px;
  font-size: 1rem;
  font-family: 'Open Sans', sans-serif;
  transition: border-color 0.3s ease;
}

.login-box input:focus {
  outline: none;
  border-color: #004e44;
}

.password-wrapper {
  position: relative;
  width: 100%;
  margin-bottom: 1rem;
}

.password-wrapper input {
  margin-bottom: 0;
  padding-right: 45px;
}

.password-toggle {
  position: absolute;
  right: 12px;
  top: 50%;
  transform: translateY(-50%);
```

```
  background: none;
  border: none;
  cursor: pointer;
  padding: 5px;
  color: #666;
  font-size: 1.2rem;
  transition: color 0.3s ease;
}

.password-toggle:hover {
  color: #004e44;
}

.login-box button[type="submit"] {
  width: 100%;
  padding: 0.9rem;
  background: #004e44;
  color: white;
  border: none;
  border-radius: 5px;
  font-size: 1rem;
  font-weight: 600;
  cursor: pointer;
  transition: background 0.3s ease;
  font-family: 'Open Sans', sans-serif;
}

.login-box button[type="submit"]:hover {
  background: #007a6e;
}

.error-message {
```

```
      background: #fee;
      color: #c00;
      padding: 0.75rem;
      border-radius: 5px;
      margin-bottom: 1rem;
      text-align: center;
      font-weight: 600;
      border: 1px solid #fcc;
    }

    .success-message {
      background: #efe;
      color: #0a0;
      padding: 0.75rem;
      border-radius: 5px;
      margin-bottom: 1rem;
      text-align: center;
      font-weight: 600;
      border: 1px solid #cfc;
    }

    @media (max-width: 768px) {
      .login-container { padding: 2rem 1rem; }
      .login-box { padding: 2rem; }
      .logo-outside { width: 200px; }
    }
  </style>
</head>
<body>

<?php include 'navbar.php'; ?>
```

```
<section class="login-container">
  <img src="img/logo2.png" alt="BottleSort Logo" class="logo-outside"
data-aos="fade-down">

  <div class="login-box" data-aos="zoom-in">
    <h2>Administrator Login</h2>

    <?php if ($success): ?>
      <div class="success-message"><?= htmlspecialchars($success) ?></div>
    <?php endif; ?>

    <?php if ($error): ?>
      <div class="error-message"><?= htmlspecialchars($error) ?></div>
    <?php endif; ?>

    <form method="POST">
      <input type="text" name="username" placeholder="Username or Email" required
autofocus>

      <div class="password-wrapper">
        <input type="password" name="password" id="password"
placeholder="Password" required>
        <button type="button" class="password-toggle" onclick="togglePassword()"
aria-label="Toggle password visibility">
          <span id="toggleIcon">👁</span>
        </button>
      </div>

      <button type="submit">Log In</button>
    </form>
  </div>
</section>
```

```
<?php include 'footer.php'; ?>

<script src="https://unpkg.com/aos@2.3.4/dist/aos.js"></script>
<script>
  AOS.init({ duration: 1000, once: true });

  function togglePassword() {
    const passwordInput = document.getElementById('password');
    const toggleIcon = document.getElementById('toggleIcon');

    if (passwordInput.type === 'password') {
      passwordInput.type = 'text';
      toggleIcon.textContent = '🙈';
    } else {
      passwordInput.type = 'password';
      toggleIcon.textContent = '👁';
    }
  }
</script>

</body>
</html>
```

style.css

```
/* ========================================================
   CRITICAL FIX: Scrollbar Always Visible
   ======================================================== */
html {
  scroll-behavior: smooth;
  overflow-y: scroll !important; /* FORCES scrollbar on ALL pages */
  height: 100%;
}
```

```
/* ---------------------- GLOBAL ---------------------- */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Open Sans', sans-serif;
  color: #004e44;
  position: relative;
  overflow-x: hidden;
  overflow-y: auto; /* Allow body to scroll */
  min-height: 100vh;
  display: flex; /* Enable flexbox layout */
  flex-direction: column; /* Stack navbar, main, footer vertically */
  background: linear-gradient(
    to right,
    #f9faf9 0%,
    #f5f8f6 50%,
    #f9faf9 100%
  );
  background-attachment: fixed; /* Keeps gradient steady on scroll */
}

/* --- Background image (gradient baked in) --- */
body::after {
  content: "";
  position: fixed;
  top: 0;
  right: 0;
  width: 100%;
```

```
  height: 100%;
  pointer-events: none;
  z-index: 0;
  background: url('../img/bg.png') center center / cover no-repeat;
  opacity: 1;
  mask-image: none;
  -webkit-mask-image: none;
  transition: opacity 0.4s ease-in-out;
}

/* Keep all content above background */
nav, section, footer, main {
  position: relative;
  z-index: 1;
  background: transparent !important;
}

/* Main content wrapper - grows to fill available space */
main {
  flex: 1 0 auto; /* Grow to fill space, don't shrink */
  width: 100%;
  display: flex;
  flex-direction: column;
}

/* ---------------------- NAVBAR ---------------------- */
nav {
  /* Positioning */
  position: sticky;
  top: 0;
  z-index: 1000;
```

```
  /* Layout */
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-shrink: 0;

  /* Box Model */
  padding: 1rem 1.5rem;
  height: 60px;
  min-height: 60px;
  max-height: 60px;

  /* Visual */
  background: #004e44 !important;
  color: white;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  backdrop-filter: none !important;

  /* Animation */
  transition: transform 0.3s ease;
}

nav.nav-hidden {
  transform: translateY(-100%);
}

/* Navbar sections alignment */
.nav-left,
.nav-center,
.nav-right {
  display: flex;
  align-items: center;
```

```
  height: 100%;
}

.nav-left {
  flex: 0 0 auto; /* Don't grow or shrink */
}

.nav-center {
  flex: 1; /* Take up remaining space and center content */
  justify-content: center;
  gap: 0; /* Remove any gap that might affect centering */
}

.nav-right {
  flex: 0 0 140px; /* Fixed width, don't grow or shrink */
  justify-content: flex-end;
}

.nav-left img {
  height: 25px;
  transform: scale(1.5);
  transform-origin: left center;
}

.nav-center a {
  color: white;
  text-decoration: none;
  margin: 0 1rem;
  font-weight: 600;
  display: inline-flex;
  align-items: center;
  transition: 0.2s;
```

```
}

.nav-center a:hover,
.nav-center a.active {
  color: #e0f8f1;
  transform: translateY(-2px);
}

/* Active link styling */
.nav-center a.active {
  border-bottom: 2px solid #e0f8f1;
  padding-bottom: 2px;
}

.nav-center img.nav-icon,
.nav-right img.nav-icon {
  width: 20px;
  height: 20px;
  margin-right: 6px;
  flex-shrink: 0;
}

.nav-right a,
.nav-right-link {
  color: white;
  text-decoration: none;
  font-weight: 600;
  display: inline-flex;
  align-items: center;
  justify-content: center;
  transition: 0.2s;
  white-space: nowrap;
```

```
  padding: 0.5rem 0.75rem;
}

.nav-right a:hover,
.nav-right-link:hover {
  color: #e0f8f1;
  transform: translateY(-2px);
}

/* Mobile Menu Toggle - HIDDEN ON DESKTOP */
.menu-toggle {
  display: none; /* Completely hidden on desktop */
}

/* ---------------------- HERO ---------------------- */
#intro {
  position: relative;
  height: 100vh;
  display: flex;
  align-items: center;
  justify-content: flex-start;
  color: white;
  overflow: hidden;
  background: rgba(0,60,50,0.4); /* Slight overlay for text readability */
}

.intro-content {
  position: relative;
  z-index: 1;
  padding: 0 4rem;
  max-width: 750px;
  text-align: left;
```

```
}

.intro-content img {
  width: 350px;
  margin-bottom: 1rem;
}

.intro-content h1 {
  font-size: 2.8rem;
  line-height: 1.2;
  margin-bottom: 1rem;
}

.intro-content p {
  font-size: 1.1rem;
  line-height: 1.7;
  margin-bottom: 2rem;
  color: #eafaf4;
}

.cta-btn {
  background: #004e44;
  color: white;
  padding: 0.9rem 2rem;
  border-radius: 30px;
  font-weight: 600;
  text-decoration: none;
  transition: background 0.3s ease;
  display: inline-block;
}

.cta-btn:hover {
```

```
  background: #007a6e;
}

/* ---------------------- PAGE SECTIONS ---------------------- */
section {
  padding: 3rem 6%;
  color: #004e44;
}

/* ---------------------- FOOTER ---------------------- */
footer {
  flex-shrink: 0; /* Prevent footer from shrinking */
  margin-top: auto; /* Push footer to bottom on short pages */
  background: #004e44 !important; /* Solid color */
  padding: 1rem;
  text-align: center;
  font-size: 0.9rem;
  color: white;
  box-shadow: 0 -2px 8px rgba(0,0,0,0.1);
  backdrop-filter: none !important; /* Remove blur */
}

/* ---------------------- MEDIA QUERIES ---------------------- */
@media (max-width: 900px) {
  body::after {
    opacity: 0.7;
  }

  /* Navigation adjustments for mobile */
  nav {
    height: auto;
    min-height: 60px;
```

```
  max-height: none;
  padding: 1rem;
  flex-wrap: wrap; /* Allow wrapping on mobile */
}

.nav-left {
  order: 1;
  flex: 0 0 auto;
}

.nav-left img {
  height: 12px;
  transform: scale(1.3);
}

.menu-toggle {
  display: flex; /* Show hamburger on mobile */
  order: 2;
  margin-left: auto;
}

.nav-center {
  order: 4;
  width: 100%;
  flex-direction: column;
  gap: 0.5rem;
  max-height: 0;
  overflow: hidden;
  opacity: 0;
  transition: all 0.3s ease;
  padding: 0;
  flex: 0 0 100%; /* Take full width */
```

```
}

.nav-center.active {
  max-height: 500px;
  opacity: 1;
  padding: 1rem 0;
}

.nav-center a {
  width: 100%;
  justify-content: center;
  margin: 0;
  padding: 0.7rem 1rem;
}

.nav-right {
  order: 3;
  width: 100%;
  max-width: 100%;
  min-width: 100%;
  flex: 0 0 100%; /* Take full width */
  justify-content: center;
  margin-top: 0.5rem;
}

.nav-right a {
  width: auto;
  min-width: auto;
  max-width: none;
  padding: 0.7rem 1rem;
}
```

```
/* Hero section adjustments */
#intro {
  height: auto;
  min-height: 100vh;
  justify-content: center;
}

.intro-content {
  padding: 2rem;
  text-align: center;
  max-width: 100%;
}

.intro-content img {
  width: 250px;
  max-width: 90%;
}

.intro-content h1 {
  font-size: 2rem;
}

.intro-content p {
  font-size: 1rem;
}

section {
  padding: 2rem 4%;
}

footer {
  font-size: 0.85rem;
```

```
  }
}

@media (max-width: 768px) {
  .intro-content img {
    width: 220px;
  }

  .intro-content h1 {
    font-size: 1.8rem;
  }

  .intro-content p {
    font-size: 0.95rem;
  }

  .cta-btn {
    padding: 0.8rem 1.8rem;
    font-size: 0.95rem;
  }
}

@media (max-width: 480px) {
  nav {
    padding: 0.8rem;
  }

  .intro-content {
    padding: 1.5rem;
  }

  .intro-content img {
```

```
    width: 180px;
  }

  .intro-content h1 {
    font-size: 1.5rem;
  }

  .intro-content p {
    font-size: 0.9rem;
  }

  .cta-btn {
    padding: 0.7rem 1.5rem;
    font-size: 0.9rem;
  }

  section {
    padding: 1.5rem 3%;
  }

  footer {
    font-size: 0.8rem;
    padding: 0.8rem;
  }
}
```

RESEARCHERS' PROFILE

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EDUCATIONAL ATTAINMENT

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JUNIOR HIGH LEVEL	St. Vincent's High School Bonifacio, Misamis Occidental (March 2020)
ELEMENTARY LEVEL	Bonifacio Central School Bonifacio, Misamis Occidental (April 2016)

SKILLS

- Proficient in C++, C#, Java, and Python
- Software development
- Hardware and circuit design
- Database management (MySQL, phpMyAdmin)
- Basic machine learning and AI tools

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